

Basics of Agriculture

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COURSE INTRODUCTION

Agriculture is still the backbone of the Indian economy providing livelihood, food security and employment to a large section of people. Many advancements are being made continuously in different sectors of agriculture. In the meantime, the climate changes have also created many new challenges to sustain agriculture in India. The conservation of resources and increasing farm income becomes a tough task, particularly in light of the changing climate scenario. Thus, to understand the complexities of agriculture and find solutions to emerging problems, it becomes important to understand the fundamentals of agriculture. The course “Basics of Agriculture” embodies six blocks covering various aspects that are the frame of Indian Agriculture.

Block 1 on ‘Introduction to Agriculture’ includes 5 different units covering the evolution and development of agriculture, soil and water conservation, and management of soil fertility, irrigation & drainage, and pests and diseases, *etc.* The important developments relating to agriculture that have occurred before and after the independence of India are discussed briefly. The emphasis is given to conserving natural resources, such as soil and water.

Block 2 deals chiefly with the cultivation of important ‘**Field Crops**’. There are five different units in this block. It starts with an introduction to the cropping system concepts. Different terms pertaining to the cropping systems have been described to avoid any confusion in the minds of the students. The prevailing and recent varieties of different crops are listed in respective units. It further covers cultivation practices of major field crops, such as major cereals, coarse grains, oilseeds and pulses.

Block 3 describes the production of ‘**Horticultural Crops**’ for profits and diversification in three different units. It includes scientific practices for the production of fruits, vegetables and flowers. The fruits and vegetables have been classified into different classes or categories. The production practices focus on the prevailing varieties, planting methods input management, *etc.*

Block 4 emphasizes the ‘**Allied Sectors**’ of agriculture to achieve the diversification of farm income. The block details the management of livestock, fishery and many subsidiary enterprises in three different units. Several practices to achieve higher and sustainable productivity from these enterprises are discussed. The production of poultry, goats, sheep and pigs, *etc.* is also covered in the unit of livestock enterprise.

Block 5 covers different aspects related to ‘**High Value Agriculture**’ in four units. It includes units on organic farming, GAP & GAHP, precision and hi-tech farming. The latest information is covered under the units of GAP (Good Agricultural Practices) and GAHP (Good Animal Husbandry Practices). It is also focused on protected cultivation, soilless cultivation, hydroponics, aeroponics, vertical farming, terrace farming, *etc.* Attempts have been made to describe the latest developments in all these fields.

Block 6 is on ‘**Post Harvest Management**’ covering three units. This block incorporates the units on post-harvest handling, and processing of agricultural and animal produce to achieve the overall goal of value addition. It includes

methods and techniques to reduce the post-harvest losses of agricultural produce. The use of modern tools, gadgets and computers is included in the unit on precision farming to save natural resources and farm inputs.

The course assessment as per IGNOU norms will be through assignments and term-end examinations. You must prepare the assignments based on your understanding of the contents of this course and the application of the same to the proposed business activity.



BLOCK 1
INTRODUCTION TO AGRICULTURE

THE PEOPLE'S
UNIVERSITY

BLOCK 1 INTRODUCTION

This block has five units.

The Unit 1 covers the ‘**Evolution and Development of Agriculture**’. It includes topics on the history of agriculture in India before and after the independence. It covers the latest information on the national agricultural research, education and extension system. The unit describes the recent emerging issues in Indian agriculture.

Unit 2 is on ‘**Soil and Water Conservation**’ encompassing various facets to preserve these two natural resources, soil and water. It details the various types of soil erosion and the factors affecting it. Different soil and water conservation measures have also been described in this unit.

Unit 3 is entitled ‘**Irrigation and Drainage**’. It incorporates topics like irrigation demand, irrigation potential, and major irrigation projects in India. Furthermore, it includes the CADP, methods of field drainage, drainage projects in India and of course, the irrigation scheduling.

Unit 4 describes the ‘**Soil Fertility Management**’. The topics encompassed in this unit range from the current fertility status of Indian soils, macronutrients, micronutrients, organic manures, and inorganic fertilizers to biofertilizers.

Unit 5 elaborates on the topic ‘**Pest and Disease Management**’. It describes the causes of pest and disease occurrence in crops, disease triangle, epidemics, diagnostic symptoms, Integrated Pest and Disease Management (IPDM)- principles and approaches, pesticide residues and consequences, *etc.*

The material provided in this block is supplemented with various examples and activities to make the learning process simple and interesting. We have also provided Check Your Progress questions for the self-test at a few places in these units which invariably lead to possible answers to the questions set in those exercises. What perhaps you ought to do, is to go through units and jot down important points as you read, in the space provided in the margin. This will help you in assimilating the content. A list of reference books has been provided at the end of each unit for further detailed reading.

UNIT 1 EVOLUTION AND DEVELOPMENT OF AGRICULTURE

Structure

- 1.0 Objectives
- 1.1 Introduction
- 1.2 History of Indian Agriculture
- 1.3 Agriculture in Prehistoric Era
- 1.4 Agricultural Development Before Independence
- 1.5 Agricultural Development After Independence
 - 1.5.1 Land reforms
 - 1.5.2 Green revolution
 - 1.5.3 Chemical fertilizers
 - 1.5.4 Quality seeds
 - 1.5.5 Irrigation
 - 1.5.6 Farm implements
 - 1.5.7 Prices
- 1.6 Animal Husbandry
- 1.7 Agricultural Research, Extension and Education System
 - 1.7.1 Indian Council of Agricultural Research (ICAR)
 - 1.7.2 State Agricultural Universities (SAU)
 - 1.7.3 Krishi Vigyan Kendra (KVK)
- 1.8 Let Us Sum UP
- 1.9 Keywords
- 1.10 Suggested Readings/ References
- 1.11 Answers to Check Your Progress Exercises

1.0 OBJECTIVES

After reading this unit, you will be able to:

- discuss the changes in agriculture since the inception of civilization;
- explain the phases of agricultural developments in India;
- state the various programmes implemented for the development of agriculture; and
- highlight the development of education, research and extension systems in agriculture.

1.1 INTRODUCTION

The word ‘agriculture’ is derived from the two Latin words ‘Ager’ referring to the soil and ‘Cultura’ referring to the cultivation. Thus, agriculture is a very broad term encompassing all aspects of crop production, livestock

farming, fisheries, poultry, forestry, etc. All primary necessities of human life viz., food, clothing and housing are directly related to agriculture. Agriculture is the main profession and the most important human economic activity worldwide. It differs from other industries being a biological process. Agriculture channelizes the energy from incoming solar radiation into crops and/or livestock through efficient management practices. The physical resource base determines the level of use to which land can be most economically put to agriculture at a particular time. The natural variability of the physical environment makes agriculture one of the most unpredictable human activities.

Soil, climate and plants are three major components of agriculture. In the absence of any one of these, agriculture is not possible. Growing plants in the existing land base under prevailing climatic conditions is called agriculture. Both soil and climate together make up the environment, which influences the completion of the plant life cycle. Plants convert environmental inputs like, solar energy, carbon dioxide, water and soil nutrients into economic products in terms of human or animal food or industrial raw materials. This unit introduces you to agriculture, particularly the evolution and development of agriculture. Development of agriculture during various phases in India, the programmes implemented, education, research and extension systems to support the agriculture sectors, etc. are the major contents discussed in this unit.

1.2 HISTORY OF INDIAN AGRICULTURE

India was a pastoral country before agriculture started. Concurrently, the development of crops and animals took place leading to the emergence of various types of farming systems that are now seen in different parts of the country.

The most important contribution of India towards the development of agriculture is the cultivation of rice, which is now the staple food of South, South-East and East Asia. Sugarcane, several legumes and tropical fruits like mango are also natives of India. *Mohan Jodaro* to *Harappa* territory was the centre of the agricultural revolution in the Indus Valley during the period 3000-1700 BC. The scene of this revolution was in the countries between the river Nile and the Ganges. Excavations in this area have revealed huge granaries. Barely, wheat, sesame, peas, cotton, mustard, date palm and lentil were the main crops during the period. Wooden plough and wheeled cart were in use. *Harappans* knew ginning, spinning and weaving of cotton into cloth. Farmers used sling-boll for scaring birds, cattle, buffalo, goats, sheep, pigs, camel, ass, dog and cat. These have special mention in the list of domesticated animals.

About 1800-1600 BC, *Aryans* migrated to India and got mixed with the *Harappans*. Horses were the main domesticated animal besides cattle. Agriculture was the most important profession during the *Vedic* age (1500-1000 BC). The use of iron implements, particularly iron plough became prevalent. Besides, barley, wheat, beans, sesame, millet and rice are frequently referred to in *Vedas*. The importance of trees is mentioned during

the Buddhist period (600 BC). The importance of sowing season, climate, cloud, soil fertility, irrigation and weed control has been emphasized several times in *Mahabharat* and *Ramayan* during the period 1000 and 500 BC, respectively. Irrigation from rivers, lakes and reservoirs has also been mentioned.

During the first century of the Christian era and 300 AD, the cultivation of rice with irrigation from the Cauvery River was the most important event in Indian agriculture. After 300 AD, during the period of Chandragupta II, *Amarkosha* was written, in which there is a mention of classification of soil and land use, irrigation, implements and use of manures. Rice transplanting was practised till that period besides the cultivation of several field crops, vegetables, fruits etc.

During the Kannauj Empire of Harsha (606-647 AD), fine-scented rice was grown in *Magadh*. Pomegranates and sweet oranges were also grown during this period. There is mention of land manuring, crop rotation, irrigation, tillage, implements, crop protection and meteorology in the book "*Krishi Parastra*" written between 950-1100 AD. Several water reservoirs were made during the period 1336-1646 AD and irrigation channels were constructed 'Anantaraja Sagar' in Vijayanagar district of A.P. (1337 AD) is one example. Later Babar laid the foundation of the Mughal dynasty in India. Except for gardens, flowers and provision of irrigation mainly in South India, there was nothing special about agricultural development during this regime. The peasants became disinterested in agriculture because of the large sum extracted from them as peasantry (tax).

1.3 AGRICULTURE IN PREHISTORIC ERA

Since agriculture fulfils the need for food for humans and animals, it appears to be a very primitive step of human civilization. The earliest man (*Homo erectus*) emerged around one and half million years ago and spread throughout the old world tropics and later to temperate zones. About 500 thousand years ago, he probably learnt to control and use fire. The earliest man was distinguished from fellow animals by his intelligence and skill in making tools and selection for self-use.

Homo sapiens, the direct ancestor of modern man lived 250 thousand years ago. *Homo sapiens*, the modern man, appeared in Africa 35 thousand years ago. He was distinguished from all other extinct species of genus *Homo*, by a large brain, small teeth and chin and capacity for making and using tools. He hunted a variety of animals and cooked their meat on fire. The weapons for hunting were boulders and spears of wood tipped with blades of flint. He also used stone-tipped arrows. Later, he domesticated the dog, which greatly helped him in hunting. Apart from the meat, he gathered a variety of seeds, leaves, fruits, roots, and other parts of plants from the forest for food. He had no control over the food supply and was unable to cloth and shelter adequately. During the period from 8700 BC to 7700 BC, he domesticated animals and turned them into a *herdsman*. He first domesticated sheep and later goats. Between the period 7500 BC to 6500 BC, man gradually shifted from hunting towards agriculture. Stone axes were used for cutting trees and

fire for burning forests.

Agriculture has no single and simple origin. It was started in different parts of the world during different periods. The evolution of agriculture took place gradually with the changes in human civilization, mostly based on experiences. The exact evidence is not known about the shifting of man into agriculture from hunting and gathering animals for food. The hunting and gathering of food was a difficult job to collect the food for each individual, while pastoral life and agriculture could support a group. Thus, residing in groups or camps and dependency of food on plants started in human civilization. It is expected that pre-historic agriculture might have developed in several steps:

- a) **Conservation of desirable native plants:** By using various parts of plants like roots, seeds, leaves, flowers, fruits, etc. probably man would have identified the desirable plants, which he referred to as crop plants. Later on, he might have tried to conserve those plants from danger.
- b) **Collection and preservation of seeds:** After eating the fruits, man would have thrown the seeds elsewhere and they have seen the germination of the same type of plants from thrown seeds. With this experience, they might have learnt and decided to grow these plants by using the collected and preserved seeds at the desirable place. In this way, they would have started the collection and preservation of seeds of useful plants.
- c) **Sowing of seeds:** With the increase in population, when man would have faced a shortage of foods, they might have started sowing based on climatic suitability.
- d) **Tilling of land:** When birds or air or flow of water damaged seeds, they might have thought to lose the hard soil surface with the help of stone or wooden tools putting the seeds inside the soil to avoid the damages.
- e) **Shifting cultivation:** After continuous growing of the same plants for a long time, they might have observed poor growth in plants and finally the yield. Thus, they might have left the land by using another land for growing the plants. Initially, there was enough land for raising the crops/plants and when they found a reduction in the yield on the same land, then they might have shifted their farming to other newly acquired lands by leaving the old land and living places. This system of farming is referred to as shifting cultivation and is followed in some tribal belts of Africa and Asia.
- f) **Weed control:** Farmers in ancient ages might have seen growing several unwanted plants with the desirable plants competing with them for growth resources. With this experience, they might have started control of these unwanted plants (weeds) from the fields by uprooting or cutting etc. Women mostly performed this work from the very beginning.
- g) **Fallowing of land:** With the gradual increase in the population, people have probably started living in groups together and established villages with some economic management and then the practice of shifting

cultivation would have been avoided. On experiencing the reduction in yield due to the continuous raising of crop plants on the same soil, they might have started following some portion of land and tilling of old fallow land with unwanted green plants. Then they might have experienced good yield from the fallow land.

- h) **Use of domestic wastes as plant nourishment:** By the use of domestic wastes in agriculture as plant nourishment, people must have experienced the good growth of crop plants. Then they might have started to preserve these wastes and use them in the cropped land as manure. Meanwhile, they might have domesticated several animals like sheep, goats, cows, buffalo etc. for their support, and then they must have seen the tremendous effect of dung, urine and animal wastes in improving the soil fertility. Thus, they started the use of ashes, domestic wastes, animal dung and urine and animal wastes in the process of agricultural production.
- i) **Following crop rotation:** In due time, the cultivators must have realized that growing cereals after legume crop give better yield, which resulted in crop rotation practice. Later, they must have known that legume plants fix atmospheric nitrogen through their root nodules.
- j) **Green manuring:** Later with this experience, legumes must have been grown and then turned into the soil with the help of available equipment to improve the soil fertility.

The important events in the history of agriculture after 7500 BC can be listed as below.

Period	Event
7500 BC	Conservation of crops (wheat, barley)
6000 BC	Domestication of cattle and pigs
4000 BC	Cultivation of maize
3500 BC	Cultivation of potato
3400 BC	Wheel was invented
3000 BC	Bronze was used to make tools
2900 BC	Plough was invented and irrigation started
2700 BC	Silk moth domesticated
2300 BC	Cultivation of chickpea, pea, mustard and cotton Domestication of fowl, buffalo and elephant
2200 BC	Cultivation of rice in India
1200 BC	Cultivation of finger millet (<i>Ragi</i>)
1725 BC	Cultivation of sorghum
1700 BC	Taming of horses
1500 BC	Cultivation of sugarcane and irrigation from wells
1400 BC	Use of iron

1500 AD	Cultivation of sweet orange, sour orange, wild brinjal, pomegranate
1600 AD	Introduction of several crops in India from Portugal like sweet potato, arrowroot, cassava, tomato, chillies, pumpkin, papaya, pineapple, guava, custard apple, groundnut, cashew nut, tobacco, American cotton, rubber.

Check Your Progress Exercise 1

Note:a) Write your answer in the space given below.

b) Compare your answers with those given at the end of the unit.

1. State true/false.

a) Agriculture is different from industries being a biological process.

T/F

b) The use of iron implements, particularly iron ploughs became prevalent in the *Vedic* age.

T/F

c) *Krishi Parastra* was written during the period 1336-1646 AD.

T/F

2. Enlist any three developments that took place in pre-historic agriculture.

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1.4 AGRICULTURAL DEVELOPMENT BEFORE INDEPENDENCE

Different dynasties had promulgated rules mostly related to the collection of revenue and tax, which influenced agriculture. During the domain of Arabs, taxes were laid on land in the form of two-fifths of produce or cash. Later in the regime of Alauddin, the revenue was fixed at one-half of the produce, however, the charges were calculated based on standard yields. Agriculture suffered due to the framing of rules not being favourable to the farming community. The produce was collected and kept under storage by the rulers. Agriculture was encouraged by Ghiasuddin Tughlak (1321-1325). The taxes in the form of land revenue were reduced. The land revenues were settled upon the principle which was not more than one-tenth of the produce. Granaries were established to store grains for long periods. During the time of Mohammed bin Tughlak, the Sultan (king) used to possess the best fertile land and to enhance productivity, and resources were used officially. According to Ibn Batuta (1325-1354), work on the reclamation of wasteland was started along with State farming. Wells were dug and cultivators were given loans. The rotation of crops was also initiated. Attention was given to planting various economical fruit trees. Firoz Shah Tughlak (1351-1388) introduced the *Jagir* system by giving lands to the actual owners. The canal

system for irrigation was introduced and the Western Yamuna canal was constructed to use the flood water in 1355. The tax was introduced on irrigation water. The barren land became usable for agriculture.

During Akbar's period (1555-1605), Todar Mal made reforms in the revenue system by obtaining the correct measurement of the land; ascertaining the quality of the produce and settlement on the equivalent in money. Productivity based on fertility was also considered. Tax on fruit gardens was also introduced.

Taquavi loans were allowed to cultivators for the purchase of seed and cattle on surety for their payment. Repayment was made at the first harvest and remaining by the next. The cultivators who were cultivating the land personally were given permanent and hereditary occupancy of the land they tilled.

Later the burden of taxation increased on cultivators which resulted in revolts. As a result, the *Zamindari* system was abolished during 1669-1710 in East Punjab. Hastings in 1772 changed the revenue system according to which lands were allotted to farmers for five years and *Pattas* were given which led to permanent settlement, but the cultivators and their tenants remained to live in distress.

The Europeans, primarily the Britishers, arrived in India in the early seventeenth century and by the 18th century had made a profound impact on India. India was forced, for the first time into a coordinated role within a world system based on industrial production rather than agriculture. Many of the dynamic craft and cottage industries that had long attracted foreigners to India suffered extensively under competition with new modes of mass production fostered by the British. Modern institutions, such as Universities, technologies, rail lines, roads, and mass communication systems were developed to serve the British rather than the Indian economic interest. A country, that in the 18th century was a magnet for trade, by the 20th century, became underdeveloped and overpopulated under alien domination.

During the British regime, attempts were made to boost agricultural production. General improvements made in transport like a rail system, the promotion of export of some agricultural commodities, and the introduction of a land tenure system also helped agriculture in general. The Department of Agriculture was created by the Government of India in 1871, which was followed by the State Governments. In 1943, the All India Policy on Agriculture was drafted. As a result of the Bengal famine in 1942, efforts were made to increase agricultural production in the country. Several steps were taken to improve agriculture in the pre-independence era.

Land ownership was given to a small group of people to collect the rent from the individual farmers and pay to the Government, a system known as '*Zamindari*'. There was also the '*Ryotwari*' system in which the ruler used to collect rent directly from farmers, who had settled on the land. Farmers had no security of possessed land, hence they had no interest in land development. Irrigation schemes were initiated in major river deltas to raise additional revenue.

A separate Department of Agriculture was created at the national level in 1871. Some of the provinces (now states) also created independent Agriculture Departments. Facilities for agricultural education were created at Coimbatore in 1878 and Pune in 1890. After the great famine of 1876-77, a Famine Commission was appointed in 1880. Dr J.A. Voelekar, Consultant Chairman of the Royal Agriculture Society submitted the report on improvement in Indian agriculture in 1891.

In the first decade of the 20th century, the Imperial (Now Indian) Agricultural Research Institute was set up in 1905 at Pusa (Bihar). Subsequently, agricultural colleges were started at Coimbatore, Pune, Nagpur, and Kanpur. Royal Commission (1926) report was accepted in 1928. The Department of Food was created in 1942 to meet the food requirement. Grow More Food Campaign, Food Policy Committee, Famine Enquiry Committee and Fertilizer Production Programme were initiated. In January 1946, the Statement of Agriculture and Food Policy in India was issued with 10 objectives to promote the welfare of the people and improvement of their standard of living. Steps were taken to organize research support for agriculture through research institutions.

1.5 AGRICULTURAL DEVELOPMENT AFTER INDEPENDENCE

During the time of independence, India was in great distress to supply the food grain to feed its population. Therefore, many initiatives were taken to enhance the production and productivity of agriculture. The important initiatives are discussed as under:

1.5.1 Land Reforms

At the time of independence, three types of land tenure systems prevailed in the country; the *Zamindari*, *Mahelwari* and *Ryotwari*. They differed in the mode of payment of land revenue. *Zamindari* and *Mahelwari* exploited farmers as it created a class of *Zamindars* who did not cultivate the land themselves but got the cultivation done by labourers. Such labourers depended on *Zamindars* for dwellings, food and other needs. After independence, land reforms were taken to stop the exploitation of actual tillers to remove impediments to increase agricultural production, eliminate all types of exploitations and social injustice, provide security to the actual tillers and assure equality of status and opportunity to all sections of the rural population.

Attempts were made to abolish the role of intermediaries, reform the tenancy conditions by regulation of rent, security of tenure and ownership rights to tenants, redistributing the land, consolidating landholdings and introducing cooperative farming. Ceiling on agricultural holdings was done. A statutory limit on the area of land an individual can possess was fixed including existing holdings and future acquisition. This is based on the local conditions of the states in economic or uneconomical lands. The land ceiling helps provide land to landless persons who may work with better zeal and produce more by increasing productivity. The fallow land is brought under

cultivation. Cooperative farming has also been followed for better production.

1.5.2 Green Revolution

The need to increase agricultural production due to the continuing stagnation of production on the one hand and rapidly increasing demand, on the other hand, was realized. A team of experts sponsored by the Ford Foundation was invited by the Government of India in the latter half of the second five-year plan to suggest ways and means to increase agricultural production and productivity. The team submitted its report entitled "Indian Food Crisis and Steps to Meet" in April 1959. This report suggested increasing production and productivity in some selected regions of the country with emphasis on the use of modern inputs like fertilizers, credit, marketing facilities etc.

The government introduced an Intensive Development Programme in seven selected districts of seven states in 1960 and this programme was named IADP (Intensive Area Development Programme). The selected districts under IADP were required to have assured irrigation, minimum hazards like floods, drought and soil erosion, use of fertilizers, plant protection measures and easy availability of agricultural credit. The programme was implemented in West Godavari (AP), Shahabad (Bihar), Raipur (MP), Thanjavur (Tamil Nadu), Ludhiana (Punjab), Aligarh (UP) and Pali (Rajasthan). The first four were selected for rice, the next two for wheat and the last one for millet. Eight more districts were added in 1962-63. This programme was later extended to all states by selecting one district from each state for intensive development. In October 1965, the programme was widened to 114 districts out of 325, in the name of the Intensive Agricultural Area Programme.

The period of the mid-sixties was very significant from the point of view of agriculture. New high-yielding varieties of wheat were developed in Mexico by Prof. Norman Borlaug and his associates. As a result of this, the production of wheat per hectare rose to the high level of 5000 to 6000 kg in Mexico in 1965. Taiwan also recorded an almost similar yield. These high-yielding varieties (HYV) required proper irrigation facilities and extensive use of fertilizers and pesticides. Accordingly, these were introduced in the form of a package programme. Dwarf high-yielding varieties of wheat like 'Kalyan Sona' and 'Sonalika' were developed using the 'Norin 10' dwarfing gene. In 1965, on the initiation of C. Subramaniam, the then Minister of Food and Agriculture, 250 tonnes of wheat seed of varieties 'Lerma Rojo 64' and 'Sonora 64' were imported from Mexico. The area under dwarf high-yielding varieties of wheat was increased from 4 ha in 1964 to 4 m ha in 1971. This could give 20 to 30 per cent more yield. The farming area was continuously expanded and multiple-crop cultivation was encouraged.

The term green revolution was coined in the 1960s when improved varieties of wheat resulted in quite high yields as they were more responsive to controlled irrigation, chemical fertilizers and better plant protection measures. Later, new high-yielding varieties of rice and maize were also developed, which increased the yields tremendously. This was followed by continued expansion of farming areas and a double cropping system. The value of seed with improved genetics was realized. The Green Revolution

resulted in a record yield of 131 million tonnes in 1978-79. The yield per hectare was increased by 30 per cent.

Besides advantages, the “Green Revolution” has also created some problems. To arrange the required inputs, farmers often went into debt. Many small landholders and landless farm workers migrated to urban areas. Assured irrigation has created the problem of salinization, waterlogging and lowering of water tables in certain areas. The pesticide residue problem increased due to the increased use of pesticides. The “Green Revolution” in agriculture has affected both agricultural biodiversity and wild biodiversity negatively as it depended more on a few varieties in each crop.

1.5.3 Chemical Fertilizers

Chemical fertilizer is an important input in increasing agricultural production by providing soil-deficient nutrients to plants, which are necessary for growth and have played a vital role in the green revolution. By 1980, India had more than 100 fertilizer factories manufacturing urea, ammonium phosphate and potash. The utility of chemical fertilizers was realized, due to drought in 1950 and 1951, which created a food shortage resulting in the import of huge quantities of food grains.

Presently,

At present, there are 32 large-size urea plants in the country manufacturing urea, 19 units producing DAP and complex fertilizers and 2 units manufacturing Ammonium Sulphate as a by-product. The production of Urea during the year 2019-20 was 244.55 lakh metric tonnes (LMT) and the production of DAP and complex fertilizers was 132.11 LMT. The estimated production of Urea during 2020-21 would be 262.61 LMT, which is higher than the previous year and the estimated production of DAP and complex fertilizers would be 133.43 LMT.

The fertilizer consumption has increased from 2 kg/ha in 1961 to 134.4 kg/ha in 2019-20, which helped increase the foodgrain production from 83 million tonnes to 297.5 million tonnes during the same period. Emphasis is being given on bio-fertilizers like nitrogen-fixing bacteria, phosphorus-solubilising bacteria, blue-green algae, vermin-compost etc. Organic farming is also being given the thrust these days.

1.5.4 Quality Seeds

The seeds of pathogen-free high yielding quality is required in any crop production programme. Farmers used to select, save and store seeds for sowing. Seed Review Committee in 1968 and subsequent committees reviewed the issues related to seeds. For quality seed production, the National Seed Corporation (1963) and State Farms Corporations (1969) were established. Indian Seed Act was passed in 1966 and seed rules were framed in 1968 to control the quality of seed. Indian Minimum Seed Standards were fixed in 1971, which were revised in 1988. The parameters of seed quality were genetic purity, free from other crop seeds, weed seeds, inert matter, seed moisture, seed germination and seed-borne pathogens.

The seeds are classified into Nucleus seed, Breeder seed, Foundation seed and certified seed. Nucleus seed is the basic seed developed by the concerned breeder. The breeder seed is produced from the nucleus seed by the breeders. From the breeder seed, the foundation seed is produced and from the foundation seed, the certified seed is produced. The breeder seeds are produced by state agricultural universities (SAUs), and agricultural institutes. The foundation and certified seeds are produced by the State Seed Corporation as well as SAUs and Institutes. Seed certification is done by the State Seed Certification Agencies. Efforts are being made to increase the seed replacement rate.

1.5.5 Irrigation

Water is an essential component in the crop production system. The selection of crops in a region is based on the availability of water. Collection of rainwater and its utilization at distant places and at different times through canals is an age-old practice. The area under irrigation has increased and the efforts are continuing to increase it further. In 1950, the net irrigated area was 20.6 million hectares, which was about 27 per cent of the total irrigation potential of the country. Large dams were constructed like Bhakra Nangal dam (1948-63), Mayurakshi scheme on Mayurakshi River (1946-56), Hirakund project on Mahanadi River (1948-57); Damodar Valley project on Damodar River (1950-58); Rihand dam on Sone River (1952-62), Tungabhadra project on Tungabhadra River (1945-54); Kakrapara project on Tapti River (1949-55), etc. Several large dams are still under construction on important rivers. They not only provide water for irrigation but are also used for electricity generation.

Irrigation is being done through open wells and tube wells. Small and large watersheds are made in villages or a group of villages to collect rainwater and use it in *Kharif* and *rabi* seasons as per need. Sprinkler and drip irrigation systems have evolved to have better water use efficiency

In 2022-23, out of the 141 million hectares of cultivated land in the country, nearly 73 million hectares, or 52%, had irrigation access. For the first time, more than half of India's cultivated land now has access to assured irrigation led by an expansion in micro-projects, which have higher water-use efficiency. However, the large area (48%) is still rainfed and efforts are continued to use available water further. Projects on the joining of rivers are being planned to have better utilization of available water.

1.5.6 Farm Implements

In the early days, farmers used to develop implements as per their needs. They used ploughs, hoes, sickles, etc. which were modified over time. Their place has been taken over by farm machines like tractors, cultivators, seed drills, harvesters, threshers, combine harvesters etc. Machines have been developed looking to the need for special crops like sugarcane, potato, groundnut etc. for planting as well as harvesting. For the seed industry, special machinery has been developed for processing, cleaning, storage, packaging, etc. Production of tractors started in 1961-62 in India and later several manufacturing units started working. The mechanization enabled

farmers to change their cropping pattern preferring more profitable crops.

1.5.7 Prices

The cultivators were exploited by merchants as they offered minimum prices for their produce. At harvest, the merchants used to dictate the price, purchased at a cheaper price, retained the produce and sold at a higher price later. Looking at the input cost of fertilizers, improved seeds, farm equipment, intercultural operations, etc. agriculture became a commercial enterprise. The Agricultural Prices Commission was established in 1965, now renamed as Commission for Agriculture Costs and Prices (CACP) with a function to determine a minimum support price in advance of the harvest season for all major crops. Based on the increase in the cost of inputs, the support price is increased accordingly year-wise. The Food Corporation of India was formed in 1965 to procure, store, transport, distribute and control the sale of food grains. This helped the cultivators in selling their produce at a reasonable price.

Check Your Progress Exercise 2

Note: a) Write your answer in the space given below.

b) Compare your answer with those given at the end of the unit.

1. Name any two programmes initiated for the development of Indian agriculture after the independence.

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2. What is the role of chemical fertilizers in crop production?

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3. How much area could be covered under irrigation?

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1.6 ANIMAL HUSBANDRY

Animal Husbandry is a part of the integrated agricultural system. Dairy, poultry, fisheries, goats and sheep are closely associated with agriculture.

- i. **Dairy:** Cattle were indispensable in the past and even today; they are important as they are used in agricultural practices, like ploughing, sowing, interculture, threshing operations, transport through bullock carts, etc. The development of cattle was both for milk as well as for draught capacity. Some Indian breeds of cattle are excellent. They are classified into the milch breed, general utility breed and draught breed.

India is the largest milk-producing country in the world. About 70 million households are engaged in dairy. The dairy depended on the crop residues and other leftover, by-products as feed, but now grains and other required nutrients are being provided. The livestock efficiency has been increased by introducing new breeds and hybridization programmes. The upkeep of livestock has experienced vast advancement. The handling of milk is highly scientific and is transported to long distances. National Dairy Development Board (NDDB) through the Operation Flood programme not only made India self-sufficient in milk and milk products but started exporting to other countries. This is named as “White Revolution”. The Operation Flood programme was implemented in phases through cooperatives from 1970 when the production of milk was 20 million tonnes. In 1991-92, it increased to 55.7 and in 2020-21 to 221.1 million tonnes. The dairy industry has developed most successfully.

- ii. **Poultry:** Poultry includes domestic fowls that are reared for their flesh, eggs and feathers like chickens, ducks, geese, turkeys, guinea fowls, pigeons, etc. However, usually, chickens have gained greater importance as poultry. Much advancement has been made in chickens to home breeds with more egg-laying capacity and tender flesh. Poultryies are now hand-managed as well as fully mechanized.

India is the second-largest producer of eggs and one of the largest broiler producers in the world. The country has developed the essential technology and capability to provide the required inputs for poultry farming.

- iii. **Fisheries:** Fisheries have received great attention as a large population depends on the fishes of seashores, riverbanks, large tanks, etc. Excellent development has taken place in the advancement of fisheries from egg management to full development and post-harvest technology. The fishes are of various kinds, like catfish, mullets, dorab, herrings, pomfrets, tuna, etc. The total marine and inland fish production in India during the year 2021-22 was 162.48 million tonnes.

- iv. **Goats and sheep:** Goats and sheep are an important component in some farming systems. These are also raised exclusively by certain farmers on a small scale. Goats reproduce fast and can live on sparse vegetation and extreme climatic conditions. They are good for arid and semi-arid zones.

They provide meat, skin, milk and manure. Sheep are useful for getting wool. 'Chegu' an inhabitant of temperate regions like Himachal Pradesh and Uttarakhand is famous for its very soft, fine and warm fibre called 'Pashmina'. Another breed 'Changthangi' of the Leh region also produces similar fine fibre. The long fibres of 'Gaddi' and 'Marwadi' sheep are used for making rugs, ropes etc. In India, more than 20 well-defined breeds of goats exist besides some regional goats. 'Barbari' and 'Jamunapari' from UP, 'Beetal' from Punjab, 'Surati' from Gujarat, 'Sirohi' from Rajasthan, 'Kannaiaadu' from Tamil Nadu are some important breeds used for different purposes. Efforts are being made for further advancements and improvements.

Check Your Progress Exercise 3

Note: a) Write your answer in the space given below.

b) Compare your answer with those given at the end of the unit.

1. How animal husbandry is a part of the integrated agricultural system?

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2. What are the advances made in the fisheries sector?

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1.7 AGRICULTURAL RESEARCH, EXTENSION AND EDUCATION SYSTEM

During the post-independence, the extension of agricultural education in the country has witnessed a sea change both in content and quality, because of the changed scenario of agriculture. On the recommendation of the Education Commission (1948), Joint Indo-American teams of 1955, 1959 and 1961, and Education Commission (1964-66), State Agricultural Universities (SAU) were established to integrate agricultural education, research and extension activities in the country. Presently, there are 62 State Agricultural Universities, 5 Deemed to be Universities, 2 Central Agricultural Universities and 4 Central Universities with Agriculture Faculty. These institutions include agriculture, veterinary science, agricultural engineering, forestry, dairy technology, food technology, horticulture, fisheries, home science and agricultural marketing subjects. The seventh schedule of the constitution of India allocated the responsibility of agricultural education, research and extension functions to the Union and State Governments. The Indian Council of Agricultural Research (ICAR), as the apex agency, is responsible for the growth and development of agricultural education and research in India.

1.7.1 Indian Council of Agricultural Research (ICAR)

The Imperial Council of Agricultural Research was constituted to promote, guide and coordinate agricultural research in the country, to impart training to researchers and to provide scholarships for students. It included animal husbandry as well. The resolution was passed on 23 May 1929. After independence, the Imperial word was replaced by Indian.

At present, ICAR is headed by the Director General, supported by Deputy Director Generals and Assistant Director Generals. The ICAR coordinates research in the country through its wide network of National and Central Research Institutes (65), National Research Centres (14), Project Directorates (13), National Bureaux (6), All India Coordinated Research Projects (60), network projects (19) and other programmes (10). The list of important Institutions is given below:

Institutions - 65	
1.	ICAR-Central Island Agricultural Research Institute, Port Blair
2.	ICAR-Central Arid Zone Research Institute, Jodhpur
3.	ICAR-Central Avian Research Institute, Izatnagar
4.	ICAR-Central Inland Fisheries Research Institute, Barrackpore
5.	ICAR-Central Institute Brackishwater Aquaculture, Chennai
6.	ICAR-Central Institute for Research on Buffaloes, Hissar
7.	ICAR-Central Institute for Research on Goats, Makhdoom
8.	ICAR-Central Institute of Agricultural Engineering, Bhopal
9.	ICAR-Central Institute for Arid Horticulture, Bikaner
10.	ICAR-Central Institute of Cotton Research, Nagpur
11.	ICAR-Central Institute of Fisheries Technology, Cochin
12.	ICAR-Central Institute of Freshwater Aquaculture, Bhubneshwar
13.	ICAR-Central Institute of Research on Cotton Technology, Mumbai
14.	ICAR-Central Institute of Sub Tropical Horticulture, Lucknow
15.	ICAR-Central Institute of Temperate Horticulture, Srinagar
16.	ICAR-Central Institute on Post Harvest Engineering and Technology, Ludhiana
17.	ICAR-Central Marine Fisheries Research Institute, Kochi
18.	ICAR-Central Plantation Crops Research Institute, Kasargod
19.	ICAR-Central Potato Research Institute, Shimla
20.	ICAR-Central Research Institute for Jute and Allied Fibres, Barrackpore
21.	ICAR-Central Research Institute of Dryland Agriculture, Hyderabad
22.	ICAR-National Rice Research Institute, Cuttack
23.	ICAR-Central Sheep and Wool Research Institute, Avikanagar, Rajasthan
24.	ICAR- Indian Institute of Soil and Water Conservation, Dehradun

25.	ICAR-Central Soil Salinity Research Institute, Karnal
26.	ICAR-Central Tobacco Research Institute, Rajahmundry
27.	ICAR-Central Tuber Crops Research Institute, Trivandrum
28.	ICAR-ICAR Research Complex for Eastern Region, Patna
29.	ICAR-ICAR Research Complex for NEH Region, Barapani
30.	ICAR-Central Coastal Agricultural Research Institute, Ela, Old Goa, Goa
31.	ICAR-Indian Agricultural Statistics Research Institute, New Delhi
32.	ICAR-Indian Grassland and Fodder Research Institute, Jhansi
33.	ICAR-Indian Institute of Agricultural Biotechnology, Ranchi
34.	ICAR-Indian Institute of Horticultural Research, Bengaluru
35.	ICAR-Indian Institute of Natural Resins and Gums, Ranchi
36.	ICAR-Indian Institute of Pulses Research, Kanpur
37.	ICAR-Indian Institute of Soil Sciences, Bhopal
38.	ICAR-Indian Institute of Spices Research, Calicut
39.	ICAR-Indian Institute of Sugarcane Research, Lucknow
40.	ICAR-Indian Institute of Vegetable Research, Varanasi
41.	ICAR-National Academy of Agricultural Research & Management, Hyderabad
42.	ICAR-National Institute of Biotic Stresses Management, Raipur
43.	ICAR-National Institute of Abiotic Stress Management, Malegaon, Maharashtra
44.	ICAR-National Institute of Animal Nutrition and Physiology, Bengaluru
45.	ICAR-National Institute of Natural Fibre Engineering and Technology, Kolkata, Kolkata
46.	ICAR-National Institute of Veterinary Epidemiology and Disease Informatics, Hebbal, Bengaluru
47.	ICAR-Sugarcane Breeding Institute, Coimbatore
48.	ICAR-Vivekananda Parvatiya Krishi Anusandhan Sansthan, Almora
49.	ICAR-Central Institute for Research on Cattle, Meerut, Uttar Pradesh
50.	ICAR-National Institute of High Security Animal Diseases, Bhopal
51.	ICAR-Indian Institute of Maize Research, Ludhiana
52.	ICAR- Central Agroforestry Research Institute, Jhansi
53.	ICAR-National Institute of Agricultural Economics and Policy Research, New Delhi
54.	ICAR- Indian Institute of Wheat and Barley Research, Karnal
55.	ICAR- Indian Institute of Farming Systems Research, Modipuram
56.	ICAR- Indian Institute of Millets Research, Hyderabad
57.	ICAR- Indian Institute of Oilseeds Research, Hyderabad

58.	ICAR- Indian Institute of Oil Palm Research, Pedavegi, West Godavari
59.	ICAR- Indian Institute of Water Management, Bhubaneswar
60.	ICAR-Indian Institute of Rice Research, Hyderabad
61.	ICAR- Central Institute for Women in Agriculture, Bhubaneswar
62.	ICAR-Central Citrus Research Institute, Nagpur
63.	ICAR-Indian Institute of Seed Science, Mau
64.	ICAR-Indian Agricultural Research Institute, Post Box No. 48, Hazaribag 825 301, Jharkhand
65.	ICAR-National Institute for Plant Biotechnology, New Delhi

National Research Centres - 14	
1.	ICAR-National Research Centre for Banana, Trichi
2.	ICAR-National Research Centre for Grapes, Pune
3.	ICAR-National Research Centre for Litchi, Muzaffarpur
4.	ICAR-National Research Centre for Pomegranate, Solapur
5.	ICAR-National Research Centre on Camel, Bikaner
6.	ICAR-National Research Centre on Equines, Hisar
7.	ICAR-National Research Centre on Meat, Hyderabad
8.	ICAR-National Research Centre on Mithun, Medziphema, Nagaland
9.	ICAR-National Research Centre on Orchids, Pakyong, Sikkim
10.	ICAR-National Research Centre on Pig, Guwahati
11.	ICAR-National Research Centre on Seed Spices, Ajmer
12.	ICAR-National Research Centre on Yak, West Kemang
13.	ICAR-National Centre for Integrated Pest Management, New Delhi
14.	Mahatma Gandhi Integrated Farming Research Institute ,Motihari
National Bureaux - 6	
1.	ICAR-National Bureau of Plant Genetics Resources, New Delhi
2.	ICAR-National Bureau of Agriculturally Important Micro-organisms, Mau, Uttar Pradesh
3.	ICAR-National Bureau of Agricultural Insect Resources, Bengaluru
4.	ICAR-National Bureau of Soil Survey and Land Use Planning, Nagpur
5.	ICAR-National Bureau of Animal Genetic Resources, Karnal
6.	ICAR-National Bureau of Fish Genetic Resources, Lucknow
Directorates/Project Directorates - 13	
1.	ICAR-Directorate of Groundnut Research, Junagarh
2.	ICAR-Directorate of Soybean Research, Indore
3.	ICAR-Directorate of Rapeseed & Mustard Research, Bharatpur
4.	ICAR-Directorate of Mushroom Research, Solan
5.	ICAR-Directorate on Onion and Garlic Research, Pune

6.	ICAR-Directorate of Cashew Research, Puttur
7.	ICAR-Directorate of Medicinal and Aromatic Plants Research, Anand
8.	ICAR-Directorate of Floricultural Research, Pune, Maharashtra
9.	ICAR-Directorate of Weed Research, Jabalpur
10.	ICAR-Project Directorate on Foot & Mouth Disease, Mukteshwar
11.	ICAR-Directorate of Poultry Research, Hyderabad
12.	ICAR-Directorate of Knowledge Management in Agriculture (DKMA), New Delhi
13.	ICAR-Directorate of Cold Water Fisheries Research, Bhimtal, Nainital

1.7.2 State Agricultural Universities (SAU)

There are 62 State Agricultural Universities (SAU), 5 Deemed to be Universities and 3 Central Agricultural Universities (CAU). Some states have more than one Agricultural university. Agricultural universities have colleges with different functions in different parts of the states. All the agricultural universities have faculties as per need and facilities to cover different aspects of agriculture.

1.7.3 Krishi Vigyan Kendra (KVK)

Krishi Vigyan Kendras (KVKs) has been established to cater for the immediate needs of farmers as well as to keep them updated with new technology developed on various crops and allied areas. There are 731 KVKs in the country. Almost all the districts in the country have at least one KVK. Most KVKs are with agricultural universities, whereas some are with other agencies including Non-Government Organizations.

Check Your Progress Exercise 4

- Note:** a) Write your answer in the space given below.
b) Compare your answer with those given at the end of the unit.

1. When ICAR was set up?

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2. How many agricultural universities are established in India?

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1.8 LET US SUM UP

Agriculture is a broad term comprising crop production, horticulture, animal husbandry, fisheries, poultry, forestry, sericulture, etc. Agriculture channelizes the environmental inputs into economic products through cultivation and animal raising practices. The development of agriculture has occurred in several phases. In the pre-historic era, conservation of desirable native plants, collection and preservation of seeds, sowing of seeds, tilling of land, shifting cultivation, weed control, use of domestic waste as a plant nutrient, crop rotation, green manuring, etc. have been the major developments in agriculture. Cultivation of several crops was started. Domestication of animals was done.

Different ruling dynasties in India have formulated rules related to taxes, which influenced agriculture. Britishers arrived in India during the seventeenth century. They attempted to boost agricultural production. Department of Agriculture was created in India in 1871. A system of *Zamindari* and *Ryotwari* was created to collect the rent from the farmers. During the independence, India was short of foodgrains to feed its population and there was an immediate need to enhance the foodgrains production in the country. For this purpose, several initiatives such as land reforms, the use of high-yielding varieties and chemical fertilizers, development of irrigation facilities were promoted. Schemes were started to promote the dairy, fisheries and other allied sectors. The Operation Flood programme played a vital role in the growth of the dairy sector. To promote the research, education and extension in agriculture to enhance the agriculture production various institutions in the area of crop production, veterinary sciences, agricultural engineering, forestry, dairy technology, food technology, horticulture, fisheries, home science and agricultural marketing were established.

1.9 KEYWORDS

DAP: Di-ammonium phosphate

Homo sapiens: The direct ancestor of modern man.

IADP: The selected districts under IADP (Intensive Area Development Programme) were required to have assured irrigation, minimum hazards like flood, drought and soil erosion, use of fertilizers, plant protection measures and availability of agricultural credit.

Nucleus seed: It is a basic seed developed by the concerned breeder.

SSP: Single super phosphate

Taquavi loan: Loan given to the cultivators for purchase of seed and cattle on surety for their payment.

1.10 SUGGESTED READINGS/ REFERENCES

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1.11 ANSWERS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress Exercise 1

1. a). T b). T c) F
2. Conservation of desirable native plants, collection and preservation of seeds, sowing of seeds, tilling of land, shifting cultivation, weed control, fallowing of land, use of domestic wastes as plant nourishment, following crop rotation, green manuring, etc.

Check Your Progress Exercise 2

1. Green revolution, land reforms, etc.
2. Fertilizers provide nutrients that are deficient in the soil and are essential for the growth of plants.
3. By constructing the irrigation infrastructure and by adopting modern technologies like drip and sprinkler systems of irrigation.

Check Your Progress Exercise 3

1. Animals are used in agricultural practices like ploughing, sowing, interculture, threshing operations, transport through bullock carts, etc. On the other hand, animals depend on crop residues and other leftover by-products as feed.
2. Advancement in fisheries is made from egg management to full development and post-harvest technology.

Check Your Progress Exercise 4

1. 1929
2. 62 State Agricultural Universities (SAU), 5 Deemed to be Universities and 3 Central Agricultural Universities (CAU).

UNIT 2 SOIL AND WATER CONSERVATION

Structure

- 2.0 Objectives
- 2.1 Introduction
- 2.2 Soil Erosion
 - 2.2.1 Types of Soil Erosion
 - 2.2.2 Mechanism of Soil Erosion
 - 2.2.3 Effect of Soil Erosion
 - 2.2.4 Factors Affecting Soil Erosion by Water
- 2.3 Water Erosion
- 2.4 Soil and Water Conservation Measures
 - 2.4.1 Agronomical Measures
 - 2.4.2 Mechanical Measures
- 2.5 Let Us Sum Up
- 2.6 Keywords
- 2.7 Suggested Readings/ References
- 2.8 Answers to Check Your Progress Exercises

2.0 OBJECTIVES

On completing this unit, you shall be able to:

- explain the meaning of soil erosion and its importance;
- differentiate amongst various types of erosion;
- discuss the process of initiation of soil erosion and the mechanism of erosion; and
- describe the soil conservation mechanism.

2.1 INTRODUCTION

Soil erosion results in a huge loss of nutrients in suspension or solution, which are washed away from one place to another, thus causing depletion or enrichment of nutrients. Besides the loss of nutrients from the topsoil, there is also degradation through the creation of gullies and ravines, which makes the soil inappropriate for agricultural production. Costly reservoirs constructed under the river-valley projects are being silted up at an alarming rate owing to the denudation of forest vegetation, and the cultivation of steep slopes without adopting any conservation practices.

In dry climates, blowing wind is the main cause of soil erosion. The government is trying out various measures of soil conservation in India to check the damages caused by this phenomenon. Though soil erosion through natural agencies can hardly be checked, human-induced erosion by mining, deforestation, and other activities is being strictly brought under the umbrella

2.2 SOIL EROSION

“Detachment (breaking of soil particles from their original places) and transportation (shifting some of the loose particles from their original places to any other place) of soil particles is referred to as *Soil Erosion*. Soil erosion is the weathering away of land surface by the action of some erosive agents viz.; natural (water, wind, glacial, earthquakes, and volcano eruption) and artificial or manmade (developmental activities, agriculture, or deforestation).

The Global Assessment of Land Degradation (GLASOD) study estimated that around 15 per cent of the Earth's ice-free land surface is afflicted by all forms of land degradation. Of this, accelerated soil erosion by water is responsible for about 56 percent and wind erosion is responsible for about 28 percent. This means that the area affected by water erosion is, very roughly, around 11 million square km., and the area affected by wind erosion is around 5.5 million square km. The area affected by tillage erosion is currently unknown.

Soil Erosion in India is a major cause of concern. In India, almost 130 million hectares of land, i.e., 45 per cent of the total geographical surface area, is affected by serious soil erosion through gorges and gullies, shifting cultivation, cultivated wastelands, sandy areas, deserts, and waterlogging. It is estimated that out of 305 million hectares reported area (the total geographical area is 329 million hectares), 145 million hectares require conservation measures.

Soil erosion by rain and river that takes place in hilly areas causes severe landslides and floods while cutting trees for agricultural implements, firewood, and timber; grazing by livestock over and above the carrying capacity of grasslands. Excessive soil erosion with a resultant high rate of sedimentation in the reservoirs and decreased fertility has become a solemn environmental problem for the country with disastrous economic consequences.

2.2.1 Types of Soil Erosion

Soil may be detached and moved by water, wind, tillage, or human activities. These three, however, differ greatly in terms of; where and when they occur, what happens to the area that is being eroded (on-site impacts), how far the eroded soil is moved, and if the soil is moved away from the place where it was eroded, what happens as a result (off-site impacts). Soil erosion can be classified based on the erosion intensity and based on the types of erosive agents.

A. Soil erosion based on erosion intensity

The following types of erosion based on the erosion intensity are observed in India:

1. **Natural/normal or geologic erosion:** Soil is naturally removed by the action of water or wind: such natural (or 'geological') soil erosion has

been occurring for some 450 million years since the first land plants formed the first soil this may be called as the 'Natural Rate of Erosion'. Even before this, natural processes moved the loose rock, or regolith, off the Earth's surface. Normally the 'Natural Rate of Erosion' or the background erosion removes soil at roughly the same rate as the soil is formed and this is considered quite natural.

2. **Accelerated rate of soil erosion:** The removal of the surface soil from areas denuded of their natural protective cover as a result of human and animal interference takes place at a much faster rate than that at which it is built up by the soil-forming process. This accelerated detachment rapidly ravages the land and it is with this type of soil erosion that we are so seriously concerned. It has impacts that are both on-site (at the place where the soil is detached or *in situ*) and off-site (wherever the eroded soil ends up or *ex-situ*). Soil erosion is just one form of soil degradation. Other kinds of soil degradation include salinization, nutrient losses, and soil compaction, etc. But 'accelerated' soil erosion *i.e.*; the loss of soil at much faster rates than its rate of formation is a far more recent problem that needs attention for restoration and management.

B. Soil erosion based on erosive agents

Based on the types of erosive agents, soil erosion can be classified into the following categories:

1. Wind erosion
2. Water erosion
3. Coastal and sea erosion
4. Fertility erosion (salt effect/waterlogging)

Now we will discuss these various types of soil erosion in detail:

1. **Wind erosion:** Wind erosion takes place normally in arid and semi-arid areas devoid of vegetation, where the wind velocity is high. The soil particles on the land surface are lifted and blown off as dust storms. When the velocity of the dust-bearing winds is retarded, coarser soil particles are deposited in the form of dunes and thus fertile lands are rendered unfit for cultivation. In other places, fertile soil is blown away by winds and the subsoil is exposed, as a result, the productive capacity of the soil is considerably reduced.
2. **Water erosion:** Soil erosion caused by water can be distinguished in various forms, viz. (i) splash or raindrop erosion, (ii) sheet erosion, (iii) rill erosion, (iv) gully erosion, (v) ravine erosion, (vi) landslides or slip erosion and (vii) stream-bank erosion. We will discuss in more detail soil erosion caused by water in section 2.3.
3. **Coastal and sea erosion:** This category of erosion falls on those lands that are adjoining the sea coast. Often during high tides or Tsunamis, a sizeable fertile area either gets washed away or becomes highly salty rendering the cultivation non-feasible till the same gets drained and reclaimed by natural rainfall or human interventions.

4. **Fertility erosion (salt effect/waterlogging):** Due to increased numbers of canals that bring water from the other areas to a new area and consequently, increased water use especially for irrigation by farmers; large tracts in the country have become salt-affected.

2.2.2 Mechanism of Soil Erosion

Water and wind are two primary sources of soil erosion. The mechanism of soil erosion through water and wind is described below.

1. **Water erosion:** Soil erosion caused by rainfall is the result of the application of energy from two distinct sources, namely (i) the falling raindrops, and (ii) the surface flow. The energy of a falling raindrop is applied slantingly or vertically from above, whereas that of a surface flow is applied more or less horizontally along the surface of the ground. The chief role of the falling raindrop is to detach soil particles, whereas that of the surface flow (outside the rills and gullies) is to transport the soil. The falling raindrop also makes a major contribution to the movement of the soil on unprotected sloping lands during the period of heavy-impact storms, by splashing large quantities downslope and by imparting transporting capacity to the surface water by keeping it turbid. More than 100 tonnes of soil per hectare can sometimes be lost yearly in this fashion from bare and highly detachable soil on sloping land. You will learn more about water erosion in section 2.3.
2. **Wind erosion:** Wind is responsible for three types of soil movement in the process of wind erosion. They are known as (i) saltation, (ii) suspension, and (iii) surface creep.
 - i) *Saltation:* The major portion of the soil carried by the wind is moved in a series of short bounces called "saltation". The soil carried in saltation consists of fine particles ranging from 0.1 to 0.5 mm in diameter. Saltation is caused by the direct pressure of wind on soil particles and their collision with other particles. After being pushed along the ground surface by the wind, the particles leap almost vertically in the first stages of saltation. Some grains rise only a short distance; others leap 30 cm or higher, depending on the velocity of the rise from the ground.
 - ii) *Suspension:* Very fine soil particles, less than 0.1 mm in diameter, are carried into suspension, being kicked up into the air by the action of particles in saltation. The movement of fine dust in suspension is completely governed by the characteristic movement of the wind. Suspended material is carried long distances from its original location and is thus a complete loss to the eroded area, especially when erosive winds are from different directions.
 - iii) *Surface creep:* Soil particles, larger than about 0.5 mm in diameter but smaller than 0.1 mm, are too heavy to be moved in saltation but are pushed or spread along the surface by the impact of particles in saltation to form a surface creep.

About 90 per cent of the total soil movement in wind erosion is below the height of 30 cm, and about 50 per cent of it is within 5 cm of the ground

level. The control of wind erosion is mainly based on the reduction or elimination of movement in saltation.

The process of wind erosion is the same as water erosion because both water and wind are termed fluids. The density makes all the differences. The mechanism of detachment, transportation, and deposition are the same in both cases.

Figure 2.1 (a) & (b) depict the various processes related to wind erosion.

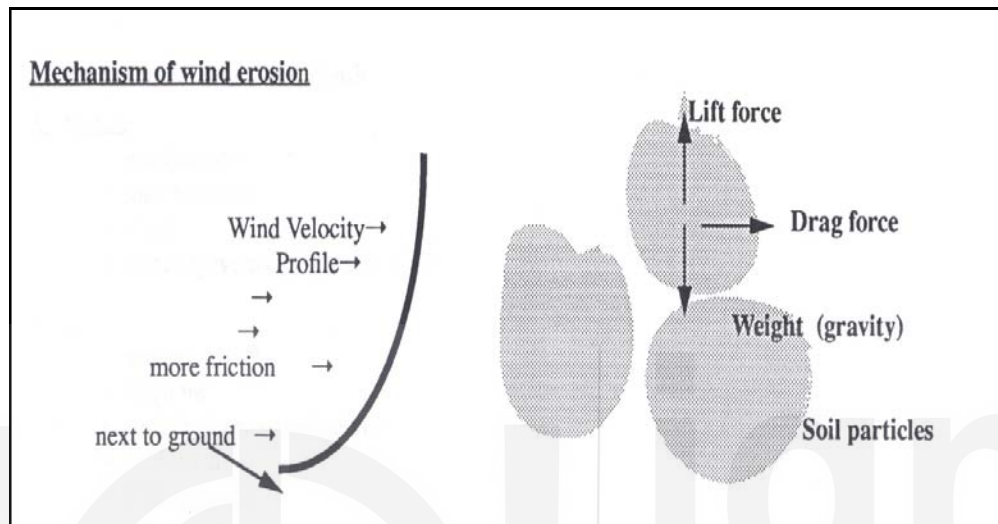


Fig.2.1 (a): Mechanism of wind erosion.

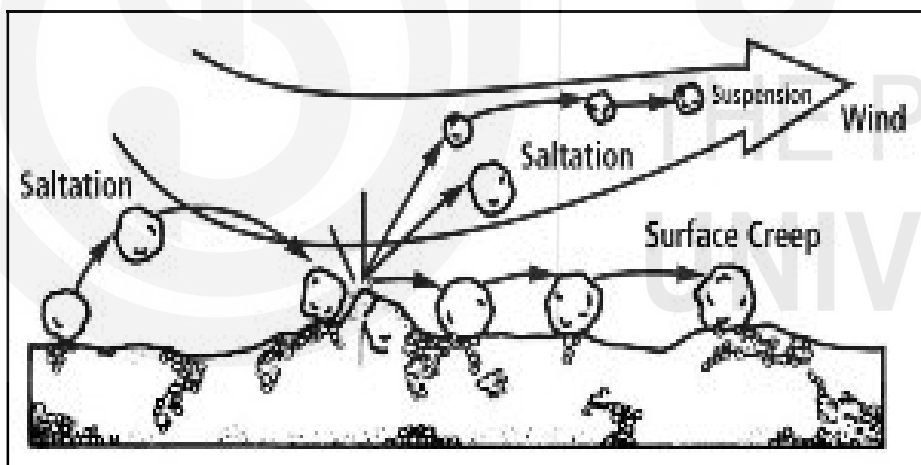


Fig. 2.1 (b): Mechanism of wind erosion.

2.2.3 Effect of Soil Erosion

Soil erosion causes both *in-situ* and *ex-situ* effects. Let us discuss these effects in detail:

1. **On-site (*in-situ*) effects of soil erosion:** The main on-site (*in-situ*) impact of soil erosion is the reduction in soil quality which results from the loss of the nutrient-rich upper layers of the soil, and the reduced water-holding capacity of many eroded soils. Loss of soil quality is a long-term problem; globally, soil erosion's most serious impact may well be its threat to the long-term sustainability of agricultural productivity, which results from the 'on-site' damage that it causes.



Fig. 2.2: On-site impact of erosion: severe rill erosion on a hillside
(Source: John Boardman through WWW)

Crops are particularly reliant on the upper horizons of the soil, which are the most vulnerable to erosion by water and wind. In this sense, erosion removes 'the cream of the soil'. Agricultural tillage also redistributes soil, resulting in thinner soils on topographically convex areas within a field. The damaging on-site effects of erosion, in terms of decreased agricultural yields, are well known in the developing countries of Africa and Asia. But even in the developed world, there is cause for concern. Water erosion is locally severe in Australia, New Zealand, parts of the US, Southern Europe, and Eastern Europe (often as a result of the former large state-controlled farms). This strategy is, however, infeasible concerning erosion in developing countries (Fig. 2.2).

2. **Off-site (*ex-situ*) effects of soil erosion:** In addition to its *in-situ* effects, the soil that is detached by accelerated water or wind erosion may be transported considerable distances. This gives rise to *ex-situ* effects too. Water erosion's main *ex-situ* effect is the movement of sediment and agricultural pollutants into watercourses. This can lead to the silting-up of dams, disruption of the ecosystems of lakes, and contamination of drinking water. A more minor off-site effect can occur in situations, where eroded soil has a decreased capacity to absorb water; increased runoff may lead to downstream flooding and local damage to property (Fig. 2.3).



Fig. 2.3: Red discharge in China's Yangtze River at the Three Gorges, in Hubei province is due to the sediment-rich water.

(Source of photo: Internet www)

Another major *ex-situ* impact results from the agricultural chemicals that often move with eroded sediment. These chemicals move into and pollute, downstream watercourses and water bodies. Where inputs of agricultural chemicals are high - as in the more affluent nations - costs of removing such pollutants from drinking water can be considerable. Therefore the on-site impacts of soil erosion are a present-day problem for many of the developing nations.

2.2.4 Factors Affecting the Soil Erosion by Water

The erosion rate is very sensitive to both climate and land use, as well as to the detailed conservation practices at the farm level. The Himalayan region of India is particularly more prone to erosion because it is subject to heavy bursts of erosive rain, falling on steep slopes with fragile soils. This contrasts with Central India where soil erosion is less because of lower rainfall on gentle slopes. Consequently, the area affected by erosion is less extensive than in the Himalayan region of North Eastern (NE) or North Western (NW) parts of the country. However, erosion is still a serious problem in NW and NE India along with the Eastern and Western Ghats and Malabar Coastal region and is on the increase over time. With a very slow rate of soil formation, any soil loss of more than 1t/ha/year can be considered irreversible within a time of 50-100 years. The rate and magnitude of soil erosion by water are controlled by the following factors.

1. **Rainfall intensity and rainfall erosivity factor (R):** Both rainfall and runoff factors must be considered in assessing a water erosion problem. The impact of raindrops on the soil surface can break down soil aggregates and disperse the aggregate materials. Lighter aggregate materials such as very fine sand, silt, clay, and organic matter can be easily removed by the raindrop splash and runoff water; greater raindrop energy or runoff amounts might be required to move the larger sand and gravel particles.
2. **Soil erodibility factor (K):** The soil erodibility factor (K) is an estimate of the ability of soils to resist erosion, based on the physical characteristics of each soil. Generally, soils with faster infiltration rates, higher levels of organic matter, and improved soil structure have a greater resistance to erosion. Sand, sandy loamy and loam textured soils tend to be less erodible than silt, very fine sand, and certain clay textured soils.
3. **Slope gradient and length (L-S factor):** Naturally, the steeper the slope of a field, the greater the amount of soil loss from erosion by water. Soil erosion by water also increases as the slope length increases due to the greater accumulation of runoff. Consolidation of small fields into larger ones often results in longer slope lengths with increased erosion potential, due to increased velocity of water which permits a greater degree of scouring (carrying capacity for sediment).
4. **Vegetation:** Soil erosion potential is increased if the soil has no or very little vegetative cover of plants and/or crop residues. Plant and residue cover protects the soil from raindrop impact and splash, tends to slow

down the movement of surface runoff, and allows excess surface water to infiltrate. The erosion-reducing effectiveness of plant and/or residue covers depends on the type, extent, and quantity of cover.

Check Your Progress Exercise 1

- Note:** a) Use the space below for writing your answers.
b) Compare your answer with those given at the end of the unit.

1. How much soil on the earth is suffering from soil erosion?

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2. How much area in the world is suffering from water erosion?

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3. What is the extent of soil erosion in India?

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4. How much area in India is under waterlogging?

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5. What are the adverse impacts of soil erosion?

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6. List the main factors that affect soil erosion by water in India.

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2.3 WATER EROSION

Water is the agent to removes the surface material from the earth's crust and transport the eroded materials from the point of removal. Due to water following types of soil erosion are observed:

- 1. Splash erosion:** Although considerable quantities of soil may be moved by rain splash, it is all merely redistributed back over the surface of the soil (on steep slopes, however, there will be a modest net downslope movement of splashed soil). Thus a more descriptive term might be 'rain splash redistribution'. Fig. 2.4 (a) depicts the original size of the water particles at the initiation of falling and after the splash has taken place when water particles strike the ground. A picturesque depiction of the splash is given in Fig. 2.4 (b) and the processes of detachment, transportation, and deposition have been explained graphically in Fig. 2.4 (c).

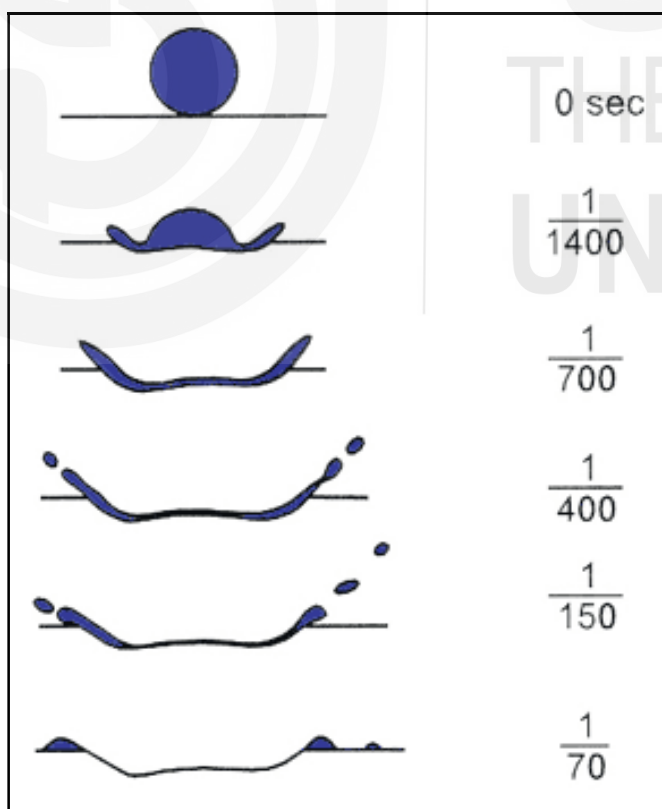


Fig. 2.4 (a): Water and soil splashed following a single raindrop impact.

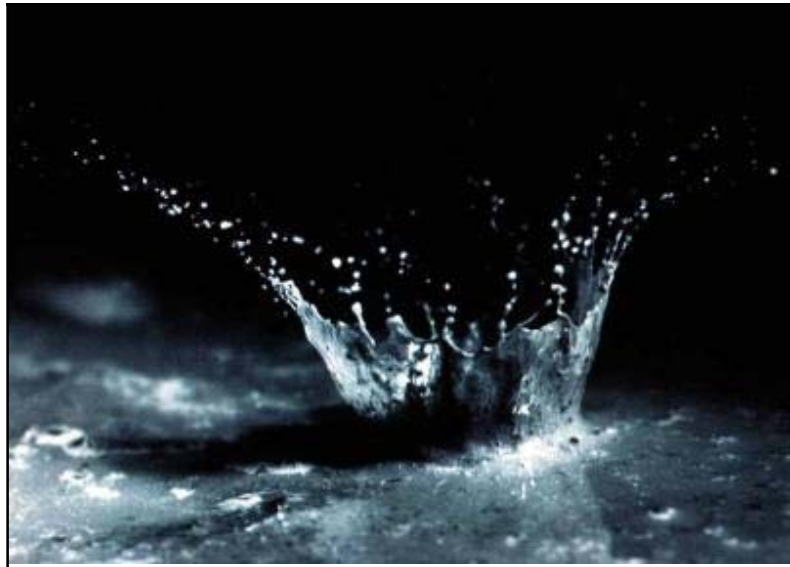


Fig. 2.4 (b): Water and soil splashed following a single raindrop impact.
(Source: The WEPP 95 CD-ROM)

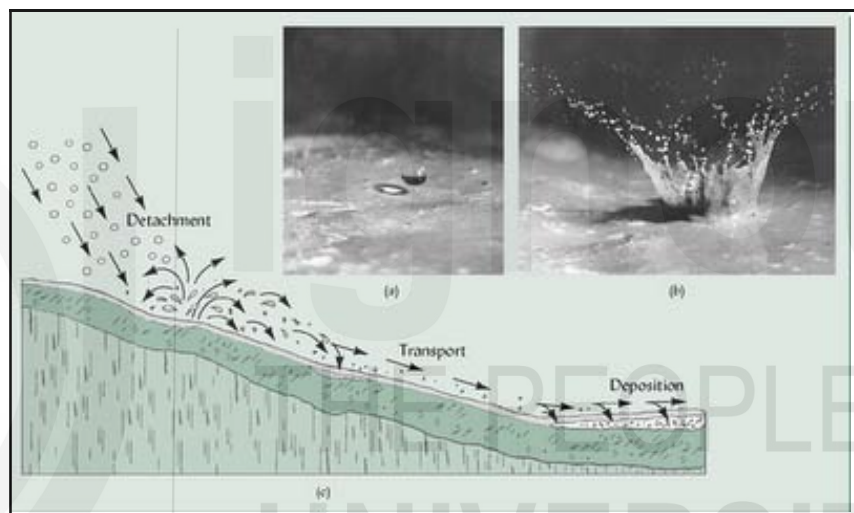


Fig.2.4 (c): The graphical representations of the processes of detachment, transportation, and deposition in water erosion.

The rain splash requires high rainfall intensities; hence, it is most destructive under convective rainstorms in the world's equatorial regions. Rain splash is relatively ineffective where rain falls with low intensity.

2. **Sheet erosion:** After the rainfall intensity becomes more than the basic infiltration rate the water build-up starts taking place which starts flowing like a thin sheet in the direction of the general slope. Water is known to flow along the path of least resistance. For flowing a small distance the sheet flow assumes the form of channel flow. During this process, two types of erosions are possible; i) sheet erosion and ii) rill erosion. Sheet erosion removes a thin covering of soil from large areas, often from entire fields, more or less, uniformly during every rain that produces run-off. Fig. 2.5 (a) depicts the overland flow conditions leading to sheet erosion [Fig. 2.5 (b)]. This form of soil erosion is the most prominent which is visible to our eyes, but the quantities being eroded by such types of erosion processes are highly unimaginable.



Fig. 2.5 (a): Diffuse overland flow. Note the raindrop impacts.
(Source: <http://www.freefoto.com>)



Fig. 2.5 (b): Initiation of the sheet erosion leading to the formation of rills and gullies

3. Rill erosion

If the sheet erosion is unchecked and allowed to continue for a while the rainfall may also move soil indirectly, through runoff in rills (small channels) or gullies (larger channels, too big to be removed by tillage). In many parts of the world, rill and gully erosion is the dominant form of water erosion. Fig. 2.6 (a & b) depicts the soil loss that has taken place due to rill erosion. If the rill erosion is not checked properly this leads to gullies.



Fig. 2.6 (a): Initiation of finer rills after the sheet erosion has been completed on a relatively flat land on the border of the ridges.



Fig. 2.6 (b): A denuded small hillock subjected to rill erosion
(Source: <http://www.art.com>)

4. **Gully erosion:** When rill erosion is neglected, the tiny grooves develop into wider and deeper channels, which may assume a huge size. This is called 'gully' erosion. The gullies tend to deepen and widen with the intensity of rainfall every heavy rainfall. They cut up large fields into small fragments and, over the course of time, make them unfit for cultivation (Fig. 2.7).



Fig. 2.7: A deep gully formation due to water erosion

It is estimated that about 2.3 m ha (5.75 m acres) of land in India is affected by severe gully erosion. These gullies are an indication of very bad management of the land resources. Apart from the fact that the gullied land has been destroyed, the gullies are a menace to the adjoining tablelands.

Control of gullies and their reclamation for various land uses

The practices described below are arranged in the order in which they should be adopted; the cost of their execution increases progressively.

- i) **Closure to grazing and other biotic interference:** A study of the natural flora of the eroded areas has shown that only poor types of annual and unpalatable grasses are growing in place of the desirable climax associations which should normally exist in the prevailing soil-climatic environment. Under the upper Damodar catchment conditions, it has been observed that closure to grazing reduced the soil loss from 3.3 tonnes/ha (under overgrazing conditions) to only 0.6 tonnes/ha.
- ii) **Contour and peripheral bunding:** After the closure of the ravine lands, the immediate problem to be tackled is to stop the devastating rate of soil erosion and the growth of gullies, and the conversion of the cultivated tablelands into wastelands for this purpose. It is thus essential to retain as much precipitation as possible on the land in semi-arid and sub-humid areas and safely dispose of the excess run-off in humid areas. In the semi-arid and sub-humid areas of India, contour and peripheral bunds of various cross-sections have been found ideal for this purpose.

- iii) **Gully plugging:** The eroding and deepening of gully beds can be prevented with gully plugs. Gully plugs protect the gully beds by reducing the speed of run-off water, redistributing it, increasing its percolation, encouraging silting, and improving the soil moisture regime for establishing a plant cover. Gully plugs of various materials, e.g. brushwood live hedges, earth, sandbags, brick masonry, and boulders have been tried in India. Earth is the cheapest and most readily available material. Boulder gully plugs are equally effective if the material is available.

Reclamation of gullies for cultivation

The belief that all the gullied land can be reclaimed for cultivation by making level benches in gully beds is not feasible. This may be possible if no limit is fixed on expenditure. It has been observed that small and medium gullies can be conveniently, safely, and economically reclaimed for cultivation, whereas deep and narrow gullies should be retired to a permanent vegetative cover of grasses and trees.

- a. **Reclamation of small gullies:** Small gullies are reclaimed by clearing, minor levelling with bulldozers, and constructing diversion-cum-check bunds of 1.5 sq m cross-section spaced at horizontal intervals of 30-45 m. Grassed ramps are provided for the disposal of excess run-off at the ends of bunds near the gully sides.
- b. **Reclamation of medium gullies:** A small gully gets transformed into a medium gully along the length of the main drainage system. A medium gully is reclaimed by clearing and levelling the bed and constructing a series of composite earth and brick masonry check dams at vertical intervals of 1.2 m (which gives a horizontal interval of 120 m on a 1% slope of the gully bed) and terracing the side slopes.

The uneven side slopes of the medium gullies having 8 to 15% slopes are bench-terraced into level terraces at 0.9 to 1.2 m vertical intervals. Bench terraces are constructed when the gully sides have a uniform slope for a length of at least 120 m to justify the cost of terracing. The terrace faces, grassed outlets and earthen-checked dams are stabilized by sodding or growing them with suitable grasses. *Dicanthium annulatum* and *Cenchrus ciliaris* have been found suitable for this purpose, especially in Gujarat.

- c. **Reclamation of deep and narrow gullies:** As mentioned earlier, the best land use of ravine lands is to retire them to permanent vegetation comprising grasses and trees. This type of land use is a must for deep and narrow gullies. The natural tree species of the ravine lands of Gujarat comprise *Acacia nilotica*, *A. senegal*, *A. leucophloea*, *Azadirachta indica*, *Albizia lebbeck*, *Feronia elephantum*, *Prosopis spicigera*, etc.

5. **Stream bank erosion:** Torrents are defined as hill streams characterized by wide-spreading beds on emergence from the hills with ill-defined

banks, flashy flows, and swift currents. Usually, they are dry watercourses, except during the rainy season when with every downpour in their catchment, they get very much swollen with flood and subside almost to their normal tiny size immediately after the storm is over. These sudden and violent flows are responsible for moving immense quantities of detritus, comprising boulders, shingles, sand, and silt, depending upon the geology of the terrain. The rivers change their courses due to this type of widely prevalent form of erosion. The stream bank erosion has been depicted in Fig. 2.8.



Fig. 2.8: Stream bank erosion

6. **Landslides and landslip erosion:** A landslide is defined as an outward and downward movement of the slope-forming material, composed of natural rocks, soil, artificial fills, etc. The fundamental causes of landslides are the topography of the region and geological structure, the kinds of rocks, and their physical characteristics. The immediate cause of a slide may be an earthquake, or a heavy rainfall, which unduly saturates the ground or a part of a road. However, these are accidents rather than fundamental causes.

2.4 SOIL AND WATER CONSERVATION MEASURES

Conservation refers to retaining extensive vegetation on the soil. Vegetation is the protective cover against the forces of wind and water, which protects the soil from being washed or blown away and preserves the physical and hydrographic balance of nature. Forests, for instance, provide the most effective protection against erosion on hill slopes.

Broadly speaking, the practical methods of soil and water conservation fall into two important classes, viz. Agronomic measures and mechanical measures (engineering procedures). Agronomic measures include contour farming, off-season tillage, deep tillage, mulching, and providing vegetative

barriers on the contour. These measures mainly prevent soil erosion but will also help in improving moisture availability in the soil. The engineering measures adopted differ with location, the slope of the land, soil type, amount, and intensity of rainfall. Depending on these parameters, the methods commonly used are contour trenching, contour stone walls, construction of temporary and permanent check dams, and gully plugging structures. Additionally, percolation ponds, silt detention tanks, and irrigation tanks are constructed to harvest water and recharge it to the groundwater for use in agriculture (irrigation). Farm ponds can also be constructed for every 4-5 ha in the watershed to provide protective/supplemental irrigation.

2.4.1 Agronomical Measures

Agronomic practices for soil and water conservation help to intercept raindrops and reduce the splash effect, help to obtain a faster intake of water into the soil by improving the content of organic matter and soil structure, help to retard and reduce the overland run-off through the use of contour cultivation, mulches, dense-growing crops, strip-cropping, and mixed cropping. These measures are discussed in detail as under:

1. **Contour farming:** During intense rainstorms, the soil cannot absorb all the rain as it falls. The excess water flows down the slope under the influence of gravity. If farming is done up and down the slope, the flow of water is accelerated, because each furrow serves as a rill. The major part of the rain is drained away without infiltrating into the soil. The top fertile soil, along with plant nutrients and seeds, is washed off resulting in scanty and uneven growth of the crop.
2. **Mulching:** Surface mulches are used to prevent soil from blowing and being washed away, to reduce evaporation, to increase infiltration, to keep down weeds, to improve soil structure, to moderate soil temperature and eventually to increase crop yields. Interculture kills weeds and produces five to seven cm thick soil mulch which helps to reduce evaporation from the topsoil. It also breaks the surface crust which forms after each downpour. Studies on mulching carried out in India under rainfed agriculture have concentrated on the measurement of crop responses rather than on how crop responses are influenced.
3. **Growing of crops that provide the maximum cover, and reduce run-off and soil loss:** Cultivation of legumes, in general, furnishes a better cover and hence better protection to cultivated land against erosion than ordinary cultivated crops. The crops and the cropping systems will naturally vary from region to region, depending on the soil and climatic conditions.
4. **Strip cropping:** Strip cropping is essentially a form of intercropping. Its importance in controlling run-off erosion and thereby maintaining the fertility of the soil is now universally recognized. Strip cropping, in effect, employs several good farming practices, including crop rotation, contour cultivation, proper tillage, stubble mulching, cover cropping, etc. Strip Cropping is of the following different forms:

- i) Contour strip Cropping
- ii) Field strip Cropping
- iii) Wind strip Cropping
- iv) Permanent or temporary buffer strip Cropping

- i) *Contour strip cropping*: Contour strip Cropping is the growing of soil-exposing and erosion-permitting crops in strips of suitable widths across the slopes on contour, alternating with a strip of soil-protecting and erosion-resisting crops. Contour strip-cropping shortens the length of the slope, checks the movement of run-off water, helps to desilt it, and increases the absorption of rainwater by the soil. Further, the dense foliage of the erosion-resistant crops prevents the rain from beating the soil surface directly. It is advisable to rotate the strip planting by sowing a non-resistant crop, following an erosion-resistant crop, and *vice versa*.
- ii) *Field strip cropping*: It is the planting of farm crops in more or less parallel strips across fairly uniform slopes, but not on exact contours.
- iii) *Wind strip cropping*: It consists of planting tall-growing crops such as *jowar*, *bajra* and maize, and low-growing crops in alternately arranged straight and long, but relatively narrow, parallel strips laid out right across the direction of the prevailing wind, regardless of the contour.
- iv) *Permanent or temporary buffer strip cropping*: In the case of permanent or temporary buffer strip cropping, the strips are established to take care of critical, i.e. steep or highly eroded, slopes in fields under contour strip cropping. These strips do not form part of the rotation practised in normal strip cropping, and they are generally planted with perennial legumes, grasses, or shrubs on a permanent or temporary basis.

5. **Mixed cropping**: The main reason for the lack of interest in strip cropping in India is the small size of the land holdings of the farmers. Instead, the Indian farmers have been interested in mixed cropping which is very extensively adopted by them. Some of the important objectives of mixed cropping are a better and continuous cover of the land, good protection against the beating action of the rain, almost complete protection against soil erosion, and the assurance of one or more crops to the farmer. The roots of various species in a mixed crop feed at different depths in the soil.

2.4.2 Mechanical Measures

Mechanical measures (also called engineering measures) play a very vital role in controlling soil erosion from agricultural. They are adopted to supplement the agronomical practices when the latter alone are not adequately effective. The main objectives of the mechanical measures for controlling erosion are: (i) to increase the time of concentration by intercepting the run-off and thereby providing an opportunity for the

infiltration of water, and (ii) to divide a long slope into several short ones to reduce the velocity of the run-off and thus prevent erosion. The various mechanical structures for controlling soil and water conservation are discussed as under:

1. **Stormwater diversion channel:** This is the drain or direction, which diverts the stormwater. It is the first line of defence and vital to the protection system for all lower structures.
2. **Channel terrace:** To avoid soil dislodgement and scouring action, some form of earthwork at right angles to the steepest slope is used to interrupt the surface run-off. These works are known as terraces, or ridges, or bunds. In the cross-section, the feature common to all is an excavated channel, hence the name channel terrace (Fig. 2.9).

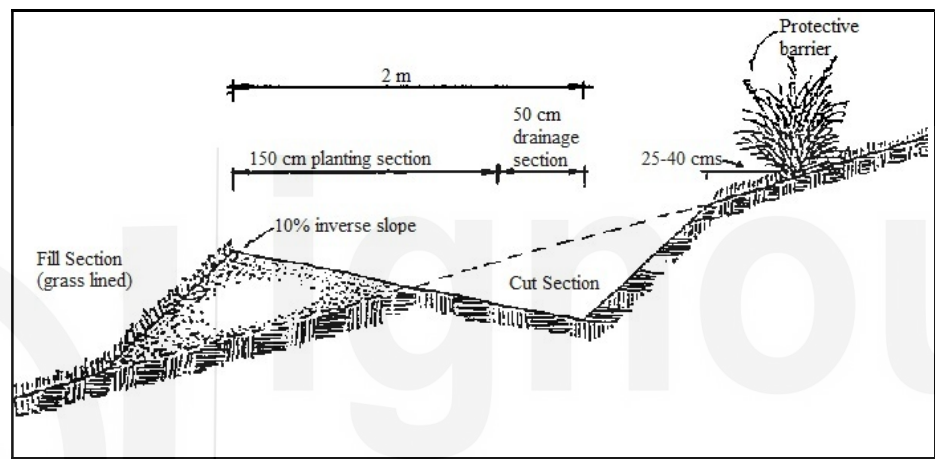


Fig. 2.9: Design details of a sloppy inward bench terrace on a slope with the provision of protective barriers.

3. **Artificial watercourse:** To discourage the water from channels, the frequent natural watercourse is not available. So it must be artificially made. They will normally run straight to the steepest slope and have a bank on each side to contain the water.
4. **Bench terrace:** Bench terracing is converting a steep slope into a series of steps. They may be constructed on the contour, to minimize run-off. When regular cultivation is not required, as far as tree crops, smaller terraces are effective (Fig. 2.10).



Fig. 2.10: A hill slope converted into level bench terraces

5. **Irrigation terrace:** When a flat bench terrace has a raised lip at the outer edge to retain irrigation water, this type of terrace is known as an irrigation terrace (Fig. 2.11).



Fig. 2.11: Wet terrace rice cultivation in high rainfall hill zone of Sikkim Himalayas

6. **Orchard terrace:** For too sloppy areas and where the soil is too shallow then the land may be developed for tree crops by the use of intermittent terraces, called orchard terraces (Fig. 2.12).

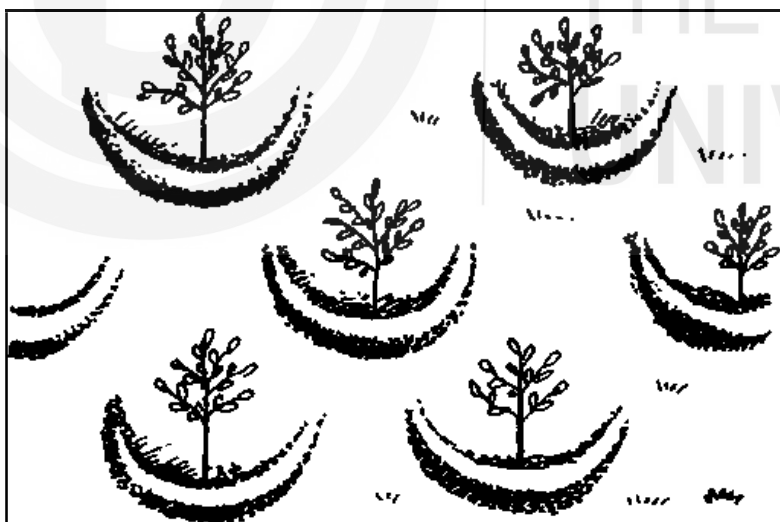


Fig. 2.12: Half-moon terraces created on the steep slopes for establishments of orchards

7. **Contour bund:** This practice consists of making a comparatively narrow-based embankment at intervals across the slope of the land on a level that is along the contour. It is an important measure that conserves soil and water in arid and semi-arid areas with high infiltration and permeability and is commonly adopted on agricultural land up to a slope of about 6 per cent.

It is a rudimentary type of channel terrace. Somewhere, it is taken as an

erosion-controlling bank, which has grasses or shrubs, planted on them (Fig. 2.13).

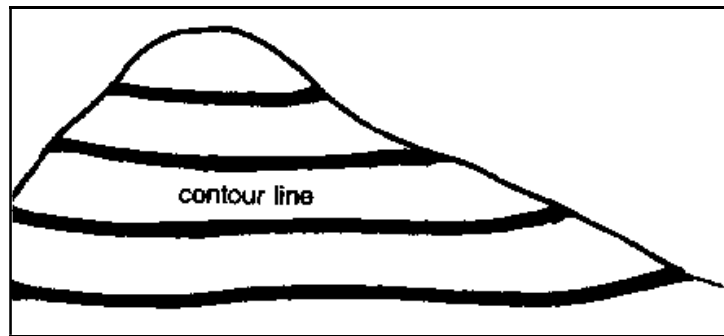


Fig. 2.13: Contour line is the imaginary line at equal elevation on the ground

8. **Contour trenches:** Contour trenches are the excavated or dug-out channels along the contour lines so that they can harvest the excess runoff along with the soil being washed away. This technology is being implemented in a big way in Mizoram (Fig. 2.14).



Fig. 2.14: Installation of a drainage trench for the safe disposal of excess rainwater

9. **Pasture furrow:** It is a small shallow drain whose function is to spread out surface water and allow it to concentrate in the minor watercourse. The primary purpose is to conserve water by increasing infiltration and the technique is frequently used on grassland, hence the name pasture furrows.
10. **Tied ridging:** It consists of covering the whole surface with closely spaced ridges in two directions at a right angle so the ground is formed into a series of rectangular depressions.
11. **Contour cultivation or grass strips:** On gentle slopes, it may be sufficient to slow down surface runoff by carrying out all tillage operations on the contour.
12. **Ridge and furrow:** This combines an element of erosion control with surface drainage. This is useful for gentle slopes and large areas. The ground is tilted into wide parallel ridges of the order of 10m wide with intervening furrows about 1/2m deep.

13. **Check dams:** The structure constructed to control the velocity of running water along the gullies is called check dams. They reduce the steep gradient of the gully into a series of steps with low rises and long flat treads, reducing the velocity of running water. This makes water deposit and increases the percolation of water in the soil. Check dams may be a temporary structure constructed with locally available materials or a permanent structure constructed with stone masonry (Fig. 2.15).



Fig. 2.15: A check-dam constructed in a gully for controlling the soil erosion

14. **Percolation pond:** The percolation pond is a multipurpose conservation structure depending on its location and size. It stores water for livestock and recharges the groundwater. It is constructed by excavating a depression, forming a small reservoir, or constructing an embankment in a natural ravine or gully to form an impounded type of reservoir.
15. **Irrigation tank:** The main function of this storage structure is irrigating crops. It is constructed below the above-mentioned structures in a watershed. In Tamil Nadu, each tank irrigates from 10 to 5000 hectares of land. In south India, there are about 2000000 tanks, irrigating about 3.5 million hectares.



Fig. 2.16: A polyethylene lined water harvesting tank for conservation and utilization of excess rainwater

16. In addition to the above measures and structures, small storage structures with a water storage capacity for an area of about 0.4 to 0.5 ha can be constructed in large numbers one for every 10 to 20 ha catchment or watershed at the foothills slopes and hilly areas. These storage facilities would attenuate the floods during storms. These measures will also ensure soil moisture for good growth of trees grown downstream recharging the groundwater in the region and making available more water for drinking and irrigation.

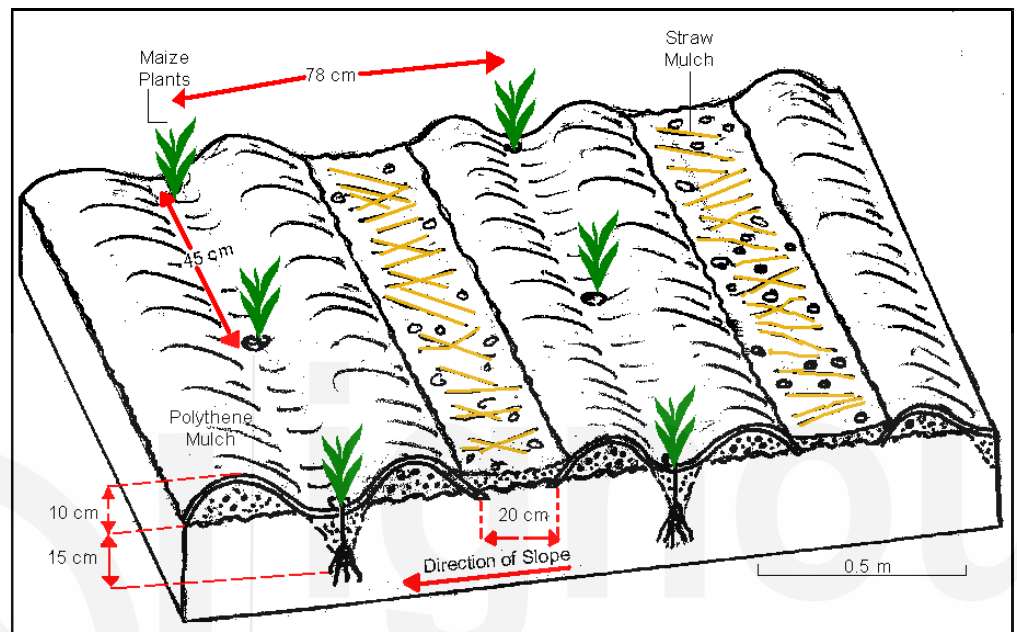


Fig. 2.17: Depiction of an experimental plot with polythene and straw mulching on ridges and furrows with maize as a test crop

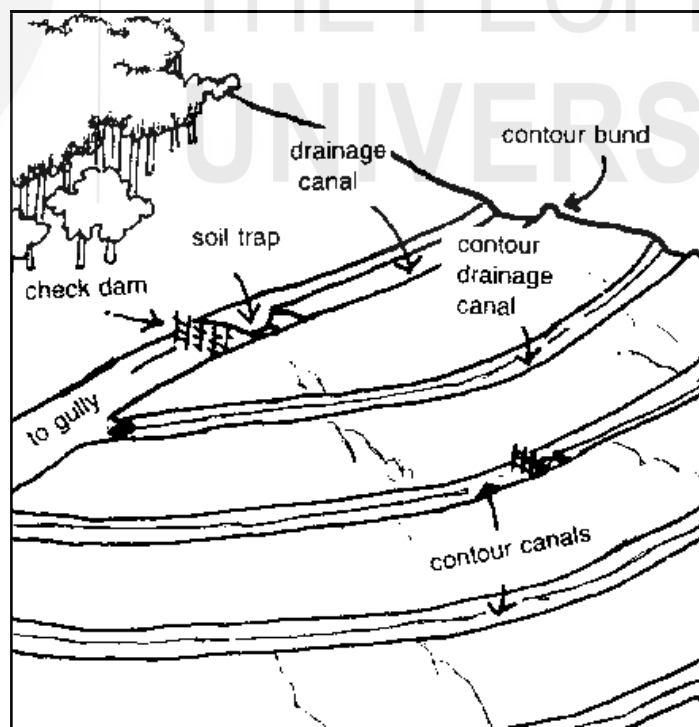


Fig. 2.18: Topo-sequential cultivation of different crops on a slope land

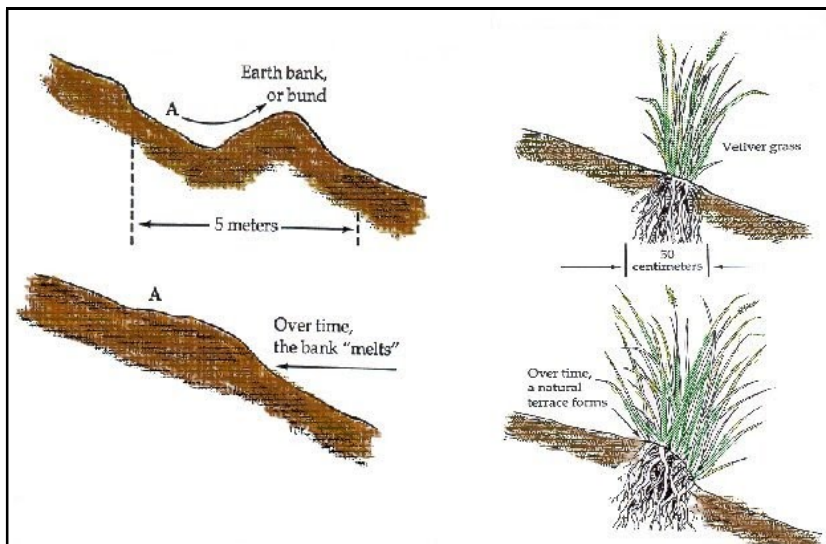


Fig. 2.19: Establishing the vegetative barriers in grassed waterways for prevention of soil loss from higher elevations

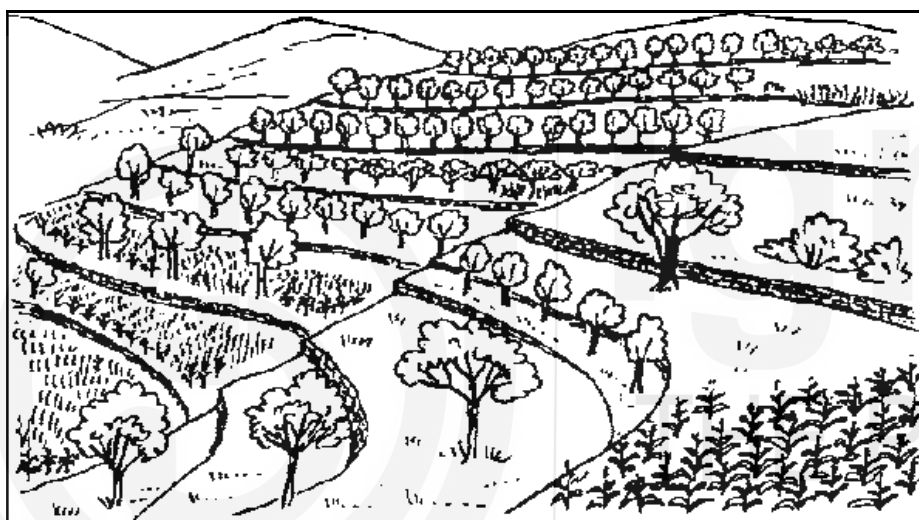


Fig. 2.20: A pictorial depiction of topo-sequential cultivation of different crops on a slope on bench terraces

Check Your Progress Exercise 2

Note: a) Use the space below for writing your answers.

b) Compare your answer with those given at the end of the unit.

1. What do you understand about conservation measures?

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2. How many types of conservation measures have been adopted to prevent soil erosion?

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3. What are the objectives of mechanical conservation measures?

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4. What is a check dam?

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2.5 LET US SUM UP

In this unit, you have learned about the process of soil erosion, various soil erosion agents, types of soil erosion, and the mechanism of erosion. There are several other ways of enhancing the process of soil erosion such as developmental activities for catering to the needs of the rapidly growing population of our country. I see in my near vicinity that a park was being leveled by a laser machine and was run over by a tractor resulting in the displacement of a huge amount of soil soon after the heavy rainfall took place in a pulverized sports ground. The engineers and workers must be trained to analyze and tackle such problems by being careful. Students must now be able to develop their independent thinking about many more ways in which soil erosion may occur. A brief idea for conservation measures has been given but one has to be vigilant at every time that unnecessary loss of soil must be prevented at all costs because our basic survival depends on this precious resource and its preservation is the moral responsibility of one and all.

2.6 KEYWORDS

Best Management Practice(BMP): Planning, design, and construction practices that minimize the production and transport of sediments.

Design storm: The storm duration and intensity for which a BMP or other structure is designed.

Embankment: A man-made deposit of soil, rock, or other material, often used to form an impoundment.

Erosion: The wearing away of soil by water, wind, and gravity.

Erosion and sediment control plan: A plan specifying the minimum level of erosion and sediment control to be installed on a site during each phase of development activity.

Evapotranspiration: The combined loss of water from an area by

evaporation from the soil surface and by transpiration of plants.

Gabion: A wire mesh cage, usually rectangular, filled with rock and used to protect channel banks and other sloping areas from erosion.

Geographic information system (GIS): A computerized system for storing, analyzing, retrieving, and displaying geo-referenced (mapped) information.

Gully erosion: Erosion resulting in a relatively deep incision of the soil surface, caused by concentrated overland runoff.

Mass wasting: Movement of earth materials caused by gravity alone, without a transporting medium such as wind or water. Mass wasting includes imperceptible processes such as soil creep and rapid processes such as debris slides and rockfall.

Rill erosion: Small eroding channels produced by surface runoff.

Runoff: It is the proportion of precipitation that flows over the land surface, in open channels, or stormwater conveyance systems.

Sedimentation: The deposition of eroded material (sediments).

Sediment delivery ratio: The fraction of the soil eroded from an upland source that reaches a stream channel, waterbody, or watershed outlet.

Splash erosion (or rain splash erosion): The dispersal of soil particles resulting from direct raindrop impact on the soil surface.

Suspended sediment: Sediment suspended in runoff water, in a stream, or any other water body.

Turbidity: A measure of the ability of a water sample to transmit light. High turbidity (poor light transmission) is normally caused by the presence of suspended matter such as clay, silt, fine organic matter, and microscopic organisms.

USLE: The Universal Soil Loss Equation. An equation was developed in the United States in the 1970s to express estimated soil loss per unit area as a function of rainfall, soil type, slope length, slope steepness, land cover, and land management characteristics.

Watershed: The area of land draining to a common outlet.

Water table: The surface of the water under the ground that is at atmospheric pressure. The water table is approximately that depth in the ground at which saturated soil is first encountered. The water table generally rises and falls seasonally.

2.7 SUGGESTED READINGS/ REFERENCES

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2.8 ANSWERS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress Exercise 1

1. Around 15 per cent of the Earth's ice-free land surface is afflicted by all forms of land degradation.
2. Around 11 million square km.
3. In India, almost 130 million hectares of land, i.e., 45 per cent of the total geographical surface area, is affected by serious soil erosion.
4. 17.9 million hectares
5. Loss of nutrients from the topsoil makes the land inappropriate for cultivation.
6. *Rainfall intensity and Rainfall erosivity factor, soil erodibility factor, slope gradient, and length factor, and vegetation*

Check Your Progress Exercise 2

1. Conservation refers to retaining extensive vegetation on the soil. Vegetation is the protective cover against the forces of wind and water, which protects the soil from being washed or blown away and preserves the physical and hydrographic balance of nature.
2. Two types: Agronomic and mechanical
3. The main objectives of the mechanical measures for controlling erosion are: (i) to increase the time of concentration by intercepting the run-off and thereby providing an opportunity for the infiltration of water, and (ii) to divide a long slope into several short ones to reduce the velocity of the run-off and thus prevent erosion.
4. The structure constructed to control the velocity of running water along the gullies is called a check dam.

UNIT 3 IRRIGATION AND DRAINAGE

Structure

- 3.0 Objectives
- 3.1 Introduction
- 3.2 Irrigation
- 3.3 Major Irrigation Projects in India
- 3.4 Irrigation Methods
 - 3.4.1 Surface Irrigation
 - 3.4.2 Sprinkler Irrigation
 - 3.4.3 Drip Irrigation
- 3.5 Irrigation Scheduling
 - 3.5.1 Irrigation Scheduling Criteria
 - 3.5.2 Advantages of Irrigation Scheduling
 - 3.5.3 Irrigation Scheduling Methods
- 3.6 Command Area Development and Water Management
- 3.7 Participatory Irrigation Management (PIM)
 - 3.7.1 Objectives of PIM
 - 3.7.2 Provisions in PIM Acts
 - 3.7.3 Constraints in Implementation of PIM
- 3.8 Drainage
- 3.9 Let Us Sum Up
- 3.10 Keywords
- 3.11 Suggested Readings/ References
- 3.12 Answers to Check Your Progress Exercises

3.0 OBJECTIVES

After going through this unit, you shall be able to:

- explain the importance of irrigation in Indian agriculture;
- describe the development of irrigation facilities in India;
- discuss the irrigation scheduling and irrigation methods; and
- define the drainage and comprehend the drainage requirements and methods.

3.1 INTRODUCTION

Irrigation is crucial for agricultural production due to its complementarity with other yield-enhancing inputs such as high-yielding variety seeds, fertilizers, and chemicals. India is generously endowed with water resources by nature. But paradoxically, water scarcity and inefficiency in its use co-exist in India's water resource management system. Irrigation retains its

crucial role in productivity-led future agricultural production, alleviating poverty and reducing inequality in income distribution in rural areas. In the past, agricultural development in general and irrigation development, in particular, has evolved around productivity and food security concerns.

3.2 IRRIGATION

The history of irrigation development in India can be traced back to prehistoric times. Vedas and ancient Indian scriptures made references to wells, canals, tanks, and dams which were beneficial to the community and their efficient operation and maintenance was the responsibility of the state. Civilization flourished on the banks of the rivers and harnessed the water for the sustenance of life. According to ancient Indian writers, the digging of a tank or well was amongst the greatest of the meritorious acts of a man. Irrigation has played a major role in the production process in a monsoon climate and an agrarian economy like India. There are pieces of evidence of the practice of irrigation since the establishment of settled agriculture during the Indus Valley Civilization (2500 BC).

These irrigation technologies were in the form of small and minor works, which could be operated by small households to irrigate small patches of land and did not require cooperative effort. Nearly all these irrigation technologies still exist in India with little technological change and continue to be used by independent households for smallholdings. In southern India, perennial irrigation may have begun with the construction of the Grand Anicut by King Raj Raja Chola and his son Rajendra Chola as early as the second century to provide irrigation from the Cauvery River. In northern India also there are many small canals in the upper valleys of rivers which are very old.

Definition of Irrigation

“Irrigation is the artificial application of water to the cropped fields (to plants) to ensure their survival and proper growth”. Irrigation is provided if there is not enough soil moisture in the absence of rain to produce usable commodities; the supplemental application of water to crops may be referred to as irrigation. There may be other definitions of irrigation in literature, but the ultimate objective of irrigation is to apply water to the fields, which do not have adequate moisture for supporting the proper plant growth.

3.3 MAJOR IRRIGATION PROJECTS IN INDIA

Several irrigation projects have been developed on various rivers to enhance the irrigated areas as well as to produce hydropower in India. A list of irrigation projects developed in India along with the area served by these is given below:

1. **Nagarjunasagar (Andhra Pradesh):** on the Krishna River near Nandikona village (about 44 km from Hyderabad.)
2. **Tungabhadra** (Joint project of Andhra Pradesh and Karnataka): on the Tungabhadra river.
3. **Gandak** (joint project of Bihar and Uttar Pradesh): Nepal also derives irrigation and power benefits from this project.

4. **Kosi** (Bihar): A multipurpose project, which serves Bihar and Nepal.
5. **Sone high-level canal** (Bihar): An extension of the Sone barrage project.
6. **Kakrapara** (Gujarat): on the Tapti River near Kakrapara, in Surat district.
7. **Ukal** (Gujarat): A multipurpose project, across the Tapti River near Ukai village.
8. **Mahi** (Gujarat): A two-phase project, one across the Mahi River near Wanakbori village and the other across the Mahi River near Kadana.
9. **Sabarmati** (Gujarat): A storage dam across the Sabarmati River near Dhari village in Mehsana district and Wasna barrage near Ahmedabad.
10. **Panam** (Gujarat): A masonry dam across the Panam River near Keldezar village in Panchmahal district.
11. **Karjan** (Gujarat): A masonry dam across the Karjan River near Jiotgarh village in Nandoo Taluka of Bharuch district.
12. **Bhadra** (Karnataka): A multipurpose project across the river Bhadra.
13. **Upper Krishna** (Karnataka): A project consisting of a Narayanpur dam across the Krishna River and a dam at Almati.
14. **Ghataprabha** (Karnataka): A project across Ghataprabha in Belgaum and Bijapur districts.
15. **Malaprabha** (Karnataka): A dam across the Malaprabha in Belgaum district.
16. **Tawa** (Madhya Pradesh): A project on the Tawa River, a tributary of the Narmada in Hoshangabad district.
17. **Chambal** (joint project of Madhya Pradesh and Rajasthan): The project comprises the Gandhi Sagar Dam, Rana Sagar Dam, and Jawahar Sagar Dam.
18. **Mahanadi Reservoir Project** (Madhya Pradesh): It has three phases:
 - i. Ravishankar Sagar project and feeder canal system for the supply of water to Bhilai steel plant and Sandur dam across Sandur village,
 - ii. Extension of Mahanadi feeder canal
 - iii. Pairi dam.
19. **Hasdeo Bango Project** (Madhya Pradesh): It is the third phase of the Hasdeo Bango Project complex and envisages the construction of a masonry dam across the Hasdeo River. The first and second phases have been substantially completed.
20. **Bargi Project** (Madhya Pradesh): It is a multipurpose project consisting of a masonry dam across the Bargi River in the Jabalpur district and a left-bank canal.

21. **Bhima** (Maharashtra): Comprises two dams, one on the Pawana River near Phange in Pune district and the other across the Krishna River near Ujjaini in Sholapur district.
22. **Jayakwadi** (Maharashtra): A masonry spillway across the river Godavari.
23. **Kukadi Project** (Maharashtra): Five independent storage dams, i.e., Yodgaon, Manikdohi, Dimba, Wadaj, and Pimpalgaon jog. The canal system comprises (1) Kukadi left bank canal, (2) Dhimba left bank canal, (3) Dhimba right bank canal (4) Meena feeder, and (5) Meena branch.
24. **Krishna Project** (Maharashtra): Dhom dam near Dhom village on Krishna and Kanhar Village of Varna River in Satna district.
25. **Upper Penganga** (Maharashtra): Two reservoirs on the Penganga River at Isapur in the Yavatmal district and the other on the Rayadhu River at Sapli in Parbhani district.
26. **Hirakud** (Odisha): The World's longest dam is located on the Mahanadi River.
27. **Mahandi Delta Scheme** (Odisha): The irrigation scheme will utilize releases from the Hirakud reservoir.
28. **Bhakra Nangal** (Joint project of Haryana, Punjab, and Rajasthan): India's biggest multipurpose river valley project comprises a straight gravity dam across the Sutlej at Bhakra, the Nangal dam, the Nangal hydel channel, two powerhouses at Bhakra dam, and two power station at Ganguwal and Kotla.
29. **Beas** (Joint venture of Haryana, Punjab, and Rajasthan): It consists of the Beas-Sutlej link and the Beas dam at Pong.
30. **Thein Dam** (Punjab): The project envisages the construction of a dam across the river Ravi and a power plant on its left bank.
31. **Rajasthan Canal** (Rajasthan): The project uses water released from the Pong dam and provides irrigation facilities to the northwestern region of Rajasthan, i.e. a part of the Thar Desert. It consists of the Rajasthan feeder canal (with the first 167 km in Punjab and Haryana and the remaining 37 km in Rajasthan) and the 445 km Rajasthan main canal entirely in Rajasthan.
32. **Parambikulam Allyar** (Joint venture of Tamil Nadu and Kerala): The project envisages the integrated harnessing of eight rivers, six in the Annamalai hills and two in the plains.
33. **Sarda Sahayak** (Uttar Pradesh): The project envisages the construction of a barrage across the river Ghagra, a link channel, a barrage across river Sarda, and a feeder channel involving constructions of two major aqueducts over Gomti and Sai.
34. **Ramganga** (Uttar Pradesh): A dam across Ramganga, a tributary of the Ganga River located in Garhwal district. The project has, besides

reducing the intensity of floods in central and western Uttar Pradesh, provided water for the Delhi water supply scheme.

35. **Left Bank Ghagra Canal** (Uttar Pradesh): A link channel taking off from the left bank of Ghagra River of Girja barrage and joining with Sarju River. Also a barrage across Sarju.
36. **Tehri Dam** (Uttar Pradesh): Earth and rock-fill dam on Bhagirathi River in Tehri district.
37. **Madhya Ganga Canal** (Uttar Pradesh): A barrage across Ganga in Bijnor district.
38. **Farakka** (West Bengal): The project was taken up for the preservation and maintenance of Kolkata port and for improving the navigability of the Hooghly. It comprises a barrage across the Ganga at Farakka, a barrage at Jangipur across the Bhagirathi, and a feeder channel taking off from the Ganga at Farakka and tailing at the Bhagirathi below the Jangipur barrage.
39. **Mayurakshi** (West Bengal): Irrigation and hydroelectric projects comprise the Canada Dam.
40. **Kangsabati** (West Bengal): The project envisages the construction of dams on the Kangsabati and Kumari rivers.
41. **Damodar Valley Project** (West Bengal and Bihar): A multipurpose project for the unified development of irrigation, flood control, and power generation in West Bengal and Bihar. It comprises multipurpose dams at Konar, Tilaiya, Maithon, and Panchet, hydel power stations at Tilaiya, Konar, Maithon, and Panchet barrage at Durgapur; and thermal powerhouses at Bokaro, Chandrapura, and Durgapur. The project is administered by the Damodar Valley Corporation (DVC).

3.4 IRRIGATION METHODS

Irrigation is an artificial application of water to the soil to enable crops to have favourable growth, particularly in areas where the amount of rainfall is not adequate and it is not uniformly distributed. The three basic irrigation methods used are surface irrigation, sprinkler irrigation, and micro or drip irrigation depending on the type of crop, soil, topographic and climatic conditions. An irrigation method should suit the local conditions. All methods have their advantages and disadvantages. The basic purpose of irrigation is to apply irrigation water uniformly so that each plant gets the desired quantity of water, neither too much nor too little. The suitability of the various irrigation methods, i.e. surface, sprinkler, or drip irrigation depends mainly on the following factors:

- soil type
- crop
- topography
- climatic conditions

- source of water and its quality
- power availability
- cost and benefits

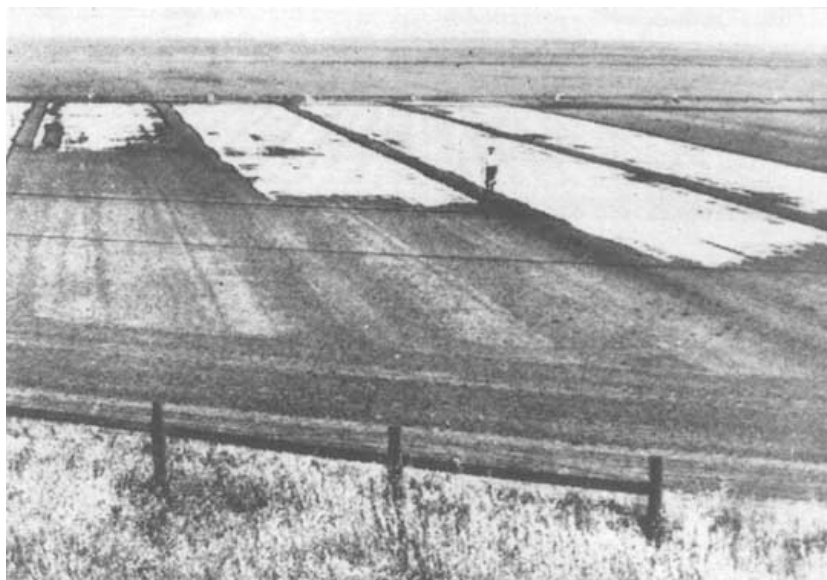
The three irrigation methods are discussed below.

3.4.1 Surface Irrigation

In the surface irrigation method, water moves over and across the land by simple gravity flow to wet it and infiltrate into the soil. Historically, this has been the most common method of irrigating agricultural land due to its simplicity and lower application cost. On loam or clay soils, all three irrigation methods can be used, but surface irrigation is more commonly found. Clay soils with low infiltration rates are ideally suited to surface irrigation. All soil types, except coarse sand with an infiltration rate of more than 30 mm/hour, can be used for surface irrigation. If the infiltration rate is higher than 30 mm/hour, sprinkler or drip irrigation should be practised.

Surface irrigation includes the following types:

- a) Continuous flood or paddy irrigation, in which small basins are flooded during essentially the growing season.
- b) Basin irrigation confines water to a given area by ponding over the area but remains ponded in orchards. A separate basin is formed for each tree and water is supplied through a supply ditch.
- c) As shown in Fig.3.1, border-strip irrigation applies water to one end of a rectangular strip of sloping land so that water advances downslope and either runs off the end or ponds behind a dike. The border strip method wherein the farm is divided into a series of strips 5 to 10 m wide is commonly used for all close-growing crops including cereal crops like wheat.
- d) As shown in Fig.3.2, Furrow irrigation uses furrows made between crops planted in rows to control and guide water for either steep land or very level land. This method of irrigation is very commonly used for row crops like maize, *jowar*, sugarcane, cotton, tobacco, and groundnut.



Where, water levels from the irrigation source permit, the levels are controlled by dikes, usually plugged by soil. This is often seen in terraced rice fields (rice paddies), where the method is used to flood or control the level of water in each distinct field. In some cases, the water is pumped, or lifted by human or animal power to the level of the land.

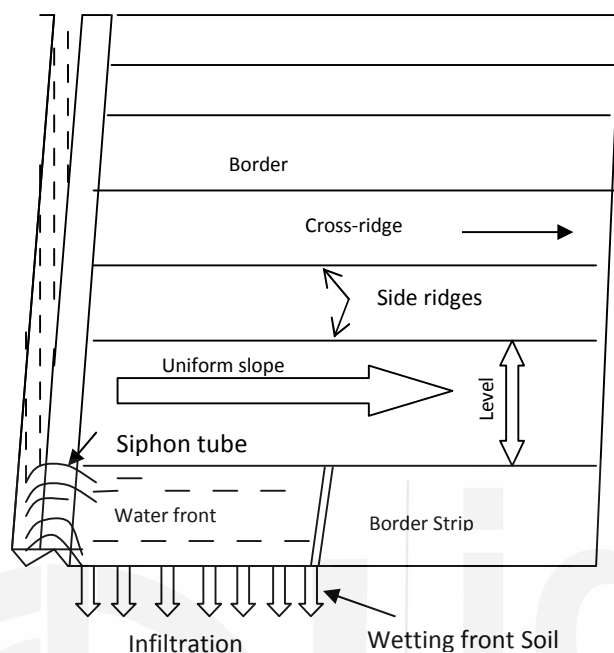


Fig. 3.1: Border irrigation

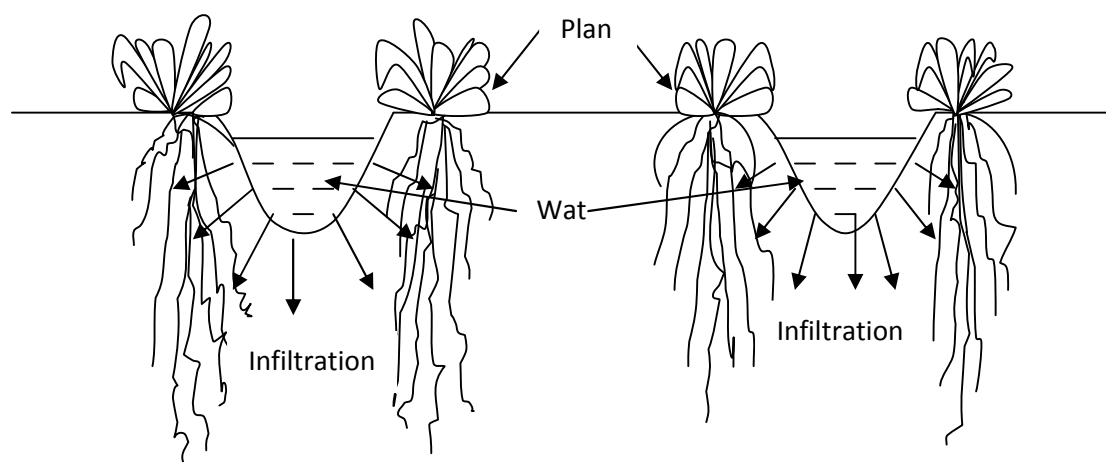


Fig. 3.2: Furrow irrigation

Advantages

- Simple, cheap, and easy to operate;
- Suited to most of the close-growing (border) and row crops (furrow); and
- Adopted for most of the soils.

Limitations

- Irrigation efficiency (ratio of water gainfully used by the crop and water applied from the source) is very low (30-40%) as a result of which scarce and precious water resource is wasted;
- A considerable area used by the water channels;
- Periodic maintenance is required; and
- Leaching of fertilizer and nutrients.

3.4.2 Sprinkler Irrigation

Sandy soils have a low water storage capacity and a high infiltration rate. They, therefore, need frequent but small irrigation applications, in particular when the sandy soil is also shallow. Under these circumstances, sprinkler irrigation is more suitable than surface irrigation as shown in Fig. 3.3. The sprinklers operate at a considerably higher pressure of $2-3 \text{ kg/cm}^2$ ($1 \text{ kg/cm}^2 = 10 \text{ m head of water}$). The sprinklers apply water in the form of rainwater, which helps in creating a better microclimate and also protects plants from frost.

The system is suited to the following soil, crop, topographic and climatic conditions:

- Sandy soils with a high infiltration rate;
- Undulating topography;
- Close growing crops; and
- Moderate wind speed



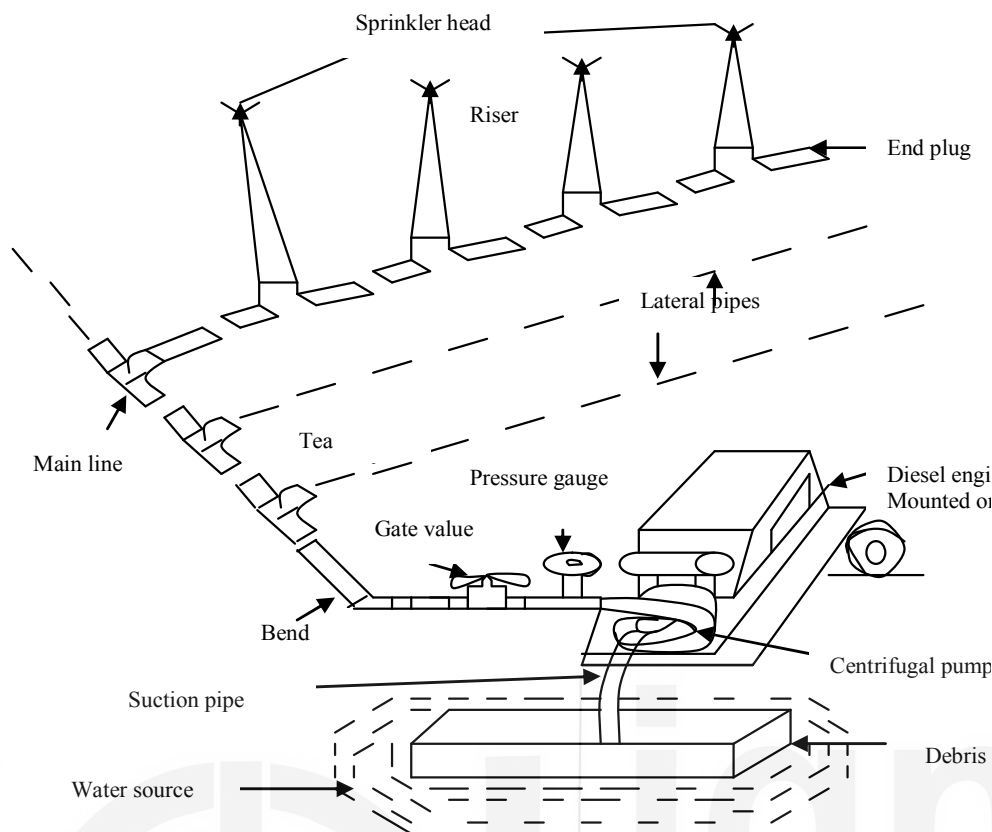


Fig. 3.3: Sprinkler irrigation system

Sprinkler irrigation systems are divided based on whether the structure moves in the field during the irrigation event. There are four basic categories:

- Permanent, solid-set sprinklers are fixed on risers from buried lines or lines suspended above a crop or over trees.
- Hand-move sprinklers are fixed sprinklers that are disassembled, moved, and reassembled between irrigations.
- Continuous-move sprinkler systems move continuously during irrigation, and
- Center-pivot irrigation systems supply water at a central point and a lateral line rotates around this center and covers every area at a time.

Advantages

- Uniform water application as per the requirement in the field;
- Improves microclimate;
- High irrigation efficiency (70-80%); and
- High fertilizer efficiency.

Limitations

- High initial investment and operating costs;
- Needs high energy; and
- High wind velocity affects the uniformity of water application.

3.4.3 Drip Irrigation

In drip irrigation, also known as trickle irrigation, water is delivered at or near the root zone of plants, drop by drop under low pressure. As you can see in Fig. 3.4, the system comprises plastic components such as pipes and drippers in addition to a filter and fertilizer applicator. This method can be the most water-efficient method of irrigation if managed properly since evaporation and runoff are minimized. In this method, fertilizer can also be applied, known as fertigation. If a drip system is operated for too long or if the delivery rate is too high, it may result in deep percolation. As water is applied in the vicinity of the plants, only a part of the area is wetted. As a result of this, there is a considerable saving of water. The fertilizer application is also highly efficient due to its use with irrigation water. Due to the optimum use of water and nutrients, not only crop productivity is high, but its quality is also superior as compared with surface irrigated crops. The drip method is suited to the following conditions:

- Widely spaced crops, such as orchard crops;
- Undulating lands with high infiltration rates;
- Widely spaced crops; and
- Saline irrigation water

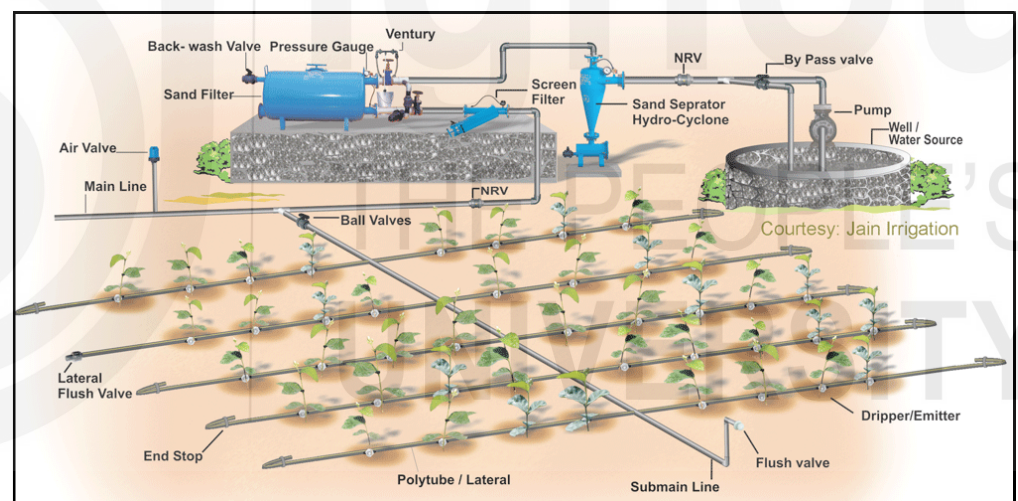


Fig. 3.4: Layout of drip irrigation

(Source: Jain Irrigation Systems Ltd.)

Advantages

- Considerable saving of water as only part of the area wetted;
- Less amount of fertilizer used;
- Relatively saline water usable;
- High irrigation efficiency (>90%)
- High fertilizer efficiency;
- High yields and better quality of produce; and
- Less weed infestation.

Limitations

- High initial investment.

3.5 IRRIGATION SCHEDULING

Irrigation scheduling is the decision of when and how much water should be applied to a field. The objective of irrigation scheduling is to maximize irrigation efficiencies by applying the exact amount of water needed to replenish the soil moisture to the desired level. It saves water and energy. The irrigation scheduling procedures consist of monitoring indicators that determine the need for irrigation. The amount of water applied is determined by using a criterion to determine irrigation needs and a strategy to prescribe how much water to apply in any situation.

3.5.1 Irrigation Scheduling Criteria

Irrigation criteria are the indicators used to determine the need for irrigation. The most common irrigation criteria are soil moisture content and soil moisture tension. Irrigation scheduling maximizes yield, saves water and hence increases the net return. The final decision depends on the irrigation criterion, strategy, and goal. Irrigators need to define a goal and establish an irrigation criterion and strategy. To illustrate irrigation scheduling, consider a farmer whose goal is to maximize yield. Soil moisture content is the irrigation criterion. Different levels of soil moisture trigger irrigation. For example, when soil water content drops below 70 per cent of the total available soil moisture, irrigation should be given. Soil moisture content to trigger irrigation depends on the irrigator's goal and strategy.

The importance of irrigation scheduling is that it enables the irrigator to apply the exact amount of water to achieve the goal. This increases irrigation efficiency. A critical element is the accurate measurement of the volume of water applied or the depth of application. A farmer cannot manage water to the maximum efficiency without knowing how much water was applied. Also, uniform water distribution across the field is important to derive the maximum benefits from irrigation scheduling and management. Accurate water application prevents over-or-under irrigation. Over irrigation wastes water, energy, and labour; leaches expensive nutrients below the root zone, out of reach of plants; and reduces soil aeration, and thus crop yields. Under irrigation stresses, crop yields may be reduced.

3.5.2 Advantages of Irrigation Scheduling

Irrigation scheduling offers several advantages:

1. It enables the farmer to schedule water rotation among the various fields to minimize crop water stress and maximize yields.
2. It reduces the farmer's cost of water and labour through less irrigation, thereby making maximum use of soil moisture storage.
3. It lowers fertilizer costs by holding surface runoff and deep percolation (leaching) to a minimum.
4. It increases net returns by increasing crop yields and crop quality.

5. It minimizes water-logging problems by reducing the drainage requirements.
6. It assists in controlling root zone salinity problems through controlled leaching.
7. It results in additional returns by using the "saved" water to irrigate non-cash crops that otherwise would not be irrigated during water-short periods.

3.5.3 Irrigation Scheduling Methods

All irrigation scheduling methods consist of an irrigation criterion that triggers irrigation and an irrigation strategy that determines how much water is to be applied. Irrigation scheduling methods differ by the irrigation criterion or by the method used to estimate or measure this criterion. A common and widely used irrigation criterion is soil moisture status. Table 3.1 compares different methods of irrigation scheduling by monitoring soil moisture content or tension. The methods described in Table 3.1 measure or estimate the irrigation criterion.

Table 3.1: Methods of Irrigation Scheduling

Method	Measured parameter	Equipment needed	Irrigation criterion	Advantages	Disadvantages
Hand feel and appearance of soil	Soil moisture content by feel	Hand probe	Soil moisture content	Easy to use; simple; can improve accuracy with experience	Low accuracies; fieldwork involved taking samples
Gravimetric soil moisture sample	Soil moisture content by taking samples	Auger, caps, oven	Soil moisture content	High accuracy	Labor intensive including fieldwork; time gap between sampling and results
Tensiometers	Soil moisture tension	Tensiometers including vacuum gauge	Soil moisture tension	Good accuracy; instantaneous reading of soil moisture tension	Labour to read; needs maintenance; breaks at tensions above 0.7 atm
Electrical resistance blocks	Electric resistance of soil moisture	Resistance blocks AC bridge (meter)	Soil moisture tension	Instantaneous reading; works over a larger range of tensions; can be used for remote reading	Affected by soil salinity; not sensitive at low tensions; needs some maintenance and field reading
Water budget	Climatic parameters	Weather station or	Estimation of	No field work	Needs calibration

approach	: temperature, radiation, wind, humidity, and expected rainfall, depending on the model used to predict ET.	available weather information	moisture content	required; flexible; can forecast irrigation needs in the future; with same equipment can schedule many fields	and periodic adjustments, since it is only an estimate; calculations cumbersome without a computer
Modified atmometer	Reference ET	Atmometer gauge	Estimate of moisture content	Easy to use, direct reading of reference ET	Needs calibration; it is only an estimation

Check Your Progress Exercise 1

Note: a) Use the space below for writing your answers.

b) Compare your answer with those given at the end of the unit.

1) Define irrigation.

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2) List different methods of irrigation water application.

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3) Which is the most commonly used irrigation method?

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4) List the factors governing the choice of irrigation methods.

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5) Under what condition is the sprinkler method of irrigation adopted?

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6) What do you understand by irrigation scheduling? List any two advantages of irrigation scheduling.

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3.6 COMMAND AREA DEVELOPMENT AND WATER MANAGEMENT

The Centrally sponsored Command Area Development (CAD) Programme was launched in 1974-75 with the main objectives of improving the utilization of created irrigation potential and optimizing agriculture production and productivity from irrigated agriculture through a multi-disciplinary team under an Area Development Authority.

Initially, 60 major and medium irrigation projects were taken up under the CAD Programme, covering a Culturable Command Area (CCA) of about 15.00 million hectares. From 1974-75 till now 314 projects with a CCA of 28.95 million ha have been included under the programme. After the inclusion of new projects, deletion of completed projects, and clubbing of some projects, there are now 136 projects under implementation. The programme was restructured and renamed as Command Area Development & Water Management (CADWM) Programme w.e.f. 1-4-2004.

The following components are covered under the CADWM Programme:

- i) Soil & Topographic Survey, planning and designing of On-Farm Development (OFD) works;
- ii) Correction of system deficiencies above the outlet up to distributaries of 4.25 cumec (150 cusec capacity);
- iii) Construction of field channels, with a minimum of 10% beneficiary contribution of total cost in cash or labour;
- iv) Full package OFD works including construction of field channels, realignment of field boundaries, land levelling and shaping also with a minimum of 10% beneficiary contribution of total cost in cash or labour;
- v) Warabandi (with hardware assistance covered under item (ii) and software assistance covered under item (ix);
- vi) Construction of field drains, intermediate and link drains for letting out

surplus water;

- vii) Reclamation of waterlogged areas of irrigated commands using conventional techniques and including bio-drainage, wherever applicable, with a minimum of 10% beneficiary contribution of total cost in cash or labour;
- viii) Renovation and desilting of existing irrigation tanks including the irrigation system and control structures within the designated irrigation commands (a minimum of 10% contribution by beneficiary farmers of total cost in cash or labour is necessary for this activity. This money is to be deposited in the accounts of WUAs as a maintenance fund, the interest on which will be used for the maintenance of tanks and the system by tank WUA).
- ix) State-sponsored software components such as training of farmers and field functionaries & officials, adaptive trials & demonstrations, action research for Participatory Irrigation Management, seminars/conferences/workshops, monitoring & evaluation of the programme, etc. through Water and Land Management Institutes (WALMI) and other states/central institutions.
- x) Institutional support to Water Users' Associations;
- xi) Establishment cost - 20% of OFD works on items (iii) or (iv), (vi), and (vii) and;
- xii) R&D activities, including training of senior-level officers, conferences, workshops, seminars, etc. arranged directly by the Ministry.

The scheme is now being implemented as a state sector scheme during the XI Five Year Plan (2008-09 to 2011-12). The evaluation made in the past has revealed that the CAD Programme made a positive impact on various important indicators, like an increase in the irrigated area, productivity, and production, irrigation efficiency, etc. Despite efforts for efficient irrigation water management, the problem of waterlogging has surfaced in many irrigated commands.

3.7 PARTICIPATORY IRRIGATION MANAGEMENT (PIM)

The National Water Policy 1987 advocated the involvement of farmers in the management of irrigation. The irrigation potential increased nearly four times since the beginning of the planned era but brought in several problems in the management of irrigation in its wake. These included the unreliable and inequitable supply of water, especially at the tail-end, improper organization and monitoring (O&M) of the systems, poor recovery of water rates, indiscipline distribution of water, and the problem of waterlogging due to seepage from the canal network on the one hand and over-irrigation on the other. To address these problems, it has been recognized that the participation of beneficiaries would help greatly for the optimum upkeep of the irrigation system and utilization of irrigation water. Keeping this aspect in view, Participatory Irrigation Management (PIM) was the thrust area under the programme during the Ninth Five Year Plan period.

Despite efforts by the Ministry of Water Resources (MOWR), the pace of PIM has been slow as several states have yet to enact legislation on PIM. Because of the problems associated with the O&M of the deteriorated irrigation systems, PIM is considered to be a necessity. The Ministry, therefore, circulated a model act on PIM in the year 1998 with the view to facilitate action by the state governments. Subsequently, conferences on PIM were also organized by the Ministry. The Ministry has also been organizing National level training programmes on PIM in various parts of the country for Command Area Development (CAD) Programme functionaries. In addition, a grant is also provided to states for organizing state and project-level training programmes for farmers and field functionaries.

3.7.1 Objectives of PIM

The main objectives of PIM are:

- i) To create a sense of ownership of water resources and the irrigation system among the users, to promote economy in water use and preservation of the system.
- ii) To improve service deliveries through better operation and maintenance,
- iii) To achieve optimum utilization of available resources through sophisticated deliveries, precisely as per crop needs,
- iv) To achieve equity in water distribution,
- v) To increase production per unit of water, where water is scarce and to increase production per unit of land where water is adequate,
- vi) To make the best use of natural precipitation and groundwater in conjunction with flow irrigation for increasing irrigation and cropping intensity,
- vii) To facilitate the users to have a choice of crops, cropping sequence, the timing of water supply, period of supply and also the frequency of supply, depending on soils, climate and other infrastructure facilities available in the commands such as roads, markets cold storages, etc., to maximize the incomes and returns,
- viii) To encourage collective and community responsibility on the farmers to collect water charges and payments to irrigation agencies,
- ix) To create a healthy atmosphere between the Irrigation Agency personnel and the users.

3.7.2 Provisions in PIM Acts

Recognizing the need for a sound legal framework for PIM in the country, the Ministry brought out a model act to be adopted by the state legislatures for enacting new irrigation acts/amending the existing irrigation acts for facilitating PIM. The legal framework provides for the creation of organisations at different levels of the irrigation system as under:

- i) **Water Users' Association (WUA):** It will have a delineated command area on a hydraulic basis, which shall be administratively viable. Generally, a WUA would cover a group of outlets or a minor.
- ii) **Distributary Committee:** It will comprise 5 or more WUAs. All the presidents of WUAs will comprise the general body of the distributary committee.
- iii) **Project Committee:** It will be an apex committee of an irrigation system and presidents of the Distributary committees in the project area shall constitute the general body of this committee.

The Associations at different levels are expected to be actively involved in (i) maintenance of irrigation system in their area of operation; (ii) distribution of irrigation water to the beneficiary farmers as per the warabandi schedule; (iii) assisting the irrigation department in the preparation of water demand and collection of water charges; (iv) resolve disputes among the members and WUA and (v) monitoring flow of water in the irrigation system, etc.

3.7.3 Constraints in Implementation of PIM

There are several constraints in making the PIM sustainable in the long run. Some of these are:

- i) **Lack of legal backup and policy changes:** In many states, there is no or very little legal backup and a lack of policy decisions at the government level to take up PIM, which is a big impediment in the implementation of PIM.
- ii) **System deficiency:** In older projects, there are many problems like deterioration of old control and measuring structures, leakages, and seepage at various places, erosion of banks and beds, siltation, and weed infestation. These are serious problems, hindering farmers from taking over the system management on technical and financial considerations.
- iii) **Uncertainty of water availability:** This is another important aspect, as farmers will understandably be reluctant to take on the responsibility for managing the system unless deliveries of water are made reliable, flexible, practical, and responsive to their needs. The Engineers on their part may not be confident about ensuring the supply of the requisite quantity of water to the WUAs, as would be obligatory in terms of the MOU signed between the irrigation agency and WUA.

Further, the farmers who have their holdings at the head of the canal tend to appropriate more water than required, whereas the farmers at the tail end often fail to get their apportioned share of water. Head-enders, therefore, have a vested interest in continuing the existing arrangements. The tail-enders may not be keen to form WUAs as water supply in such areas remains inadequate and erratic and they remain apprehensive that the situation will not be materially altered if an association is formed. These differences in perceptions and conflicts of interest inhibit the coming together of the head-end and tail-end farmers.

- iv) **Fear of financial viability:** Maintenance and operation of the system demand huge finances. Farmers have apprehension that in the absence of surety of finance, it would be difficult for them to fulfil the requirement of funds for operation and maintenance. They feel that when the government is not able to handle the system with huge amounts of money available to them, how farmers would be able to do justice?
- v) **Lack of technical knowledge:** Apart from the financial uncertainty, lack of technical input is one of the inhibiting factors to take over the system.
- vi) **Lack of leadership:** On account of the limited exposure of the farmers to the rest of the world and PIM in particular, potent leadership is lacking, rather on account of limited knowledge. At times so-called local leaders give the negative or unclear version before other farmers which further creates misunderstanding among the farmers bringing them sometimes into a fix.
- vii) **Lack of publicity and training:** Seeing is believing; and knowledge brings confidence in people. This aspect is lacking and there is a constraint to the adoption of PIM.
- viii) **Demographic diversity:** Due to variations in economic, ethnic, education levels, etc. diversity of farmers, PIM is taking much time in this country. To handle this aspect deep study, analysis and solution need to be found out.
- ix) **Mega irrigation projects:** The world scenario indicates that there are smaller projects in the countries of the world, where irrigation project transfer has taken care of PIM. In India, huge projects have a very large distribution system and culturable command area sometimes more than 20 lakh hectares. The larger the project, the more complex would be its maintenance, operation, and management aspects and so the formation and functioning of farmers' associations for different necessary activities.
- x) **WUAs v/s Panchayats:** In many of the areas, where WUAs have been formed, there is a clash of interest among Panchayats and WUAs on who is to own the system, particularly when watershed schemes are being handed over to the Panchayats.

3.8 DRAINAGE

Drainage is the removal and disposal of excess water from the field. Excess water in the root zone restricts soil aeration, affects soil temperatures, hinders tillage operations, etc. which should be removed by adopting proper drainage methods. When the water table comes near the surface and affects the crop growth such a condition is called waterlogging. It affects the root development of the crops and can bring harmful substances to the root zone. Waterlogging problems are widespread in irrigated or high-rainfall areas. Drainage is necessary not only for the removal of excess water but also for

removing the salts from the root zone.

Reasons for waterlogging

Both natural and man-made (artificial) conditions may cause waterlogging. Important natural causes for waterlogging may be:

- i) The poor natural drainage of the subsoil due to the existence of hardpan or rock near the soil surface;
- ii) submergence under floods; and
- iii) deep percolation from rainfall.

The important artificial causes of waterlogging are:

- i) High intensity of irrigation irrespective of the soil and subsoil;
- ii) Heavy seepage from unlined canals, and farm watercourses;
- iii) Enclosing irrigated fields with embankments and choking up natural drainage; and
- iv) Non-maintenance of natural drainages or blocking of natural drainage channels by roads and railways.

Benefits of drainage

The important benefits of drainage in the agriculture field are given below:

1. Provides a better environment for plant growth
2. Improves the soil structure and infiltration characteristics of soil
3. Provides optimum condition for tillage for a longer range of time
4. Maintain desirable soil temperature
5. Promotes increased leaching of salts and prevents their accumulation in soil
6. Larger crop growing season
7. The weed problem is minimized
8. Reduced diseases which thrive on wetland
9. Opportunity for desirable soil micro-organisms to develop through aeration and higher soil temperatures.

Methods of Drainage

Drainage problems may be caused by one or a combination of several factors mentioned above. Control of the sources of excess water and provision for removal of the excess water is required to solve the drainage problem. Based on the severity and the causes of the drainage problem, we may need to apply one or a combination of measures to solve the problem. Period and the amount of water standing, number of days for which soil is workable for agricultural operations, the salinity of topsoil, depth of water table, etc should be considered in deploying the drainage method.

The surface method and subsurface method are two broad types of drainage methods.

1. **Surface drainage methods:** Surface drainage is the safe removal of excess water through land shaping and improved or constructed channels from the land surface. Surface drainage in agricultural lands is needed to remove the excess rainfall as well as collect and dispose of the excess surface runoff. This problem mainly arises in soils with low infiltration rates and heavy soils. In deep heavy soils, wherein vertical movement of water is restricted and also the inadequate hydraulic conductivity of the substrata, subsurface drainage systems may not be feasible. Surface drainage is the only possible method in such an area. The systems are also known as shallow drainage systems.

In surface drainage, water moves due to land elevation to provide a hydraulic gradient and the system can be considered in three functional parts, viz. (1) collection system, (2) conveyance or disposal system, and (3) outlet. Water from the individual fields is collected through the collection system and moves through the disposal system to the outlet.

There are four types of drainage systems used in flat areas (less than 2 per cent slope). These are: (1) Random drain system (2) Parallel filed drain system, (3) Parallel open ditch system, and (4) Bedding system.

2. **Sub-surface drainage methods:** A subsurface drainage system is required for water table control and for maintaining a favourable salt balance in the crop root zone to create a favourable environment for crop production. In sub-surface drainage, water moves under the influence of gravity to suitable outlets. This can be accomplished by tile drains including perforated pipes, mole drains, drainage wells, deep open drains, and a combination of tile and open drains. Deep open drains serve the purpose of removing excess surface water and lowering the groundwater table. However, their main functions are the removal of excess surface water and to serve as an outlet for underground pipe drains, called tile drains. Sub-surface drainage may be obtained mainly by a sub-surface drainage system called tile drainage and vertical subsurface drainage comprising a system of drainage wells.

Check Your Progress Exercise 2

Note: a) Use the space below for writing your answers.

b) Compare your answers with those given at the end of the unit.

- 1) When Command Area Development (CAD) Programme was launched? Give its objectives.

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2) What do you understand by the Participatory Irrigation Management?

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3) What do you understand by drainage problem?

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3.9 LET US SUM UP

Water is the most essential and crucial input for production. If the appropriate quantities are not available at the right time, nothing can be produced. Due to the uneven distribution of rainfall, the country needed to develop its irrigation potential and spread the canal networks far and wide. Consequently, there has been a continuous increase in the irrigation potential through the canal and water distribution system created under major & medium irrigation projects and minor irrigation schemes utilizing surface water. Water being a state subject, planning and implementation of water resources projects are undertaken by respective state governments. Several schemes have been planned for implementation by the respective state governments. The government of India provides central grants to the state governments under the Accelerated Irrigation Benefits Programme for early completion of the schemes. The Ministry of Water Resources has also prepared a National Perspective Plan for optimal utilization of the water resources particularly through the diversion of surplus flood water to water-deficient areas. National Perspective Plan envisages various links to divert surplus flood water to water-deficient areas. Ministry of Water Resources has also launched schemes for repair, renovation, and restoration of water bodies under which assistance is provided to States.

3.10 KEYWORDS

Canal: A water-carrying channel or conduit.

Created potential: Provisions for making water available for the successful growing of the crop.

Drainage: Removal of water from the root zone.

Irrigation: Artificial water application for crop production.

Ultimate potential: Area that can be supplied with water for producing crops.

Water balance: A total of all the water inflows and outflows.

Water demand: Requirement of water during the crop growing season.

Water logging: Saturation of the crop root zones due to water stagnation that hampers production and productivity.

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3.12 ANSWERS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress Exercise 1

- 1) Irrigation is provided if there is not enough moisture in the absence of rain to produce usable commodities for humanity at large; hence the supplemental application of water to crops may be referred to as irrigation.
- 2) Continuous flood, basin irrigation, border-strip irrigation, furrow irrigation, sprinkler irrigation methods, drip irrigation methods, etc.
- 3) Border-strip irrigation
- 4) Soil type, crop, topography, climatic conditions, source of water and its quality, power availability, cost, benefits, etc.
- 5) Sandy soils have a low water storage capacity and a high infiltration rate. They, therefore, need frequent but small irrigation applications, in particular when the sandy soil is also shallow.
- 6) Irrigation scheduling is the decision of when and how much water to apply to a field. It enables the farmer to schedule water rotation among

the various fields to minimize crop water stress and maximize yields. It reduces the farmer's cost of water and labour through less irrigation, thereby making the maximum use of soil moisture storage.

Check Your Progress Exercise 2

- 1) The Command Area Development (CAD) Programme was launched in 1974-75 with the main objectives of improving the utilization of created irrigation potential and optimizing agriculture production and productivity from irrigated agriculture through a multi-disciplinary team under an Area Development Authority.
- 2) To address various problems in the canal network area, it has been recognized that the participation of beneficiaries would help greatly for the optimum upkeep of the irrigation system and utilization of irrigation water.
- 3) Excess water in the root zone restricts soil aeration, affects soil temperature, hinders tillage operations, *etc.*, which should be removed by adopting the proper drainage method.



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UNIT 4 SOIL FERTILITY MANAGEMENT

Structure

- 4.0 Objectives
- 4.1 Introduction
- 4.2 Soil Fertility
- 4.3 Soil Fertility Status of Indian Soils
- 4.4 Essential Plant Nutrients: Macro and Micro Nutrients
 - 4.4.1 Classification of Essential Plant Nutrients
 - 4.4.2 Role of Essential Plant Nutrients
 - 4.4.3 Deficiency/ Toxicity Symptoms of Essential Plant Nutrients
 - 4.4.4 Nutrient Mobility and its Effect on Deficiency Symptoms
 - 4.4.5 Range of Nutrient Supply from Deficiency to Toxicity
- 4.5 Evaluation/ Assessment of Soil Fertility
 - 4.5.1 Visual Plant Diagnosis
 - 4.5.2 Plant Analysis
 - 4.5.3 Field Experiment: Control Test Plot & Complex Experiments
 - 4.5.4 Soil Testing
- 4.6 Maintenance of Soil Fertility
 - 4.6.1 Balanced Plant Nutrition/ Fertilization
 - 4.6.2 Integrated Plant Nutrient Management
- 4.7 Let Us Sum Up
- 4.8 Keywords
- 4.9 Suggested Readings/ References
- 4.10 Answers to Check Your Progress Exercise

4.0 OBJECTIVES

After studying this unit, you should be able to:

- explain the meaning, groups, and components of soil fertility;
- identify the status and causes for the decline in soil fertility of Indian soils;
- describe the essential plant nutrients, their movement in the soil, and uptake by plants;
- explain the soil fertility assessment and maintenance under the Indian situation; and
- identify various nutrient sources available in India under integrated nutrient management.

4.1 INTRODUCTION

The decline in soil fertility is a major concern to the farmers, especially under intensive cropping systems. Proper soil fertility builds a solid foundation for

optimum yields. A sound soil fertility management is not an on-again-off-again practice, but rather a long-term investment. Therefore, soil fertility should be managed to get sustained crop production.

Soil fertility is a component of overall soil productivity that deals with its available nutrient status, and its ability to provide nutrients out of its reserves and through external applications for crop production. The declining soil fertility is a threat to sustainable crop production. It is quite pertinent to understand the reasons that reduce soil fertility. It is important to note that soil productivity is the crop-yielding ability of soil from a unit area. Soil fertility is vital to productive soil but, fertile soil is not always a productive one. Poor drainage, weeds, insects, diseases, drought, and other factors can limit productivity even when fertility is adequate. Fertile soils are not always highly productive, but productive soils are always fertile.

4.2 SOIL FERTILITY

Soil fertility is the overall ability of a soil to support vigorous crop growth by ensuring adequate plant nutrients and suitable conditions for water uptake, and by providing favourable conditions for root growth and development. “In other words” a fertile soil is one that contains an adequate supply of all the nutrients required for the successful production of plant life. Fertile soil is essential for optimum productivity, profitability, and, yes, even protection of groundwater quality and soil from erosion.

A. Soil fertility groups: Soil fertility is divided into 3 groups, i.e. physical, chemical, and biological soil fertility, which are interrelated with each other. Better management of physical and chemical soil fertility improves soil biological fertility, which in turn can be further enhanced by good agronomic/ cultural practices. Hence, soil fertility is a key to sustainable agriculture (Figure 4.1).

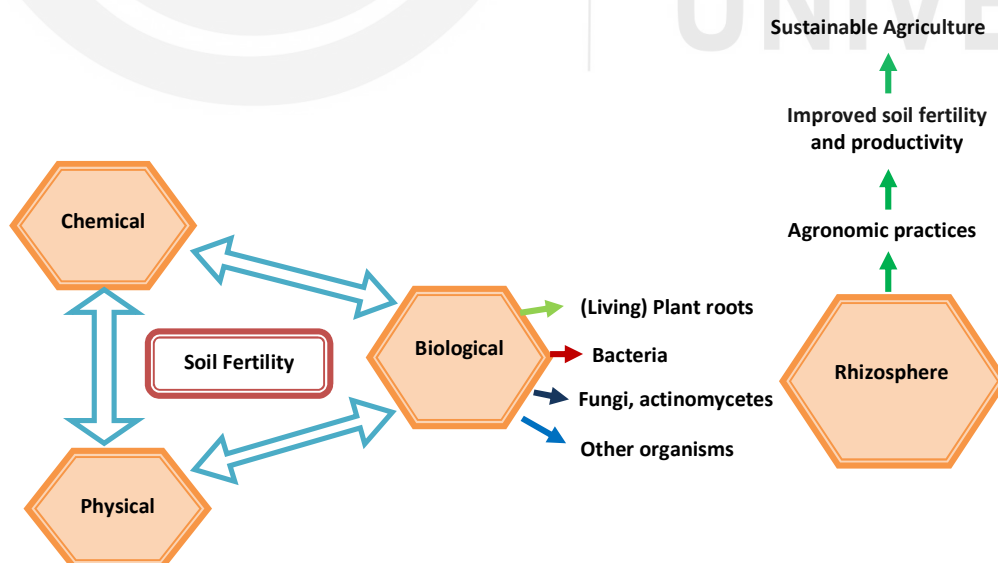


Fig. 4.1: Relationship of soil fertility to sustainable agriculture

1. Biological soil fertility: The capacity of organisms living in the soil to contribute to the nutritional requirements of plants and foraging animals for productivity, reproduction, and quality while maintaining biological

processes that contribute positively to the physical and chemical state of the soil.

2. **Chemical soil fertility:** The capacity of soil to provide a suitable chemical and nutritional environment for plants and foraging animals for productivity, reproduction, and quality in a way that supports beneficial soil physical and biological processes, including those involved in nutrient cycling.
3. **Physical soil fertility:** The capacity of soil to provide physical conditions that support plant productivity, reproduction, and quality without leading to loss of soil structure or erosion and supporting soil biological and chemical processes.

B. Components of soil fertility:

Soil fertility has the following important components:

1. **Soil depth:** It determines the volume of soil accessible to the root system. Most arable crops prefer about one meter without any obstructing layer.
2. **Soil structure:** It is based on the size distribution and aggregation of soil particles. This determines the distribution of pore sizes which is decisive for the supply of air and water to the roots.

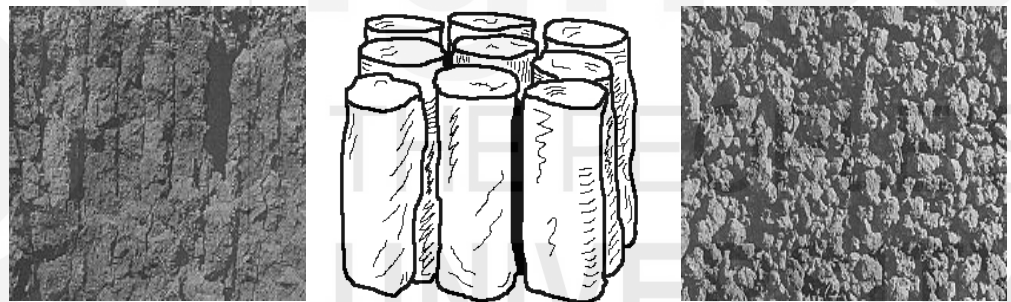


Fig. 4.2: Different kinds of soil structure

3. **Soil texture:** It designates the proportion of different particle size fractions in the soil. Of special importance to soil fertility is the percentage of soil particles of less than 2 mm in diameter, which constitutes fine soil. Soil texture influences to a large extent several components of soil fertility such as the amount of nutrient reserves and their proportion to the available nutrient fraction. It also influences several properties such as aeration, pore space distribution, Water holding capacity (WHC), and drainage characteristics.
4. **Content of nutrients:** with different degrees of availability.
5. **Storage capacity:** for soluble nutrients from the soil and fertilizers.
6. **Soil pH/ reaction:** an indicator and regulator of chemical processes and availability of nutrients.
7. **Content of detrimental or toxic substances:** either naturally occurring (e.g. salts in saline soils, aluminium in extremely acidic soils) or man-made (e.g. from pollution).

8. **Quantity and activity of soil organisms:** as agents of transformation processes.
9. **Soil organic matter (SOM):** It promotes the physical, chemical, and biological properties of soil.

4.3 SOIL FERTILITY STATUS OF INDIAN SOILS

India is largely a tropical country; hence, the organic carbon (C) content of the soils is low. Most of the nitrogen in the soil is present in organic forms. The faster decomposition of organic matter leads to a release of nitrogen, which is removed by the plants or lost to the environment. Therefore, the deficiency of N is universal in India. Most of the Indian soils are low to medium in available phosphorus. Due to intensive cropping and no replenishment, K deficiency has also become widespread in the country. Among the micronutrients, the deficiency of zinc is the most acute, followed by boron. There is a need to correct the deficiency of all these nutrients.

A. Causes for decline in soil fertility:

The causes of reduced soil fertility are given below.

1. **Loss of topsoil by erosion:** The topsoil is rich in nutrients and organic matter. Loss of the fertile topsoil components through erosion by water and wind results in decreased fertility.
2. **Nutrient mining:** The removal of more nutrients by crops than added through manures or fertilizers is called nutrient mining or depletion. Nutrient mining gets accelerated by imbalanced fertilization.
3. **Physical degradation of soil:** Physical degradation of soil (poor structure, compaction, crusting, and waterlogging) reduces its fertility. Degraded soil harms root growth.
4. **Decrease in organic matter content and soil bioactivity:** The decreased levels of organic matter cause a strong reduction in soil fertility as it plays several roles in soil. Decreased organic matter levels also result in poor physical, chemical, and biological properties.
5. **Loss of nutrients through various routes:** Ammonia volatilization from urea and some ammonium-containing fertilizers result in losses of nitrogen from the soil.
6. **Soil acidification, salinization, and alkalinization:** Acidification, salinization, or alkalinization causes a reduction in soil fertility, and eventually leads to problems of nutrient deficiencies, toxicities, and imbalances.
7. **Inefficient soil management:** Poor or inefficient soil management results in decreased soil fertility. Improper crop rotations may decrease soil fertility tremendously. Excessive soil tillage leads to erosion of the soil, which leads to reduced soil fertility.
8. **Soil pollution:** Soil pollution caused by indiscriminate use of agrochemicals and heavy metals reduces the fertility of the soil by affecting the soil's biological properties.

Check Your Progress Exercise 1

Note: a) Use the spaces given below for your answers.
b) Check your answer with those given at the end of the unit.

1. What do you mean by soil fertility?
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2. What are the three important groups of soil fertility?
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3. What is soil productivity?
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4. Give three important causes for a decline in soil fertility.
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4.4 ESSENTIAL PLANT NUTRIENTS: MACRO AND MICRO NUTRIENTS

Plants need water, air, light, suitable temperature, and seventeen nutrients to grow. Plants contain practically all natural elements but need only 17 for good growth. Essential plant nutrients absorbed by plants are necessary for the completion of their normal life cycle. These include C, H, O, N, P, K, Ca, Mg, S, Cu, Fe, Zn, Mn, B, Cl, Ni, and Mo. Plants get carbon, hydrogen, and oxygen from air and water. The other fourteen nutrients come from the soil. Nutrients are divided into two groups (macro and micro-nutrients) according to the amounts needed by plants. Some other mineral nutrient elements, e.g. Na, Si, Co, have a beneficial effect on some plants but are not essential.

A total of only 17 elements are essential for the growth and full development of higher green plants according to the criteria laid down by Arnon and Stout (1939). These criteria are:

- A deficiency of an essential nutrient makes it impossible for the plant to complete the vegetative or reproductive stage of its life cycle.
- Such deficiency is specific to the element in question and can be prevented or corrected only by supplying this element.
- The element is involved directly in the nutrition of the plant quite apart from its possible effects in correcting some unfavourable microbiological or chemical conditions of the soil or other culture medium.

4.4.1 Classification of Essential Plant Nutrients

Out of the 17 essential elements, carbon (C) and oxygen (O) are obtained from the gas CO_2 , and hydrogen (H) is obtained from water (H_2O). These three elements are required in large quantities for the production of plant constituents such as cellulose or starch. Hence, many times referred to as framework elements.

The other 14 elements are called mineral nutrients because they are taken up in mineral (inorganic) forms. They are traditionally divided into two groups, macronutrients, and micronutrients, according to the relative amounts required. Regardless of the amount required, physiologically, all of them are equally important. The 14 mineral elements are taken up by plants in specific chemical forms regardless of their source. Oxygen, carbon, and hydrogen make up 95 per cent of plant biomass, and the remaining 5 per cent is made up of all other elements. The difference in plant concentration between macronutrients and micronutrients is enormous. The relative contents of N and molybdenum (Mo) in plants are in the ratio of 10,000:1. Plants need about 40 times more magnesium (Mg) than Fe. These examples indicate the significant difference between macronutrients and micronutrients.

Macronutrients are nitrogen (N), phosphorus (P), potassium (K), calcium (Ca), magnesium (Mg), and sulphur (S). The one or two letter symbol in parentheses is the universal chemical symbol for that nutrient. N, P, and K are often referred to as primary nutrients and are the most common elements found in commercial fertilizers. Ca, Mg, and S are referred to as secondary nutrients and are also found in fertilizers and soil amendments. Micronutrients include iron (Fe), manganese (Mn), zinc (Zn), boron (B), copper (Cu), chloride (Cl), molybdenum (Mo), and nickel (Ni).

Cobalt (Co), sodium (Na), vanadium (V), and silicon (Si) are sometimes designated as beneficial plant nutrients. They are not required by all plants but appear to benefit certain groups of plants or plants growing under specific conditions. Cobalt is required for nitrogen fixation in legumes. Sodium and vanadium benefit some plants although they can complete their life cycle without them. Silicon is found in plant cell walls and appears to produce tougher cells. This increases the resistance of these plants to piercing and sucking insects and decreases the spread of fungal diseases. The classification of 17 essential nutrients is given in the following figure.

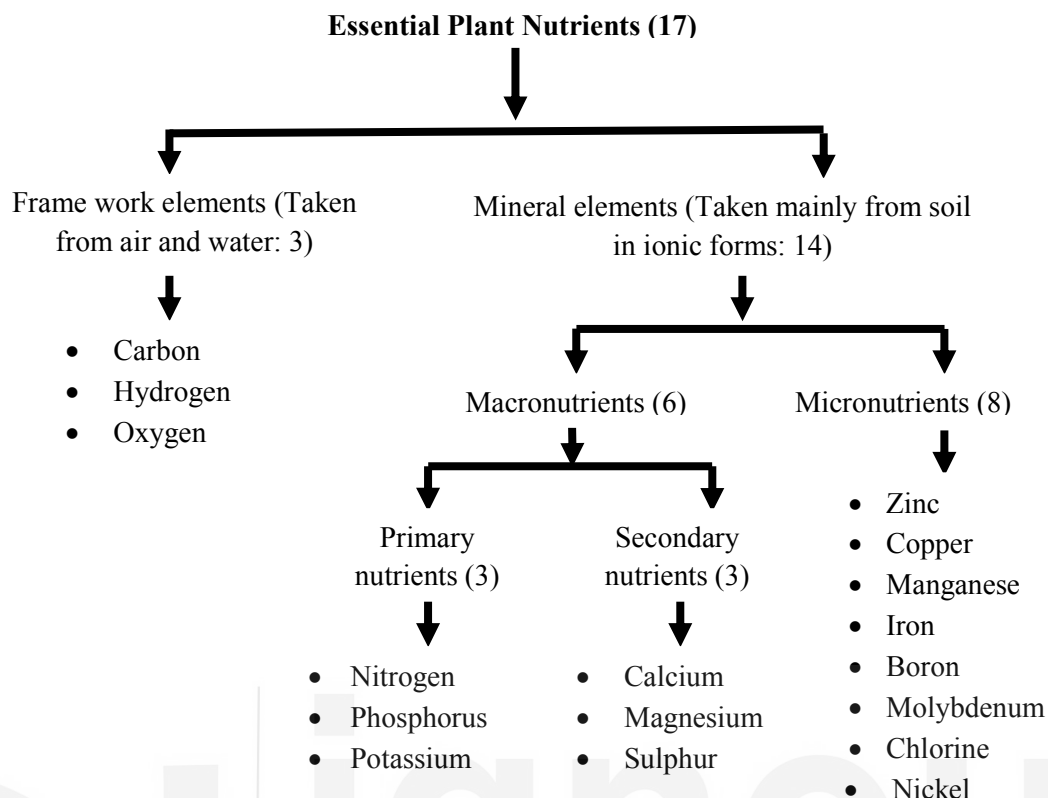


Fig. 4.2: Essential plant nutrients.

Some microelements can be toxic for plants at levels only somewhat higher than normal. In the majority of cases, this happens when the pH is low to very low. Aluminium and manganese toxicity are the most frequent ones, in direct relation to acid soils.

4.4.2 Role of Essential Plant Nutrients

Essential plant nutrients play a variety of roles/functions in plants. The important roles of different plant nutrients are described below:

1. **Nitrogen (N):** N is a part of chlorophyll (the green pigment in leaves) and is an essential constituent of all proteins. It is responsible for the dark green colour of stems and leaves, vigorous growth, branching/tillering, leaf production, size enlargement, and yield formation.
2. **Phosphorus (P):** It is essential for growth, cell division, root lengthening, seed and fruit development, and early ripening. It is a part of several compounds including oils and amino acids. The P compounds adenosine diphosphate (ADP) and adenosine triphosphate (ATP) act as energy carriers within the plants.
3. **Potassium (K):** K is involved in the working of more than 60 enzymes, in photosynthesis and the movement of its products (photosynthates) to storage organs (seeds, tubers, roots, and fruits), water economy, and providing resistance against several pests, diseases, and stresses (frost and drought).
4. **Calcium (Ca):** It is a part of the architecture of cell walls and membranes. It is involved in cell division, growth, root lengthening, and activation or inhibition of enzymes.

5. **Magnesium (Mg):** Mg occupies the centre spot in the chlorophyll molecule and, thus, is vital for photosynthesis. It is associated with the activation of enzymes, energy transfer, maintenance of electrical balance, production of proteins, metabolism of carbohydrates, etc. Mg is mobile within the plants.
6. **Sulphur (S):** It is a part of the amino acids cysteine, cystine, and methionine. Hence, it is essential for protein production. S is involved in the formation of chlorophyll and the activation of enzymes.
7. **Boron (B):** Key roles of B relate to (i) membrane integrity and cell-wall development, which affect permeability, cell division, and extension; and (ii) pollen tube growth, which affects seed/fruit set and, hence, yield.
8. **Chlorine (Cl):** It is thought to be involved in the production of oxygen during photosynthesis, in raising cell osmotic pressure, and in maintaining tissue hydration.
9. **Copper (Cu):** Cu is involved in chlorophyll formation and is a part of several enzymes such as cytochrome oxidase. It participates in lignin formation, protein, and carbohydrate metabolism, and is possibly required for symbiotic N fixation.
10. **Iron (Fe):** It plays a role in the synthesis of chlorophyll, carbohydrate production, cell respiration, chemical reduction of nitrate and sulphate, and N assimilation.
11. **Manganese (Mn):** It is known to activate several enzymes and functions as an auto-catalyst. It is essential for splitting the water molecules during photosynthesis. Mn is also important in N metabolism and CO₂ assimilation.
12. **Molybdenum (Mo):** Mo is involved in several enzyme systems, particularly nitrate reductase, which is needed for the reduction of nitrate, and nitrogenase, which is involved in BNF.
13. **Zinc (Zn):** It is required directly or indirectly by several enzyme systems, auxins, and protein synthesis, seed production, and rate of maturity. Zn is believed to promote RNA synthesis, which in turn is needed for protein production.
14. **Nickel (Ni):** Ni is a part of the enzyme urease, which breaks down urea in the soil. It also plays a role in imparting disease resistance and is considered essential for seed development.

4.4.3 Deficiency/Toxicity Symptoms of Essential Plant Nutrients

Plants show different shades of greenness, but a medium to dark green colour is usually considered a sign of good health and better growth. Chlorosis or yellowing of leaf colour can be a sign of a marginal deficiency and is often related to retarded growth. Chlorosis may be caused by a lower content of chlorophyll. A severe deficiency may result in the death of the tissue (necrosis). Necrotic leaves cannot be recovered by the addition of the missing nutrients, but the plant may survive by forming new leaves. Deficiency symptoms can serve as a guide for diagnosing limiting nutrients and the need

for taking corrective measures. The deficiency/ toxicity symptoms of essential nutrients in plants are given below:

1. **Nitrogen:** N is mobile in the phloem and therefore it can be re-translocated from older to younger leaves under N deficiency. N deficiency in plants results in a marked reduction in growth rate. In cereal crops, tillering is poor, and the leaf area is small. As N is a constituent of chlorophyll, its deficiency appears as yellowing or chlorosis of the leaves. The effects of N toxicity are less evident than those of its deficiency. They include prolonged growing (vegetative) periods and delayed crop maturity. High NH_4^+ in a solution can be toxic to plant growth, particularly where the solution is alkaline.
2. **Phosphorus:** P is readily mobile within the plant both in the xylem and phloem tissues. Plant growth is markedly restricted under P deficiency, which retards growth, tillering (cereals) and root development and delays ripening. The deficiency symptoms usually start on older leaves. A bluish-green to reddish colour develops, which can lead to bronze tints and red colour. Extremely high levels of P can result in toxicity symptoms. These generally manifest as a watery edge on the leaf tissue, which subsequently becomes necrotic.
3. **Potassium:** The general symptom of K deficiency is chlorosis along the leaf margins followed by scorching and browning of tips of older leaves. The affected area moves inwards as the severity of deficiency increases. K-deficiency symptoms occur in the older tissues because of the mobility of K. Affected plants have slow and stunted growth, weak stalks, susceptibility to lodging, greater incidence of pests and diseases, low yield, shrivelled grains, and in general, poor crop quality.
4. **Calcium:** Ca is immobile in the phloem, and hence its deficiency is seen first on growing tips and the youngest leaves. Ca-deficient leaves become small, distorted, cup-shaped, crinkled, and dark green. They cease growing, become disorganized, twisted, and, under severe deficiency, die.
5. **Magnesium:** As Mg is readily translocated from older to younger plant parts, its deficiency symptoms first appear in the older parts of the plant. A typical symptom of Mg deficiency is the interveinal chlorosis of older leaves in which the veins remain green but the area between them turns yellow. Leaves are small and break easily (brittle).
6. **Sulphur:** Unlike N deficiency, S-deficiency symptoms in most cases appear first on the younger leaves, and are present even after N application. Plants deficient in S are small and spindly with short and slender stalks. Their growth is retarded, and maturity in cereals is delayed. S toxicity can occur under highly reduced conditions, possibly as a result of sulphide (H_2S) injury.
7. **Boron:** B deficiency usually appears on the growing points of roots, shoots, and youngest leaves. Young leaves are deformed and arranged in the form of a rosette. There may be cracking and cork formation in the

stalks, stem, and fruits; thickening of stem and leaves; shortened internodes, withering or dying of growing points, and reduced bud, flower, and seed production. B-toxicity symptoms are yellowing of the leaf tip followed by gradual necrosis of the tip and leaf margins, which spreads towards the midrib (central vein). Leaves become scorched and may drop early.

8. **Chlorine:** A deficiency of Cl leads to chlorosis in younger leaves and overall wilting as a consequence of the possible effect on transpiration. Cl-toxicity symptoms are: burning of the leaf tips or margins; bronzing; premature yellowing; leaf fall; and poor burning quality of tobacco.
9. **Copper:** Cu-deficiency symptoms are first visible in the form of narrow, twisted leaves and pale white shoot tips. At maturity, panicles/ears are poorly filled and even empty where the deficiency is severe. In fruit trees, dieback of the terminal growth can occur. Cu-toxicity symptoms are more variable with species and less established than its deficiency symptoms.
10. **Iron:** Fe deficiency begins to appear on younger leaves first. Otherwise, its deficiency symptoms are somewhat similar to those of Mn, as both Fe and Mn lead to failure in chlorophyll production. Yellowing of the interveinal areas of leaves (commonly referred to as iron chlorosis) occurs. Fe toxicity of rice is known as bronzing. In this disorder, the leaves are first covered by tiny brown spots that develop into a uniform brown colour.
11. **Manganese:** Mn-deficiency symptoms resemble those of Fe and Mg deficiency where interveinal chlorosis occurs in the leaves. However, Mn-deficiency symptoms are first visible on the younger leaves whereas, in Mg deficiency, the older leaves are affected first. Mn-toxicity symptoms lead to the development of brown spots, mainly on older leaves and uneven green colour.
12. **Molybdenum:** Mo deficiency in legumes can resemble N deficiency because of its role in N fixation. Mo deficiency can cause marginal scorching and rolling or cupping of leaves and yellowing and stunting in plants. Yellow spot disease in citrus and whiptail in cauliflower is commonly associated with Mo deficiency. Fodders containing more than 5 µg/g Mo in the dry matter are suspected to contain toxic levels of Mo for grazing animals (associated with the disease molybdenosis).
13. **Zinc:** The mobility of Zn is low. The rate of Zn mobility to younger tissue is particularly depressed in Zn-deficient plants. Common symptoms of Zn deficiency are stunted plant growth; poor tillering (cereals); development of light green, yellowish, bleached spots; chlorotic bands on either side of the midrib in monocots (particularly maize); brown rusty spots on leaves in some crops, which in acute Zn deficiency as in rice may cover the lower leaves. Zn toxicity can result in a reduction in root growth and leaf expansion followed by chlorosis.
14. **Nickel:** Ni is a part of the enzyme urease, which breaks down urea in the

soil. It also plays a role in imparting disease resistance and is considered essential for seed development. Information on various aspects of Ni as a micronutrient is gradually becoming available.

Beneficial elements:

1. **Silicon:** In lowland or wetland rice that is low in Si, vegetative growth and grain production are reduced severely and deficiency symptoms such as necrosis of the mature leaves and wilting can occur. Similarly, sugarcane suffers growth reduction under conditions of low Si availability.
2. **Cobalt:** It is essential for N-fixing microorganisms, irrespective of whether they are free-living or symbiotic. Co deficiency inhibits the formation of leghaemoglobin and, hence, N₂ fixation.

4.4.4 Nutrient Mobility and its Effect on Deficiency Symptoms

Nutrients must come in contact with the root surface. All the mineral nutrients, in general, are absorbed in ionic forms (Table 4.1).

Table 4.1: Forms of essential elements absorbed by plants

Cation(3 ⁺)	Cation(2 ⁺)	Cation(1 ⁺)	Neutral(0)	Anion(1 ⁻)	Anion(2 ⁻)
<i>Macronutrients:</i>					
	Ca ²⁺ Mg ²⁺	K ⁺ NH ₄ ⁺		NO ₃ ⁻ H ₂ PO ₄ ⁻	HPO ₄ ²⁻ SO ₄ ²⁻
<i>Micronutrients:</i>					
Fe ³⁺	Fe ²⁺ , Mn ²⁺ , Zn ²⁺ , Cu ²⁺ Ni ²⁺		B(OH) ₃	Cl ⁻	MoO ₄ ²⁻

While nutrients are transported easily from roots to shoots, their redistribution from the original place of deposition is more difficult for the so-called immobile nutrients. In the event of nutrient deficiency, a partial re-activation is required to supply newly formed leaves from the reserves of older ones. The relative mobility of a nutrient within the plant helps us understand the reasons for the differential appearance of nutrient deficiency symptoms. For example:

Deficiencies indicated by symptoms appearing first on older leaves:

- chlorosis starting from leaf tips later leaves turn yellowish-brown: N;
- reddish discolouration on green leaves or stalks: P;
- leaves with brown necrotic margins, the wilted appearance of plant: K;
- stripe chlorosis, mainly between veins, while veins remain green: Mg;
- spot necrosis: greyish-brown stripe-form spots in oats (grey-speck disease): Mn;
- dark-brown spots in oats and barley, whitish spots in rye and wheat: Mn.

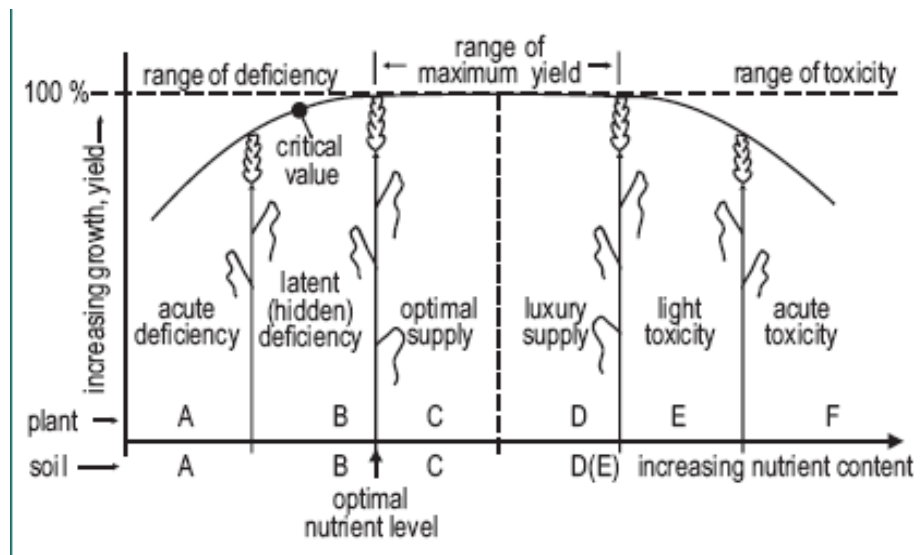
Deficiencies indicated by symptoms appearing first on younger leaves:

- completely yellowish-green leaves with yellowish veins: S;
- yellow or pale yellow to white leaves with green veins: Fe;
- the youngest leaf with white, withered, and twisted tips (oats and barley). Cu;
- yellowish leaves with brownish spots (part of acidity syndrome): Ca.

4.4.5 Range of Nutrient Supply from Deficiency to Toxicity

The nutrient status of a plant can range from acute deficiency to acute toxicity. A broad division of nutrient status into three groups namely deficient, optimal, and excess may be useful for general purposes. For a more accurate assessment of the nutritional status of plants, a detailed categorization is required in which six different ranges can be distinguished (see Figure 4.3):

- **Acute deficiency:** It is associated with definitely visible symptoms and poor growth. The addition of deficient nutrients usually results in increased growth and yields.
- **Marginal or latent deficiency (hidden hunger):** It is a small range with or without visible deficiency symptoms. However, growth and yield are reduced. An optimal nutrient supply prevents hidden hunger.
- **Optimal supply:** Here all nutrients are at the most desired level. In this range, healthy green plants, good growth, and high yields with good quality can be expected. The optimal supply is reached above the critical concentration, which is generally associated with 90 per cent of the maximum yield.
- **Luxury supply:** Although there is no definite borderline between optimal and luxury supply, it is useful to identify this range of unnecessarily high nutrient supply. Even if there may not be any negative effects on plant growth or yield, nutrient input is wasted and product quality, as well as disease resistance, may be reduced especially in the case of excess N.
- **Marginal or light (hidden) toxicity:** Here the nutrient concentration is moving towards toxicity. Above the critical toxic concentration, crop growth and yield start to decrease because of the harmful effects of a nutrient surplus, or of toxic substances on biochemical processes and imbalances.
- **Acute toxicity:** This is the other extreme of excessive supply or poor nutrient management. Plants are damaged by toxic levels resulting in toxicity symptoms, poor or no growth, poor yield, low quality, and damage to soil and plant health.



Source: Finck, 1992.

Figure 4.3: Plant growth and yield dependence on nutrient supply

4.5 EVALUATION/ASSESSMENT OF SOIL FERTILITY

The evaluation or assessment of soil fertility is perhaps the most basic decision-making tool for balanced and efficient nutrient management. It consists of estimating the available nutrient status of soil for crop production. A correct assessment of the available nutrient status before planting a crop helps in taking appropriate measures to ensure adequate nutrient supply for a good crop over and above the amounts that the soil can furnish. The techniques used include soil testing and plant analysis, the latter including related tools such as total analysis of the selected plant part, tissue testing, crop logging, and the diagnosis and recommendation integrated system (DRIS). The important ones are described below.

4.5.1 Visual Plant Diagnosis

A healthy dark-green colour of the leaf is a common indicator of good nutrient supply and plant health. The degree of “greenness” can be specified in exact terms for each crop using Munsell’s Plant Colour Chart or other such charts. Any change to light green or a yellowish colour generally suggests a nutrient deficiency where other factors are not responsible such as cold weather, plant diseases, damage caused by sprayings, air pollution, etc. A deviation from the normal green colour is easily detected. However, it is by no means always caused by N deficiency, as usually assumed. For example, it is a common but questionable practice to always relate light-green late foliar discolouring with N deficiency, but it can often be caused by other deficiencies. Even where the colour is more or less “satisfactory”, there may be a latent deficiency (“hidden hunger”) that is often difficult to establish from visual observation but can still cause yield reduction. Deficiency symptoms are only reliable where a single nutrient and no other factors are limiting.

4.5.2 Plant Analysis

The nutrient concentration of growing plants provides reliable information on their nutritional status in most cases, except in the case of Fe. It reflects the current state of nutrient supply and permits conclusions as to whether a supplementary nutrient application is required. Interpretation of plant analysis data is usually based on the total concentrations of nutrients in the dry matter of leaves or other suitable plant parts, which are compared with standard values of “critical nutrient concentrations” (“critical values”) and grouped into supply classes. The critical level is the level of concentration of a nutrient in the plant that is likely to result in 90 per cent of the maximum yields (Table 4.2). Standard tables of plant analysis interpretation are based on requirement levels for very high yields (maximum yield) or those for the more practical and realistic medium to high yields corresponding to critical values of 90 per cent.

Table 4.2: Critical nutrient concentrations for 90 per cent yield for interpretation of plant analysis data

Element	Wheat & rice	Oilseed rape	Sugarcane	Alfalfa (Lucerne)	Grass*	Citrus
(%)						
N	3.00	3.50	1.50	3.50	3.00	2.50
P	0.25	0.30	0.20	0.25	0.40	0.15
K	2.50	2.50	1.50	2.00	2.50	1.00
Mg	0.15	0.20	0.12	0.25	0.20	0.20
S	0.15	0.50	0.15	0.30	0.20	0.15
(µg/g)						
Mn	30.0	30.0	20.0	30.00	60.00	25.00
Zn	20.0	20.0	15.0	15.00	50.00	20.00
Cu	5.00	5.00	3.00	5.00	8.00	5.00
B	6.00	25.00	1.50	25.00	6.00	25.00
Mo	0.30	0.30	0.10	0.20	0.30	0.20

Note: Growth stage and plant parts used: wheat/rice: 1 - 2 nodes, whole shoots; oilseed rape: pre-flowering, youngest mature leaf; sugarcane: 5-7 months, third leaf blade from top; alfalfa (Lucerne): pre-flowering, whole shoots; grass: pre-flowering, whole shoots; citrus: 5-7 months old leaves from mid of non-fruiting branch.*Not critical but optimal concentrations for cows producing 15 litres of milk per day. (FAO, 2006)

4.5.3 Field Experiment: Control Test Plot & Complex Experiments

This is a proven and effective method for assessing nutrient needs. It is also essential for the calibration of soil test and plant analysis results with crop response. Fertilizer experiments provide direct information on the amounts of nutrients required for a crop in a specific environment at a given yield level. These are used mainly by researchers to generate background data needed for making practical recommendations for farmers. Field experiments can be simple test plots or complex replicated trials.

4.5.4 Soil Testing

Soil testing can be defined as an acceptably accurate and rapid soil chemical analysis for assessing available nutrient status for making fertilizer recommendations. Soil testing as a diagnostic tool is useful only when the interpretation of test results is based on correlation with crop response and economic considerations to arrive at practically usable fertilizer recommendations for a given soil–crop situation.

After soil analysis, the concentrations of available nutrients measured must be interpreted into ranges of nutrient supply and then into the nutrient amounts required to reach a certain yield level. In general, the lower the soil fertility status (soil test value) is, the greater the need for external nutrient application. For macronutrients, the data are generally classified into categories of supply, e.g.: very low, low, medium, high, and very high. Table 4.3 provides information on the general soil test limits used for classifying soils into different fertility classes.

Table 4.3: General soil test limits used for classifying soils into different fertility classes

Nutrient	Method/Extractant	General fertility class*		
		Low	Medium	High
N (% Organic carbon)	Organic carbon	<0.5	0.5-0.75	>0.75
N (kg/ha)	Alkaline permanganate	<280	280-560	>560
P ₂ O ₅ (kg/ha)	Sodium bicarbonate	<23	23-56	>56
K ₂ O (kg/ha)	Ammonium acetate	<130	130-335	>335
S (kg/ha)	Heat soluble, CaCl ₂	<20	20-40	>40
Zn (µg/g)	DTPA	<0.6	0.6-1.2	>1.2
Mn (µg/g)	DTPA	<3.0		
Cu (µg/g)	DTPA/ Ammonium acetate	<0.2		
Fe (µg/g)	DTPA	<2.5-4.5		

*Very general limits based on several published Indian sources (FAO, 2006).

Check Your Progress Exercise 2

Note: a) Use the spaces given below for your answers.

b) Check your answer with those given at the end of the unit.

1. Give the names of macro and micronutrients.

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2. What are the roles of iron in plant growth?

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3. What do you mean by beneficial nutrients?

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4. What do you mean by soil testing?

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4.6 MAINTENANCE OF SOIL FERTILITY

Soil fertility management is of utmost importance for optimizing crop nutrition to achieve sustainable crop production. It is the foundation on which all input-based high-production systems can be built. Soil fertility can be managed effectively with the adoption of balanced plant nutrition or fertilization and integrated nutrient management practices.

4.6.1 Balanced Plant Nutrition/Fertilization

A crop requires an adequate supply of all essential mineral nutrients for optimum growth. If more than one is in short supply, growth is determined by the nutrient, which is in the lowest supply. If several nutrients are in short supply, we need to apply all of them to get optimum production. It is well documented that unbalanced availability of nutrients not only produces low and poor quality yield but can also lead to mining of soil nutrient reserves which are in short supply and to losses of nutrients supplied in excess. The long-term fertilizer experiments (LTFEs) conducted in India have shown that soil productivity can be maintained by the balanced nutrition of crops. The grain yields of some LTFEs in India clearly show that response to N declined sharply in the absence of P and K fertilizers. Crop yields increased considerably with an application of P along with N and improved further by the inclusion of K.

Unbalanced plant nutrition has also resulted in micronutrient deficiencies in many parts of the country. Zinc deficiency is the most commonly occurring

micronutrient deficiency problem limiting crop growth and production. In India, the application of Zn has resulted in spectacular yield increases in rice-wheat growing areas. Balanced plant nutrition increases crop yields, nutrients, water use efficiencies, and of course, farmers' profit. Thus, crops should be well supplied with all the 14 essential mineral nutrients in balanced amounts to get the maximum benefit from crop production.

4.6.2 Integrated Plant Nutrient Management (IPNM)

Integrated plant nutrient management (IPNM) maintains and possibly enhances soil fertility through a balanced use of chemical fertilizers combined with organic and biological sources to improve the efficiency of plant nutrients, increase crop productivity and minimize the losses of nutrients to the environment. The possible conventional and non-conventional sources of plant nutrients for integrated supply in the Indian agriculture context are discussed below:

1. Organic manures/ sources: The use of organic manures not only provides plant nutrients but also improves soil's physical, chemical, and biological properties. The important organic manures in the Indian context are farmyard manure (FYM), different kinds of composts, vermicompost, green manures, etc. Long-term fertilizer experiments in India have demonstrated that, in addition to fertilizers, the use of organic manure is essential for sustaining crop productivity.

i) Farmyard manure: Farmyard manure refers to the decomposed mixture of dung and urine of farm animals along with the litter (bedding material) and left-over material from roughages or fodder fed to the cattle. On average, well-rotten FYM contains 0.5 per cent N, 0.2 per cent P_2O_5 , and 0.5 per cent K_2O under Indian conditions (Butterworth *et al.*, 2003). Nearly 50 per cent of the cattle dung produced in India today is utilized as cooking fuel and is thus a loss to agriculture.

FYM is an excellent soil amendment, which adds to the balance of nutrients while contributing valuable organic material to the soil. The dung and urine of livestock and their bedding materials are the major constituents of FYM. The urine of many animals is generally richer in nitrogen and potassium as compared to their dung. The urine of most animals is deficient in phosphorus. Bedding materials also add a good amount of nutrients, mainly NPK, to the FYM. As such, compared to the potential supply of major nutrients from FYM, only 8 per cent of nitrogen, 26 per cent of phosphorus, and 16 per cent of potash are added every year to Indian soils (Yawalkar *et al.*, 1996).

ii) Compost: Composting is a microbiological process, which converts organic residues into non-odoriferous blackish-brown stabilized humic substances with a narrow C/N ratio. These are prepared through microbial action on wastes such as crop residues, weeds bagasse, sawdust, kitchen, and human habitation wastes. These materials under optimum temperature, moisture, and aeration undergo intensive biodegradation into an amorphous brownish-black colloidal substance. This stabilized humified material is a valuable source of plant nutrients.

Depending on the nature of the substrate undergoing decomposition, composts are broadly classified into two groups namely the rural and urban composts. In India, the two recommended methods for compost preparation are the “Indore method” (aerobic) and the “Bangalore method” (Initially aerobic but later anaerobic). In the Indore method, a pit is dug near the cattle shed on a site free from waterlogging. After use, the wet beds consisting of shredded crop residues along with animal excreta are removed and spread in a layer in the pit. Then cattle dung is spread over this layer which is sprinkled with water to maintain optimum moisture. The process is repeated until layers of residues reach 30 cm above ground level. Residues are turned every fortnight and good quality compost is ready in 16 weeks. In the Bangalore method, greater conservation of nutrients is attempted.

- iii) **Vermicompost:** Vermicompost is organic manure produced by earthworms. It is a mixture of worm castings (faecal excretions), organic material including hummus, live earthworms and their cocoons, and other organisms. Vermicomposting is a cost-effective and efficient technique for recycling agricultural residues, industrial wastes, and animal wastes from cattle, piggery, and poultry. Earthworms consume practically all kinds of organic matter. One kg of worms can consume 1 kg of residue every day. Earthworms feed on any organic waste, and after using 5-10% of the feedstock for their growth, excrete the rest of the ingested matter coated with intestinal and cutaneous mucus with a higher C: N ratio than that of the original matter. The nitrogen of the organic matter is either turned into earthworm biomass or excreted as ammonium. The average composition of vermicompost and other composts is given in Table 4.4.

Table 4.4: Average nutrient content of vermicompost and other composts

Compost	Nutrient content (% of dry matter)		
	N	P ₂ O ₅	K ₂ O
Vermicompost	1.60	2.20	0.67
Rural compost	1.22	1.08	1.47
Urban compost	1.24	1.92	1.47
Paddy straw compost	0.90	2.05	0.90
Maize stalk compost	1.09	1.30	1.01
Cotton wastes compost	1.62	1.04	1.26
Water hyacinth	2.00	1.00	2.30
Poultry manure	2.87	2.90	2.35
Castor	5.80	1.80	1.00

(Marwaha and Jat, 2004)

- iv) **Green manures (GM):** Green manuring with N fixing leguminous crops improves soil fertility and enhances the availability of other nutrients. The sandwiching of green manuring crops in different cropping systems is considered not only an excellent source of nitrogen, but the intervention improves the physical condition of the soil and the

availability of other nutrients. Major green manure crops are: dhaincha (*Sesbania aculeata*, *Sesbania rostrata*), sunhemp/sanai (*Crotalaria juncea*), lobia (*Vigna unguiculata*) and guar (*Cyamopsis tetragonoloba*), etc. The main problem in the adoption and spread of green manure crops is that they compete with cash crops for space, time, water, and other scarce inputs. The nitrogen contribution from some of the green manure crops is given in Table 4.5.

Table 4.5: Potential N contributions of N-fixing legumes in Indian soils.

Name	Botanical name	Sowing season	Average yield of green matter (t/ha)	N(%) on green weight basis)	N added (kg/ha)
Sunn hemp	<i>Crotalaria juncea</i>	Kharif	15.2	0.43	84.0
Dhaincha	<i>Sesbania aculeata</i>	Kharif	14.4	0.42	77.1
Mungbean	<i>Vigna radiata</i>	Kharif	5.7	0.53	38.6
Cowpea	<i>Vigna unguiculata</i>	Kharif	10.8	0.49	56.3
Guar	<i>Cyamopsis tetragonoloba</i>	Kharif	14.4	0.34	62.3
Senji	<i>Melilotus alba</i>	Rabi	20.6	0.51	134.4
Khesari	<i>Lathyrus sativus</i>	Rabi	8.8	0.54	61.4
Berseem	<i>Trifolium alexandrinum</i>	Rabi	11.1	0.43	60.7

(Yawalkar *et al.*, 1996)

Crop residues

A huge quantity of crop residues such as wheat straw, cotton sticks, sugarcane trash/tops, and rice straw/ husk, *etc.* are available. But due to some economic compulsions such as the need for animal fodder and fuel, the crop residues are partially recycled in the soil or burnt to clear fields for sowing of the next crop otherwise these may contribute to improving organic matter in the soil and thus keep it productive. Research data have shown that the incorporation of rice and/or wheat straw may or may not increase crop yield, but it does improve soil fertility and quality. There are reports that there is some initial immobilization of nutrients with the incorporation of rice/wheat residue, but it can be easily overcome by adding 20-25 kg/ha additional fertilizer N at the beginning or by mixing rice/wheat residue with a legume residue. Thus in the long run crop residue helps in giving sustained production and improves soil fertility and productivity.

2. **Chemical fertilizers:** The term fertilizer is derived from the Latin word *fertilis*, which means fruit-bearing. A fertilizer can be defined as a mined, refined, or manufactured product containing one or more essential plant nutrients in available or potentially available forms. Many prefixes such as synthetic, mineral, inorganic, artificial, or chemical are often used to describe fertilizers and these are used interchangeably.

Different kinds of chemical fertilizers are available to supply nutrients to the rice crop.

Nitrogenous fertilizers

World fertilizer N production is based on the synthetic fixation of atmospheric N in the form of NH_3 . The NH_3 produced is further used for the production of inorganic fertilizers, containing either NH_4^+ , NO_3^- , a combination of both, or the amide form ($-\text{NH}_2$). In addition to these single (straight) N fertilizers, multi-nutrient (compound) fertilizers containing N together with other primary nutrients, such as phosphorous (P) and/or potassium (K) are widely used. A list of nitrogenous fertilizers along with their nutrient content is given in Table 4.6.

Table 4.6: Common nitrogenous fertilizers and their nutrient content

N fertilizer material	Percentage					
	N	P_2O_5	K_2O	Ca	Mg	S
Ammonium nitrate NH_4NO_3	34	0	0	0	0	0
Ammonium nitrate sulphate $\text{NH}_4\text{NO}_3 + (\text{NH}_4)_2\text{SO}_4$	30	0	0	0	0	5
Ammonium sulphate $(\text{NH}_4)_2\text{SO}_4$	21	0	0	0	0	24
Calcium nitrate $\text{Ca}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O}$	15	0	0	21	0	0
Crotonylidene diurea (CDU)	32	0	0	0	0	0
Isobutylidene diurea (IBDU)	31	0	0	0	0	0
Urea $\text{CO}(\text{NH}_2)_2$	46	0	0	0	0	0
Urea (sulphur coated) $\text{CO}(\text{NH}_2)_2 + \text{S}$	36- 38	0	0	0	0	13- 16
Ureaform (urea + formaldehyde)	38	0	0	0	0	0

Nitrogen use efficiency (NUE) of nitrogenous fertilizers, especially in rice under lowland conditions, is low, i.e. 30-50%. Nitrification inhibitors can be used to reduce N losses. Nitrification inhibitors retard the nitrification process in soil and thereby reduce the N losses by leaching and denitrification. Many chemicals have been reported to have nitrification inhibiting properties. The chemicals most widely tested are N-Serve (2-chloro 6(trichloromethyl) pyridine), AM (2-amino-4-chloro-6 methyl pyrimidine), DCD (dicyandiamide), and ST (sulphathiazole). Increased rice yields and increased NUE in India have been reported for N-Serve, AM, ST, DCD, and neem oil-coated urea (500-1000 ppm).

Phosphatic fertilizers

These fertilizers contain phosphorus in phosphate form. In trade, the P content is given as phosphoric anhydride or in terms of P_2O_5 . P-fertilizers can be divided into 3 groups:

Water-soluble P fertilizers: A small sample of fertilizer is extracted with distilled water for a prescribed period (AOAC, 1960) and the slurry is then filtered. The amount of P in the filtrate is determined and expressed as a % by the weight of the fertilizer. The above fraction is called water-soluble P. Examples- monocalcium phosphate [$Ca(H_2PO_4)_2$ -single superphosphate], mono ammonium phosphate [$NH_4H_2PO_4$], diammonium phosphate [$(NH_4)_2HPO_4$] and potassium phosphate [$K H_2PO_4$], etc.

Citrate-soluble P fertilizers: The fertilizer residue left after extracting water-soluble P with a neutral (pH 7.0) solution of 1N ammonium citrate for a prescribed time (AOAC, 1960), and slurry is then filtered. The P content in the filtrate is then determined and expressed as % by the weight of the fertilizer. The above fraction is called citrate-soluble P. Examples- dicalcium phosphate [$CaHPO_4$], basic-slag (a by-product of the steel industry, useful for acidic soils) and calcium metaphosphate [$Ca(PO_3)_2$], etc.

Citrate-insoluble P fertilizers: The P content in the residue left after extracting citrate-soluble P. It is unavailable to plants. Examples include phosphate, bone meal, etc. The sum of the water-soluble and citrate-soluble P represents an estimate of the fraction available to plants and is termed as available P. The sum of water-soluble, citrate-soluble, and citrate-insoluble P represents the total P in fertilizer. Citrate-insoluble P is unavailable to plants. A list of phosphatic fertilizers along with their nutrient content is given in Table 4.7.

Table 4.7: Sources of phosphatic fertilizers

Name	Formula	Content	Comments
Single superphosphate	$Ca(H_2PO_4)_2 \cdot H_2O + CaSO_4 \cdot 2H_2O$	7–9% P, 13–20% Ca, 12% S	Soluble, neutral (16–21% P_2O_5)
Triple superphosphate	$Ca(H_2PO_4)_2 \cdot H_2O$	18–22% P, 9–14% Ca, 1.4% S	Soluble, neutral (41–50% P_2O_5)
Monoammonium phosphate (MAP)	$NH_4H_2PO_4$	22% P, 11% N	Soluble, acidifying (51% P_2O_5)
Diammonium phosphate (DAP)	$(NH_4)_2HPO_4$	20–23% P, 18–21% N	Soluble, acidifying (46–53% P_2O_5)
Partly acidulated rock phosphate	$Ca_3(PO_4)_2$	10–11% P	>1/3 water-soluble (23–26% P_2O_5)

Potassic fertilizers

There are mainly two potassium fertilizers namely, muriate of potash (MOP) (KCl) containing 60% K₂O and sulphate of potash (SOP) (K₂SO₄) containing 48% K₂O. Due to high costs, mostly MOP is used in India. For plants that are tolerant to chloride and whose quality is not impaired by high Cl, the cheaper potassium chloride (MOP) is preferred. For plants that are sensitive to high Cl for quality or other reasons, potassium sulphate or potassium nitrate is a better choice. Of the crops, potatoes and tobacco and many horticultural crops belong to the chloride-sensitive group. However, the Cl component is suitable for “salt-liking” plants, such as sugarbeets and palms, and it brings extra beneficial effects. A list of potassic fertilizers along with their nutrient content is given in Table 4.8.

Table 4.8: Sources of potassic (K) fertilizers.

Name	Formula	Content	Comments
Potassium chloride (Muriate of potash)	KCl	50% K	Muriate of potash (60% K ₂ O)
Potassium nitrate	KNO ₃	37% K, 13% N	In compounds (44% K ₂ O)
Potassium sulfate	K ₂ SO ₄	40–43% K, 18% S	In compounds (50% K ₂ O)
Compound fertilizers	N + P + K	Variable	Common in rice

Secondary nutrients fertilizers

S, Mg, and Ca-containing fertilizers fall in the category. Substitution of high-analysis fertilizers such as urea and DAP for ammonium sulphate and single superphosphate has been the main cause of widespread S deficiency in Indian soils.

There are 2 kinds of sources of S. The first kind is the popular fertilizers containing S as the nutrient other than that for which fertilizer is applied. For example, single superphosphate (SSP) is marketed as phosphate fertilizer but contains 12% S. The other kind of sources of S are those materials that are marketed and used only as a source of S. These are referred to as S-fertilizers and include elemental S, gypsum, iron pyrites and phosphogypsum (a byproduct of phosphoric acid industry). Most S is applied to the soil at sowing. A list of sulphur fertilizers along with their nutrient content is given in Table 4.9.

Table 4.9: Sulphur containing fertilizers

Name	Formula	Content	Comments
Ammonium sulphate	(NH ₄) ₂ SO ₄	24% S	Quick-acting
Single superphosphate	Ca(H ₂ PO ₄) ₂ ·H ₂ O + CaSO ₄ ·2H ₂ O	12% S, 7–9 % P, 13–20% Ca	Soluble, quick-acting

Potassium sulphate	K_2SO_4	18% S	Quick-acting
Magnesium sulphate (Epsom salt)	$MgSO_4 \cdot 7 H_2O$	13% S, 10% Mg	Very quick-acting
Kieserite	$MgSO_4 \cdot H_2O$	23% S, 17% Mg	Quick-acting
Langbeinite	$K_2SO_4 \cdot MgSO_4$	18% K, 11% Mg, 22% S	Quick-acting
Gypsum	$CaSO_4 \times 2 H_2O$	17% S	Slow-acting
Elemental S	S	97% S	Slow-acting
S-coated urea	$CO(NH_2)_2 + S$	6–30% S, 30–40% N	Slow-acting

Magnesium-containing fertilizers are either quick-acting soluble salts (mainly sulphates) or in slow-acting form (magnesium carbonate) (Table 4.10).

Table 4.10: Magnesium containing fertilizers

Name	Formula	Content	Comments
Kieserite	$MgSO_4 \cdot H_2O$	17% Mg, 23% S	Soluble, quick-acting
Langbeinite	$K_2SO_4 \cdot MgSO_4$	18% K, 11% Mg, 22% S	Quick-acting
Magnesium chloride	$MgCl_2$	9% Mg	Soluble, quick-acting
Magnesium oxide	MgO	42% Mg	Slow-acting, for foliar application
Magnesite	$MgCO_3$	25-28% Mg	Slow-acting
Dolomite	$MgCO_3 \cdot CaCO_3$	13% Mg, 21% Ca	Slow-acting, content of Ca and Mg varying

Calcium is only applied other than for liming in cases of definite deficiency. Quick-acting sources are calcium chloride, solid or in solution; and the slow-acting source is gypsum. Soil application, though simple, is often disappointing because of restricted translocation within the plant; in such cases a foliar spray is preferable. A list of calcium fertilizers along with their nutrient content is given in Table 4.11.

Table 4.11: Calcium containing fertilizer sources

Name	Formula	Content	Comments
Calcium chloride	$CaCl_2 \cdot 6H_2O$	18% Ca	Soluble, quick-acting, does not raise pH
Gypsum	$CaSO_4 \cdot 2H_2O$	23% Ca, 18% S	Slightly soluble, slow-acting, for saline and alkaline soils
Dolomite	$MgCO_3 \cdot CaCO_3$	13% Mg, 21% Ca	Slow acting, the content of Ca and Mg varying
Lime	$CaCO_3$	40% Ca	Slow-acting, for acid soils

Micronutrients fertilizers

Apart from the widely recognized areas of natural deficiencies of micronutrients, attention should be given to the possibility that they may become critical "minimum factors" as other critical factors are amended and yield levels rise. They may be applied either as single-nutrient fertilizers or as supplements in macronutrient fertilizers. The quickest and most common method of correcting deficiencies is by foliar application.

Deficiencies of 3 micronutrients have been reported in rice-growing areas of India. First of all, zinc deficiency in rice fields was noticed in *Tarai* soils. Later, a deficiency of Fe was noticed on the sandy soils of Punjab. Of late, boron (B) deficiency in rice has been reported on highly calcareous soils of Bihar. Zinc sulphate and zinc oxide are the important zinc fertilizers available in India. Application of 25 kg zinc sulphate (heptahydrate)/ hectare is generally recommended in India (for rice). Dipping rice roots in 2-4% ZnO suspension has also been found to be very effective. If zinc is not applied in the soil at transplanting/ sowing and zinc deficiency symptoms appear, a foliar spray of 0.5-1.0% zinc sulphate with 0.25% lime is recommended.

Fe chlorosis in rice is better controlled by a foliar spray of 0.5% Fe-sulphate. Soil application of B should be made when a deficiency of B has been established. Indiscriminate use may result in the toxicity of plants. When the foliar application is a must, a 0.2% borax solution (with lime twice the weight of borax) is recommended to avoid injury to foliage). A list of micronutrient fertilizers along with their nutrient content is given in Table 4.12.

Table 4.12: Micronutrient containing fertilizers included in the Fertilizer Control Order (FCO) in India.

Nutrient	Fertilizer	Chemical formula	Minimum nutrient content
Zinc	Zinc sulphate heptahydrate	$\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$	21
	Zinc sulphate monohydrate	$\text{ZnSO}_4 \cdot \text{H}_2\text{O}$	33
	Chelated zinc	Zn-EDTA	12
Iron	Ferrous sulphate	$\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$	19
	Chelated iron	Fe-EDTA	12
Manganese	Manganese sulphate	$\text{MnSO}_4 \cdot \text{H}_2\text{O}$	30.5
Copper	Copper sulphate	$\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$	24
Boron	Borax (Sodium tetraborate)	$\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$	10.5
	Boric Acid	H_3BO_3	17
Molybdenum	Ammonium molybdate	$(\text{NH}_4)_6\text{Mo}_7\text{O}_{24} \cdot 4\text{H}_2\text{O}$	52

- Biofertilizers:** Biofertilizers (BF) (microbial inoculants) are the products containing living cells of different types of microorganisms (bacteria,

fungi, actinomycetes, etc.) which can fix atmospheric nitrogen and mobilize and solubilize phosphorus in the soil from unavailable form to plant usable form. They have great potential as a supplementary, renewable and environmentally friendly source of plant nutrients and are an important component of the Integrated Plant Nutrient System (IPNS). In recent times the phenomenal increase in production and use of biofertilizers in India is the result of the special attention given by the government and interest by entrepreneurs to set up biofertilizer production facilities. Farmers have also realized the benefits of the use of biofertilizers.

The use of *Rhizobium* culture in legumes is most promising among different kinds of biofertilizers. Biofertilizers have to be applied properly to the seed or soil to produce the desired benefit. However, the performance of biofertilizers is highly unpredictable due to their biological nature and susceptibility to biotic and abiotic stresses. Hence, there is a need to develop more effective, competitive, and stress-tolerant strains to increase nutrient supply from biofertilizers. The use of phosphate solubilizing bacteria (PSB) and Arbuscular Mycorrhizae (AM) to supply phosphorus and micronutrients has tremendous potential in India. Examples of some biofertilizers currently in use worldwide are shown in Table 4.13.

Table 4.13: Organisms, mode of action, crops, and producers of biofertilizers currently in use for agriculture

Type	Mode of action	Crop
<i>Rhizobium</i> spp.	N ₂ fixation	Legumes
Cyanobacteria	N ₂ fixation	Rice
<i>Azospirillum</i> spp.	N ₂ fixation	Cereals
<i>Mycorrhizae</i>	Nutrient acquisition	Conifers
<i>Penicillium bilaii</i>	P solubilization	Cereals, legumes

Check Your Progress Exercise 3

Note: a) Use the spaces given below for your answers.

b) Check your answer with those given at the end of the unit.

- Write down the three aims of balanced fertilizer use.

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- What are the main objectives of integrated plant nutrient management?

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3. Name two methods recommended for making compost in India.

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4. Give at least three examples of water-soluble P fertilizers.

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4.7 LET US SUM UP

Soil fertility is the capacity of soil to supply essential minerals to the plants in appropriate and adequate amounts. Adequate soil fertility is the foundation on which all input-based high-production systems can be built. The decline in soil fertility is a major cause of stagnating or even decreasing crop yields. The important causes for reduced soil fertility are soil erosion, loss of organic matter resulting in reduced biological activity, nutrient depletion as a result of soil erosion and mining of nutrients, etc. Under poor management, soil fertility can be seriously depleted and soils may become useless for agriculture.

A correct assessment of the available nutrient status before planting a crop is necessary for taking appropriate measures to ensure adequate nutrient supply for a good crop over and above the amounts that the soil can furnish. Balanced fertilization and integrated plant nutrition system (IPNS) or integrated nutrient management (INM) enables the adaptation of plant nutrition and soil fertility management in farming systems to site characteristics, taking advantage of the combined and harmonious use of organic, mineral, and biofertilizer nutrient resources to serve the concurrent needs of food production and economic, environmental and social viability.

4.8 KEYWORDS

Available nutrient: Form(s) of a plant nutrient in the soil or fertilizer that is immediately or potentially available for being taken up by growing plants during their growth span.

Complete fertilizer: A chemical compound or a blend of compounds containing significant quantities of nitrogen, phosphorous, and potassium. It may contain other plant nutrients.

Compound fertilizer: A fertilizer formulated with two or more plant nutrients.

Fertilizer grade: The guaranteed minimum analysis in per cent of the major

plant nutrient elements contained in a fertilizer material or a mixed fertilizer. The analysis is usually designated as N-P₂O₅ -K₂O, but it may be N-P-K where permitted or required as specified by state law.

Mixed fertilizer: Two or more fertilizer materials blended or granulated together into individual mixes. The term includes dry mix powders as well as granulated, clear liquid, suspension, and slurry mixtures.

Top-dressed fertilizer: The surface application of fertilizer to the soil after the crop has been established.

Soil fertility: The ability of soil to supply the nutrients essential to plant growth.

Soil productivity: The capacity of the soil, in its normal environment, to produce a specified plant or sequence of plants under a specified system of management.

Soil reaction (pH): The degree of acidity or alkalinity of a soil, usually expressed as a pH value.

4.9 SUGGESTED READINGS/ REFERENCES

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4.10 ANSWERS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress Exercise 1

1. Soil fertility is the overall ability of a soil to support vigorous crop growth by ensuring adequate plant nutrients and suitable conditions for water uptake, and by providing favourable conditions for root growth and development.
2. The three groups of soil fertility are biological, physical, and chemical soil fertility.
3. Soil productivity is the crop yielding ability of soil from a unit area.
4. Three important causes for a decline in soil fertility are loss of topsoil by erosion, nutrient mining, and physical degradation of soil.

Check Your Progress Exercise 2

1. Macronutrients: N, P, K, Ca, Mg, S; micronutrients: Cu, Fe, Zn, Mn, B, Cl, Ni, Mo.
2. Iron (Fe) plays a role in the synthesis of chlorophyll, carbohydrate production, cell respiration, chemical reduction of nitrate and sulphate, and N assimilation.
3. Beneficial nutrients are not required by all plants but appear to benefit certain groups of plants or plants growing under specific conditions. Cobalt (Co), sodium (Na), vanadium (V), and silicon (Si) are sometimes designated as beneficial plant nutrients.
4. Soil testing can be defined as an acceptably accurate and rapid soil chemical analysis for assessing available nutrient status for making fertilizer recommendations.

Check Your Progress Exercise 3

1. Three aims of balanced fertilizer use are; increasing yield and quality of the produce, increasing the income of the farmer and correcting the inherent soil nutrient deficiencies.
2. The main objectives of integrated plant nutrient management (IPNM) are to maintain and possibly enhance soil fertility through a balanced use of chemical fertilizers combined with organic and biological sources.
3. In India, the two recommended methods are the “Indore method” (aerobic) and the “Bangalore method” (Initially aerobic but later anaerobic) for making composts.
4. Examples of water-soluble P fertilizers are single superphosphate, monoammonium phosphate, diammonium phosphate, potassium phosphate, etc.

UNIT 5 PEST AND DISEASE MANAGEMENT

Structure

- 5.0 Objectives
- 5.1 Introduction
- 5.2 Causes of Insect Pests and Diseases in Crops
- 5.3 Pest Epidemics
- 5.4 Pest Diagnostics
 - 5.4.1 Disease Diagnosis
 - 5.4.2 Insect Damage Diagnosis
- 5.5 Integrated Pest Management (IPM)
 - 5.5.1 Principles of IPM
 - 5.5.2 Approaches of IPM
- 5.6 Pesticide Residues and Consequences
 - 5.6.1 Adverse Effects of Pesticide Residues
 - 5.6.2 Safety Precautions against Residues
- 5.7 Let Us Sum Up
- 5.8 Keywords
- 5.9 Suggested Readings/ References
- 5.10 Answers to Check Your Progress Exercises
- 5.11 Terminal Questions

5.0 OBJECTIVES

After studying this unit, you will be able to:

- explain causes of pest incidence in crops;
- discuss pest epidemics and their causes;
- describe diagnostic symptoms of pests in crops;
- explain the principles and approaches of integrated pest management; and
- discuss pesticide residues and their consequences.

5.1 INTRODUCTION

By studying the principles and practices of crop production, you must have an idea about how we can enhance our crop productivity and production. However, crop yield potential is threatened by biotic constraints, such as insect pests and diseases. Until and unless these bottlenecks are not tackled effectively, it is very difficult to harvest a high crop yield. In this unit, you will learn about causes that increase pest intensity on crops. You will also be acquainted with the concept of pest diagnostics and the principles and practices of integrated pest management. Finally, the harmful effects of pesticide residues and safety measures against residues will be discussed.

5.2 CAUSES OF INSECT PESTS AND DISEASES IN CROPS

Since man started agriculture by clearing forests, he has been disturbing the balance of nature by reducing environmental resistance through ecologically unsound methods of cultivation. Even any slight decrease in any of the processes of 'environment resistance' results in a population explosion of an insect species and the consequent epidemics. Nature too sometimes plays a role in causing pest epidemics. Favourable conditions, which reduce the natural mortality and bring down the rapid development of the insect coupled with the conditions unfavourable to the natural enemies of that insect, often result in the rapid increase of its population leading to a sudden pest outbreak. Man's interference in ecosystems thus causes intensification of insects and diseases on crops and the main causes of this are discussed below.

- i) **Monoculture:** Due to ease in agronomic operations and other economic considerations, farmers resort to monoculture. Continuous cultivation of a crop species on extensive tracts year after year provides unhindered availability of food for pests and as a result, these attain higher populations than they would in a mixed crop environment.
- ii) **Staggered planting:** Certain pests attack crops during a particular crop growth stage, *e.g.* rice gundhi bug damages crops during the milky stage. Staggered planting of rice in an area ensures food availability to the pest for a prolonged duration thereby increasing the extent of crop losses.
- iii) **Ratooning:** Growing the same crop on a piece of land continuously for more than one season during a year or subsequently ensures uninterrupted food supply to pest species, thus accentuating pest problems, *e.g.* ratooning of certain crops, *e.g.* ratooning of sugarcane and brinjal aggravates pest problems. Also, growing two similar crops of a plant family in continuity that share the same pest fauna also increases pest losses, *e.g.* in the rice-wheat cropping system, losses due to pink borer and nematode have increased.
- iv) **Improved cultivars:** The majority of improved crop cultivars are bred for better yield and quality without addressing the pest resistance issue. These respond very well to fertilizers and irrigation showing excellent vegetative growth. These are preferred by pests and diseases that reproduce profusely on them inflicting heavy crop losses.
- v) **Monogenic resistant cultivars:** Monogenic resistant cultivars may lead to the development of pest biotypes. It has happened in the case of rice where several biotypes of gall midge and brown planthopper (BPH) have evolved in response to wide-scale cultivation of monogenic resistant cultivars against these pests.
- vi) **Excessive use of fertilizers and irrigation:** Instead of applying fertilizers in a balanced manner, excessive use of nitrogenous fertilizers stimulates pest reproduction. Likewise, excessive use of irrigation also aggravates pests in crops. Heavy doses of nitrogenous fertilizers accompanied by continuous flooding of rice fields create favourable

conditions for rice brown plant hopper (BPH).

- vii) Indiscriminate use of pesticides:** Indiscriminate use of broad-spectrum pesticides leads to the death of natural enemies of pests and their continuous use evolves pests that are resistant to pesticides. In such a situation, when neither pesticide is effective nor natural enemies are there, pests wreak havoc. Likewise, excessive use of pesticides also results in the outbreak of secondary pests.
- viii) Import & export of agricultural commodities:** Due to quarantine negligence, foreign pests sometimes enter new territories where their natural enemies are not present. Due to the reduced environmental resistance, they proliferate and realize very high densities.
- ix) Introduction of crops in new areas:** If a crop is introduced into a new area where crops sharing pest fauna with the introduced crop are grown, it would increase pest problems.
- x) Climate change:** Climate change is also responsible for causing shifts in the pest fauna of crops in different regions. A northward shift in favourable zones of crops, as well as pests, is being observed with climate change. Pests might thus gain entry into new areas and in the event of reduced environmental resistance might realize higher populations.

Check Your Progress Exercise 1

Note: a) Use the spaces given below for your answers.

b) Check your answer with those given at the end of the unit.

1. How are populations of organisms maintained in balance in nature?

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2. Enlist causes of pest intensification on crops.

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5.3 PEST EPIDEMICS

Insects and pathogens (agents causing plant diseases) are integral components of forest ecosystems and normally present at a relatively low density, causing little damage and having a negligible impact on crop growth and vigour. Pest populations are frequently regulated below their potential levels of abundance by natural enemies, host resistance, or other biological

interactions. However, sporadically, in time or space, some species may attain rapidly very high populations, which may persist for a variable length of time before subsiding. Such large populations are called epidemics and may have adverse effects on the growth, yield, and quality of crops.

Factors responsible for pest epidemics

- i) **Human manipulation of the environment:** Large-scale cultivation of resistant crop cultivars against some pest species may result in a large population of some other pest species due to reduced competition. Introduction of Bt cotton resistant to bollworms experienced very high populations of cotton mealybug, *Phenacoccus solenopsis*. Lack of competition from bollworms and high fertilizers, especially nitrogenous fertilizers, stimulated population build-up of sucking pests. The unnatural conditions in the man-made environment induce unusually high and destructive pest populations.
- ii) **Foreign pests:** Despite quarantine regulations, many foreign insects species get accidentally introduced into new habitats. With their lower environmental resistance due to the absence of their natural enemies in new habitats, their populations may attain unusually high magnitude. Some of the foreign pests, which caused havoc in India during the last 1-2 decades include sugarcane woolly aphid, *Ceratovacuna lanigera*; coconut eriophyid mite, *Aceria guerreronis*; coffee berry borer, *Hypothenemus pascalivorus*; sapota seed borer, *Trymalitis margirias*; spiraling whitefly, *Aleurodicus disperses*; serpentine leaf miner, *Liriomyza trifolii*. Introduced species became serious problems when they found an abundant food supply, a favourable climate, and few or no effective natural enemies.
- iii) **Introduction of the crop into the unsuitable environment:** Many ornamental plants are chosen for their beauty and planted without proper consideration of their suitability for a particular site. Drought, nutrient, or temperature stress may affect such plants and make them more susceptible to pest attacks. Proper watering and fertilizing of plants allow them to compensate for pest damage while stressed plants may be seriously injured by the same amount of pest damage.
- iv) **Weather:** Weather plays an important role in pest outbreaks by initiating the early activity of pests, promoting growth rates, and reducing mortality rates. Favourable weather in the form of early onset of summer rains with suitable temperatures may initiate early breeding of pest population which may facilitate the completion of extra generations during the season and as a result, enormous population levels may be witnessed. Increased population levels may be 50-100 times higher than normal owing to the prolific fecundity of pest species. It has been observed that the early onset of rains in May further followed by regular monsoon rains led to a BPH outbreak on rice, leading to severe yield losses. The pest situation may further be aggravated by the heavy use of nitrogenous fertilizers, continuous flooding of rice fields, and indiscriminate use of broad-spectrum pesticides.

- v) **Pesticides:** Broad-spectrum pesticides kill pests and their natural enemies as well. Pesticides prove effective until pests develop resistance against them. Once a resistance problem arises pesticide does not kill pests hence pest population explodes. This is known as pest resurgence. Pesticides targeted against major pests may also kill natural enemies of secondary pests but not secondary pests themselves, particularly when they belong to different groups, *e.g.* pesticide used against an insect pest may not kill a mite pest. In such a situation, secondary pest attains an enormous population due to the absence of major pests as well as due to the absence of their natural enemies.

Check Your Progress Exercise 2

Note: a) Use the spaces given below for your answers.

b) Check your answer with those given at the end of the unit.

1. Define pest epidemics.

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2. Enumerate two factors responsible for causing pest epidemics.

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5.4 PEST DIAGNOSTICS

Rapid and accurate diagnosis of pests and diseases is necessary before we can adopt proper control measures. It is the first step towards the study of any pest and its management. Diagnosis is largely done based on characteristic symptoms expressed by the affected plant. Identification of causal organisms is also essential to diagnosis. Diagnosis of plant diseases is more complex as their symptoms are more varied than those of insect pests. Insect pests are visible to the naked eye and symptoms can be broadly assigned to the type of pests in the field itself and then species can be further identified.

5.4.1 Disease Diagnosis

Diagnosis of plant disease is a practical science and experience is the most important factor in the identification of a disease in the field. In preliminary disease identification, illustrated guides and charts are of paramount significance. Identification of those diseases is easier in which pathogen produces specific growth, such as, in downy mildew, powdery mildew, rusts, smuts, and many sclerotial diseases. However, in several other diseases,

symptoms do not prove as helpful in identification as they overlap among many diseases, *e.g.* chlorosis or yellowing of leaves can be caused by a nutritional disorder, fungus, virus, or bacteria.

In the case of the identification of overlapping symptoms, the first step is to determine whether the pathogen is infectious or non-infectious. Field observations on the pattern of development of disease in plant populations and the spread of symptoms in other plants help in distinguishing the two. If the disease spreads among the plants then it is infectious. An infectious disease may be caused by bacteria, fungi, viruses, and nematodes. To determine their presence, an infected plant can be observed for the presence of bacterial ooze, fungal structures, or nematode cysts or females and then confirmed by laboratory studies. If a fungus is not a biotroph (organism or fungus that survives and multiplies only in another living organism), it can be cultured on artificial media and then tested for Koch's postulates. Bacteria can also be detected similarly. Apart from the ooze often seen on the host surface in the case of bacteria, examination of cut pieces of the infected part in water under the microscope would reveal streaming of bacterial cell masses.

A. Koch's postulates comprise the following steps:

Koch's postulates constitute four criteria designed to establish a causal relationship between a causative microbe (organism) and a disease. Postulates mean the fundamental elements or basic principles. The postulates were initially formulated by Robert Koch and Friedrich Loeffler in the year 1884 and afterwards refined and published by Koch in 1890. Koch applied the postulates to establish the etiology of anthrax and tuberculosis, but they have been generalized to other diseases.

- i) The pathogen must be invariably found in infected plants or must be associated with it in some form.
- ii) The pathogen should be separated from the host and grown in artificial culture.
- iii) The pathogen from the artificial culture should be able to reproduce the disease when inoculated on a healthy plant of the same kind from which it was isolated. Symptoms produced should be identical to those seen on the plant from which the isolation was made.
- iv) The artificially produced disease should yield the same pathogen on re-isolation.

Nematodes, if present, can be seen on the host or macerated tissues examined under the microscope. Nematodes can not be grown on artificial media. However, by separating the eggs and larvae from the host and multiplying them on a susceptible host under sterile conditions, sufficient larvae can be obtained to prove Koch's postulates.

Viruses can not be seen or grown in culture. They generally produce symptoms similar to those of nutritional deficiencies. However, the latter condition is non-infectious and does not spread from plant to plant while viruses are infectious and spread among plants.

If in artificial culture no pathogen is obtained and in tissue examination, including the vessels, no fungal structures of a biotroph or structures of nematodes are seen but the disease is infectious, it can be expected that the disease is caused by a virus or mycoplasma like organism (MLO). The differentiation between the viruses and MLO can be made by electron microscopy and by spraying streptomycin antibiotics, which mask the symptoms of MLO diseases but not that of virus diseases.

B. Plant disease symptoms

- i) **Rust:** Rusts infecting thousands of economically important plants are caused by more than 4,000 species of fungi. These diseases produce rusty symptoms. The rusts appear as small pustules of spores that may be dusty or compact and red, brown, yellow, or black. Yellow, brown, and black rusts are important diseases of wheat (Fig. 5.1). During their life cycle, rust fungi parasitize either one species of plant (autoecious or monoecious rust) or two distinct species (heteroecious rust).



Fig. 5.1: Wheat affected by yellow rust

- ii) **Mildew:** In these diseases, the pathogen is seen as white, grey, brownish, or purplish growth on the host surface (Fig. 5.2). In powdery mildew, an enormous number of spores are formed on the superficial growth of fungus on the surface of leaves, buds, young shoots, fruits, and even flowers, producing a dusty or powdery appearance. This is caused by many specialized races of fungal species in the genera *Erysiphe*, *Microsphaera*, *Phyllactinia*, *Podosphaera*, *Sphaerotheca*, and *Uncinula*.

In downy mildew, the superficial growth of the fungus is tangled cottony and downy growth and grey, bluish, or violet downy patches of mildew form mostly on the undersides of leaves in damp weather. Pale-green to yellow or brown areas usually develop on the upper leaf surface opposite

the downy growth. A disease of plants, especially in cool humid regions is caused by several fungi including species of *Basidiophora*, *Phytophthora*, *Pseudoperonospora*, and *Sclerospora*.



Fig. 5.2: Powdery mildew on grapes
Courtesy: Commons.wikimedia.org

- iii) **Smut:** Smut disease infecting wheat, maize, rice, sorghum, onion, and grasses is caused by many species of fungi (Fig. 5.3). The term ‘smut’ means a sooty or charcoal-like powder. It is characterized by spores that accumulate in soot-like masses called sori, in seeds, leaves, stems, flower parts, and bulbs. The sori usually breaks up into a black powder that is readily dispersed by the wind. In some smuts, a mass of spores may be compact and held under the epidermis appearing as black streaks on the host surface.



Fig. 5.3: False smut disease in rice

- iv) **White blisters:** In crucifers and many other plants, numerous white, blister-like pustules are found on leaves. They broke open the epidermis and exposed white powdery masses of spores. Often, such symptoms have been called white rust but since there is nothing common between them and true rusts, they are more appropriately termed as white blisters.
- v) **Blotch:** This symptom consists of a superficial growth giving the fruit a blotched appearance as in the sooty blotch of apple fruit.
- vi) **Sclerotia:** A sclerotium is a compact, often hard, mass of dormant fungus. In some diseases, as in ergot of cereals, the sclerotium assumes a characteristic horn-like shape but in others, the shape may be different. An ear of rye infected with ergot exudes yellowish mucus for some time followed by loss of starch as the ear ceases growth. The ovaries are then permeated by the mycelium, which in autumn forms the spur-like purple-black sclerotium.
- vii) **Exudations:** In some fungal diseases, the invasion of the pathogen causes the exudation of a gum-like material on the stem and collar. This is known as gummosis. In most bacterial diseases, *e.g.* bacterial leaf blight and bacterial leaf streak of rice, fire blight of pear and apple, bacterial cankers of stone fruits, masses of bacterial cells come out as ooze on the infected host surface where they may be seen as drop or smear. On drying, they form a crust.
- viii) **Colour changes:** Changes of colour from the normal or discolouration is one of the most common symptoms of plant diseases. Chlorosis is yellowing due to the influence of low temperature, lack of iron, excess lime or alkali in the soil, and infection of viruses, fungi, and bacteria. On the other hand, etiolation is the disappearance of green pigment due to lack of light.

Mosaic is caused by various strains of viruses. Symptoms are variable but commonly include irregular leaf mottling (light and dark green or yellow patches or streaks). Leaves are commonly stunted, curled, or puckered while veins may be lighter than normal or banded with dark green or yellow. Plants are often dwarfed, with fruit and flowers fewer than usual, deformed, and stunted. Flowers may be blotched or streaked.
- ix) **Atrophy or hypotrophy or dwarfing:** In many diseases, there is inhibition of growth that results in stunting or dwarfing. The whole plant may be dwarfed or only certain organs may be affected. Stunting may be caused by viral, bacterial, fungal, or nematode infections and by non-infectious (abiotic) means including excess or lack of water, imbalance of soil nutrients, excess light, chemical or mechanical injuries, insect or mite feeding, and too deep planting.
- x) **Overgrowth and hypertrophy:** Many pathogens induce hormonal imbalances in plants through their biochemical activity. This results in excessive growth of host tissues and causes an abnormal increase in the size of the affected organ. It is brought about by one or both of the processes known as hyperplasia and hypertrophy. Hyperplasia is the

abnormal increase in the size of plant organs due to an increase in the number of cells while hypertrophy refers to the increased size of the organ due to an increase in the size of cells. The overgrowth and its effect are seen in the form of galls, curls, pockets, hairy roots, witches broom, intumescence, *etc.* Bakane disease of rice caused by fungus results in lanky plants as a result of elongation of tillers and reduced tillering.

- xi) Spots:** The cells are killed in limited areas and dead tissues usually become brownish (Fig. 5.4). The most common diseases in this group are the leaf spots, *e.g.* brown leaf spot of paddy. Leaf spots are caused by species of bacteria or by many fungus species.



Fig. 5.4: Black spot

Courtesy: Commons.wikimedia.org

- xii) Dieback:** This disease is characterized by the progressive death of twigs, branches, or shoots, from the tip backwards. Dieback is caused by many fungi and a few bacteria that produce cankers, anthracnose, wilts, and stem or root rots. Nematodes, stem or root boring insects, mechanical damage, paving over roots, winter injury from cold and a deficiency or excess of moisture or an essential element may cause dieback.

- xiii) Blight:** Symptoms of blight include sudden and severe yellowing, browning, spotting, withering, or dying of leaves, flowers, fruit, stems, or the entire plant, *e.g.* late blight disease of potato and tomato (Fig. 5.5). Blight is caused by bacterial or fungal infections.



Fig. 5.5: Late blight

Courtesy: Commons.wikimedia.org

- xiv) **Streaks or stripes:** These are elongated but relatively narrow lesions containing dead cells. These are initially yellowish before the death of affected cells.
- xv) **Damping-off:** In this disease, the tender stem of seedlings is attacked near the soil line. The affected portion becomes constricted due to necrosis and seedling collapse. This is caused by seed- and soil-borne fungi.
- xvi) **Rot:** Rot is characterized by plant decomposition and putrefaction. These are named after the plant part infected, *e.g.* root rot, stem rot, bulb rot, fruit rot, *etc.* They are caused by any species of soil-borne bacteria and fungi. Fungal rots are usually dry to spongy or powdery, while bacterial rots are commonly moist, soft to mushy, and foul-smelling.
- xvii) **Burn or scorch:** This is a symptom of plant disease in which tissue is burnt because of unfavourable conditions or infection by bacteria or fungi. Commonly there is a dead area along or between the veins and margins of leaves.
- xviii) **Wilt:** Wilt is characterized by drooping of leaves and stems due to water loss (Fig. 5.6). Wilt diseases are caused by a variety of fungi, bacteria, and viruses and are easily confused with root and crown rots, stem cankers, insect injuries, *etc.*



Fig. 5.6: Wilt disease in chillies

- xix) **Anthracnose:** Symptoms of anthracnose include varied coloured spots on leaves, stems, flowers, or fruits. The spots often enlarge, leading to wilting, withering, and dying of tissues.
- xx) **Canker:** A canker is a dead area in the bark or cortex of the stem and sometimes on leaves. Canker symptoms include sunken, swollen,

flattened, or cracked dead areas on the stem, twig, limb, or trunk. Cankers may enlarge and girdle a twig or branch, killing the foliage beyond it.

Disease symptoms from Sr. No. **xi-xx** are referred to as necrosis lesions because they result in the death of cells, tissues, and organs due to parasitic activity.

5.4.2 Insect Damage Diagnosis

Damage symptoms of insect pests depend upon the types of their mouthparts and plant parts affected. Accordingly, there is damage in the form of tissue boring, defoliation, leaf folding, leaf-mining, sap-sucking, *etc.*

A. Tissue boring

This includes stem boring, root feeding, and fruit boring.

- i) **Stem boring:** Stem borer larvae bore into plant stems and cause their death. Rice stem borer causes the death of the central shoot called ‘dead heart’ in young plants and the death of the panicle known as the ‘white ear’ in maturing plants (Fig. 5.7). Likewise, maize stem borer and sugarcane early shoot borer also cause dead heart symptoms. The damaged shoot can easily be pulled out occasionally with the larva still inside it.



Fig. 5.7: Stem borer damage in rice

- ii) **Root boring:** Some insects like root borers, termites, white grubs, and root weevils feed on plant roots thereby killing the plants. Affected plants can easily be pulled out. The difference between stem borer and root borer damage is that in the case of the former, the roots of the plant are not damaged, and the only above-ground portion of the stem is damaged.
- iii) **Fruit boring:** Fruit borers' larvae damage fruits, *e.g.* in tomato, ladyfinger, brinjal, and pomegranate (Fig. 5.8). Likewise, fruit fly maggots cause damage in mango, cucurbits, ber, which results in rotting of the fruit.



Fig. 5.8: Brinjal damaged by fruit borer

B. Defoliation

Defoliators, such as grasshoppers, beetles, and caterpillars damage plant leaves causing defoliation (Fig. 5.9). The extent of defoliation varies from small holes to complete defoliation and sometimes only midribs are left. In the case of grasshoppers and armyworms, damage resembles as if grazed by cattle.



Fig. 5.9: Cabbage defoliated by cabbage butterfly

C. Leaf mining

Insect larvae make mines in the leaf tissue by eating away the green tissue, *e.g.* citrus leaf miner, pea leaf miner. White zigzag pathways can easily be spotted on the upper leaf surface or both sides (Fig. 5.10).



Fig. 5.10: Leaf miner damage in soybean

D. Leaf folding/rolling

Insect larvae fold leaves and then feed on green tissue inside the fold. When the fold opens, the affected leaf looks white as in the case of rice leaf folder damage in rice (Fig. 5.11). Leaf rolling is also observed in the case of cotton.



Fig. 5.11: Leaf folder damage in rice

E. Sap sucking

Sucking pests like aphids, jassids, planthoppers, whiteflies, bugs, thrips, coccids, scales, and mites suck sap from xylem and phloem bundles of plants with their piercing and sucking mouthparts. As a result of sucking from leaves, stems, flowers, and fruits plants lose vigour and turn pale and dry up. In the case of brown planthopper of rice, hopper burn symptoms are observed which are initially small circular patches of dried plants but enlarge with an increase in damage magnitude (Fig. 5.12). In case of leafhopper damage in ladyfinger, upward cupping of leaves occurs, and eventually, they may fall. Likewise, thrips damage results in flower drops in many crops.



Fig. 5.12: Rice crop damaged due to plant hopper sucking

Check Your Progress Exercise 3

Note: a) Use the spaces given below for your answers.
b) Check your answer with those given at the end of the unit.

1. Why is pest diagnosis done?
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2. Which crop is affected by rust diseases?
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3. Expand MLO.

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4. Differentiate between hyperplasia and hypertrophy.

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5.5 INTEGRATED PEST MANAGEMENT (IPM)

The sole reliance on pesticides to control insect pests has created several problems, such as the development of resistant pests, pest outbreaks, mortality of useful organisms, adverse effect on human health, and environmental degradation. Adverse effects of pesticides prompted scientists to explore safer and environment-friendly methods of pest control and consequently integrated pest management came into vogue.

Integrated Pest Management (IPM) refers to the use of available pest control methods in a unified manner to reduce pest population below the economic injury levels and ensure favourable, ecological, and economic consequences. Pest management signifies an ecological approach to tackle pest problems because here the emphasis is on containing pests in such a way that other biotic components of the system, such as natural enemies, human beings, and wildlife are not harmed and the environment is not degraded.

Pest management works on the premise that all pest population levels are not injurious to crops and crops can tolerate some amount of pest injury. Moreover, some pest population is also required for the survival of natural enemies of the pests. This has given rise to the concept of economic injury level (EIL), which helps to avoid the unwarranted application of control measures. The EIL refers to the minimum pest population or damage that causes economic damage. On the other, economic damage signifies the damage level at which expenditure on pest control is equal to the value of crop saved from pest damage. The EIL-based pest control measures ensure a favourable cost-benefit ratio for farmers. Pest management also underlines that pest control tactics should be socially acceptable and within the reach of farmers. It is a holistic approach to pest problems in which we aim to protect our commodities against all pests and ensure the production of healthy crops.

5.5.1 Principles of IPM

- a. **Understanding of pests:** The biology of pests should be thoroughly studied to detect the weak points in the life cycle for maximum benefit

from control measures. We should know whether it is univoltine i.e. having a single generation in a year or multivoltine. In the case of univoltine species, the action during the weak link of the life cycle can reduce their populations significantly, for example in the case of rice grasshopper and mango mealybug, the eggs are laid on the bunds of paddy fields and under the mango tree in the soil, respectively. The destruction of egg pods by raking field bunds and mango tree basins greatly reduces the pest populations. In the case of fruit borers and stem borers, the destruction of newly hatched larvae is the most appropriate method to control them; otherwise, once these enter the tissue it becomes very difficult to control them. Likewise, population reduction can also be done by attracting and killing the moths. Proper identification of the pest is necessary, for knowing its natural enemies so that these can be conserved and if necessary can be produced from other places for mass multiplication and released against pest.

- b. Understanding the agroecosystem:** Ecosystems are self-sufficient habitats where living organisms and nonliving environments interact to exchange energy and matter in a continuing cycle. Ecosystems are entities, such as forests, ponds, and fields, and in general, they are self-regulating. Agro-ecosystems contain a lesser diversity of animal and plant species than do natural ecosystems, such as forests and prairies. Usually, there are few major species and numerous minor species and, in a pest outbreak, usually, only one pest species at a time is present in large numbers. A typical agricultural unit may contain only 1-4 major crop species and 6-10 minor pest species.

The agroecosystem is intensively manipulated by man through ploughing, mowing, and treatments with pesticides which harms the natural enemies. Natural enemy diversity is thus relatively low in agro-ecosystems. Therefore, agroecosystems are more susceptible to pest damage and catastrophic outbreaks. This happens due to a lack of diversity in plant and insect species and due to the sudden alternations imposed by weather and man. Often an insect can attack, establish and survive only during a short period but staggered planting may make food available for pests for a longer period and increase pest problems.

- c. Planning of agroecosystem:** Planting crops should be done in such a way as to minimize or prevent pest problems. Agronomic practices can be modified suitably for this purpose. Deep ploughing in summer exposes soil pests, such as cutworms, white grubs, mole crickets, armyworms, *etc.* to predators and extremes of weather, thereby reducing their population in the ensuing crop season. Similarly, planting time can be preponed or postponed, for example, early planting of rapeseed mustard in September in place of October reduces aphid infestation on these crops. Rice crop planted in August faces less incidence of Gundhi bug than crops planted in June or July. Sometimes changes in planting dates may have a contrasting effect on different pests. In such a situation, the planting date should be chosen based on the relative importance of the pests.

If pest incidence appears in the nursery on crop seedlings then it should be controlled in the nursery itself as it is easier to control it on a small scale with minimum contamination of the environment. Likewise, seedling dipping in pesticide solution at the time of transplanting or seed treatment before sowing also proves very effective in checking pest problems and reducing environmental pollution. Rice seedlings dipping in chlorpyrifos before transplanting prove very effective against stem borer, gall midge, and plant hopper. Seed treatments in pulses, mustard, sorghum, and cotton also prevent pest problems.

- d. **Cost /benefit:** It is the ratio between the cost of control measures and the value of the crop saved from pest attack due to the application of control measures. For a profitable situation, this ratio should be less than one. The lesser the ratio, the more the profit to the farmer due to the control of pests. When its value is greater than one, then the farmer suffers indirect loss. In this situation, the cost of control measures is more than the value of crops protected from pest attacks. Such a situation does not warrant the adoption of any control measures against the pest. To ensure that farmers do not face such loss incurring situation, the control measures should only be applied when the population reaches the economic threshold level. The economic level is worked out based on the cost of control measures, the market value of the crop, and the damage potential of the pest in the crop.
- e. **Benefit/risk:** Benefit/risk analysis provides a means for assessing the relevant economic benefits versus the risk pest control poses to the environment. The risk may include harmful effects on beneficial organisms, wildlife, and humans; soil, water, and air degradation, and development of resistant pests, and the presence of pesticide residues in food. The consideration and assessment of benefits/risks are fundamental to pest management.
- f. **Tolerance of pest damage:** To promote pest management and reduce our dependence on pesticides for pest control, we must learn to live with insects. The pests cause economic loss only when they have reached some population level on the crop. Therefore, complete freedom from pests is not necessary for obtaining higher yields because crop plants are capable of compensating for some damage due to pests. However, this is not applicable for insect vectors of plant diseases, where complete eradication is desirable. In some cases, the removal of excess foliage due to pest damage even caused an increase in yield over the healthy crops. There should not thus be any reason for panic on witnessing a few insects in the field. Moreover, some pest population is always needed for the survival of natural enemies of pests, which play a very important role in pest suppression in the field. Therefore, minimum population or damage that can be tolerated without any adverse effect on crop yield and quality should be determined to avoid unwarranted application of control measures. The determination of economic threshold level (ETL) and economic injury level (EIL) is a step in this direction. These imply a tolerant approach to pest problems and need-based application of pesticides.

- g. Leaving a pest residue:** We have discussed that complete freedom from pests is not necessary for higher yield as plants can always compensate for some pest injury. Pest management philosophy thus advocates suppression of pest population to non-economic level and not the elimination of pests from the field. With this approach, fewer insecticidal treatments are required as compared to the elimination of pests. The presence of some pests in the field will ensure the survival of natural enemies of the pests, which would reduce dependence on pesticides and curtail environmental contamination.
- h. Timing treatments:** Pest management emphasizes the need-based application of pesticides. The calendar-based application or prophylactic treatments (preventive measures) are generally discouraged. The need for the application of pesticides can be determined through regular monitoring to find out whether the population has reached the economic threshold level or not. The need-based application may reduce the pesticide requirements by at least 50%. The pesticide application should coincide with exposure to the most vulnerable stage of the pest for best results. For example in the case of stem borers, such as, rice stem borer or maize stem borer and fruit borers, regular monitoring for egg laying is very important. The application should coincide with egg hatching so that larvae come in contact with pesticides and die. Otherwise, once the larvae enter the stems or fruits, the application of pesticides will not be effective. In the case of fruit flies, an attempt should be made to control adults of these pests because only this stage is exposed whereas eggs and maggots are hidden in the fruits and pupae are in the soil.

Proper or precise placement of insecticides also helps in controlling pests more effectively and at the same time avoids contamination of the environment. For example, the rice planthoppers are mainly confined to the stems of rice plants. The pesticide application does not prove effective if the spray is directed at foliage, hence should be directed at stems.

- i) Public understanding and acceptance:** Farmers are the users of IPM programmes therefore these should be developed through their active participation. The farmers' experiences with pest problems and the techniques used by them to combat pests should be given due consideration. The indigenous technical know-how (ITK) adopted by farmers has a sound scientific basis and hence the IPM programmes should be developed through a judicious blend of indigenous and modern technologies. Such programmes will be popular among farmers as they will consider them as their programmes. Besides, these will also prove economical.

Extension workers should teach farmers that community cooperation is essential for the effective management of pests. The management of pests, such as armyworms, locusts, and white grub beetles requires coordinated effort on the part of the entire community. Similarly, planting crops around the same time in an area also reduces the incidence of several pests. They should also have the will to share their plant protection equipment.

5.5.2 Approaches of IPM

In IPM, different methods of pest control, such as resistant varieties, cultural methods, physical methods, natural enemies, and pesticides are suitably integrated to suppress the pest population without jeopardizing other components of the environment. Any pest management programme should preferably be based upon the use of resistant cultivars and natural enemies and other control methods should be blended with them for tackling pest menace. The utility of different pest management components has been described in the following discussion (Fig. 5.13).

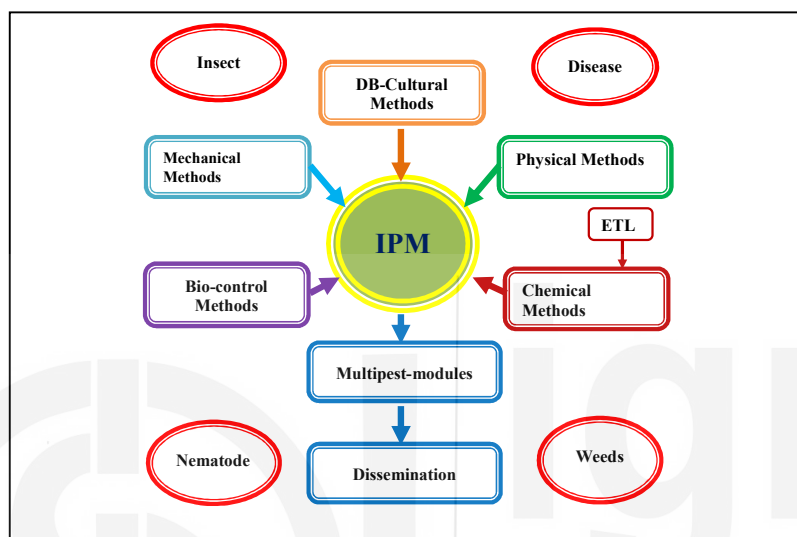


Fig. 5.13: IPM components

- a. **Resistant cultivars:** Preference should be given to the cultivation of resistant cultivars because their use greatly reduces the need for other methods of pest control (Fig. 5.14). The resistant cultivars enhance the efficiency of some natural enemies against the pests, for example, there is more predation of spiders against planthoppers on moderately resistant varieties than on susceptible varieties. Likewise, on moderately resistant cultivars, the need for pesticide application is greatly reduced. Similarly, these are also compatible with cultural methods of pest control. However, the continuous use of resistant varieties has given rise to the problem of biotype development in some insects.



Fig. 5.14: Resistant and susceptible germplasm

Biotypes of brown planthopper and gall midge have been developed due to the cultivation of monogenic resistant varieties of rice. Resistant cultivars of various crops against insect pests are presented in Table 5.1.

Table 5.1: Resistant cultivars/sources of resistance against insect pests in different crops

Crop	Pest	Cultivars
Rice	Stem borer	Deepti, Rama, Ratna, Sasyasree, TKM-6, Vikas, IR 20, IR 26
	Gall midge	Karna, Lalat, Sarasa, Ahaya, Ruchi, Kshira, Dhanyalaxmi, Pothan, Mahaveer, Divya, IR36, Kakatiya, Kunti, Phalgun, Shakti, Surekha
	Brown planthopper	Bhadra, Chandana, CO 42, Jyoti, Kanaka, Pratibha, Vajram, Chaitanya, Nagarjuna, Sonasali, Manasarovar, Krishnaveni, Kakatiya, Aruna, Makom, Remya, HKR 120
	Green leafhopper	IR 20, Vani, Vikramarya
	Leaf folder	Adt-34, Dular, Giza-171, Basmati Dwarf, Pattambi-29, IR-28, IR-50, Tainan-3, Pusa-1120-7-3
	<i>Multiple resistance</i>	
	Gall midge, BPH, WBPH	Suraksha, Shaktiman
	Gall midge, BPH, GLH	Daya, Lalat
	Gall midge, BPH, WBPH, GLH	Kshira
Wheat	Shoot fly	UPD 8, UP 85042, HD 2307, PDW 215, PDW 206, PBW 225, Raj 3214, HUW 206, WH 841, WH 283, HI 8381
	Brown wheat mite	C 306, A-9-30-1, DT 18, HDR 132, HPW 42, WH 589, Hyb 65, ISW 38, UP 2377, UP 2378, UP 2379
Barley	Aphids	DL117, DL200
Maize	Stem borer	Deccan hybrid, Ganga-5, Kundan, Chanda
	Stem borer and shoot fly	Composite-217

Sorghum	Shoot fly	SPV 1015, SPV 1413, SPV 1450, SPV 1467, SPH 1174, SPH 1077, SPH 1166, Selection 3, P 311, M 35-1, Swati, CSV 14R
	Stem borer	SPV 1155, SPV 1359, SPV 1466, SPH 1162, SPH 1078, SPH 1079, SPH 1165
	Midge	ICSV 197, ICSV 88032
	Aphids	SPV 1450, SPV 1359, SPV 1155, SPV 1380, SPV 1413, SPV 1452, SPV 1453, SPV 1462, SPV 1465, Selection 3, SPH 1162, SPH 1174, SPH 1020, SPH 1026, SPH 1079
	Multiple resistance	SPH 1161, SPV 1413, SPV 1450
Chickpea	Gram pod borer	PDE-2, ICC 738, ICC 506, ICCV 7, ICC 6663, PDE-5, N 37, GL 1002
Cowpea	Aphids	TVu 57, TVu 408, TVu 410, TVu 1937, TVu 3273, P2, R-1476, R-1473
Lentil	Pod borers (blue butterfly, gram pod borer)	DPL 7, DPL 16, DPL 26, DPL 34, PL 81-1, PL 88-54, PL 406, L 19-12, L 4136, L 4150, LL 56, LH 88-8, DHL 90-13
Pigeonpea	Gram pod borer	ICPL (Abhaya), ICPL 87088, ICPL 84060, ICPL-1, ICPL-2, ICPL 87-1, ICP 10466-E3, IPCL-6, JAT-10, UPAS-120, ICP-903E, ICP-1903E
	Pod fly	ICPL 82064-E15-E, PDA-881E, PDA-882E, PDA-883E, ICP 7946, PDA-89-2E, ICP 8102-5-6, ICP 7946, MA2, PDE-89-23
Soybean	Stem fly, grey weevil & YMV (whitefly) YMV transmitted by whitefly	PK 1069, PK 1107, MM-39-2(DS-93-39-2) PK 1180, PK 1189, PK 1092, PK 1060, DS 1016, Pusa 9702, DSK-18, SL-443, SL-444
Groundnut	Tabacco caterpillar	NCA-17840, NFG-79, EC-21898
Sugarcane	Top borer	CoJ 65, Co 859, C0 1158, Co 7224
	Internode borer	Co 975, Co 62175, Co 6806, CoC 77-1

	Scale insect White grub	CoC 671, Co 62174, CoC 6907, CoS 611, Co 1132, Co 6304, CoS 510
Cotton	Jassid	Bikaneri Nerma
	Whitefly	Surekha, PS-141, Kanchan
	Boll worms (<i>Earias</i> & <i>Helicoverpa</i>)	A-1318, Bes-9-7, FBRN-3-7
Okra	Leaf hopper	AE-30, Punjab-7, IIHR-4, AC-3375, Reshmi, Sangli-2, Okra Red, Selecton 6, Selection 8
	Shoot and fruit borer	AE-79, AE-22, AE-71, Narnaul special, Perkins Long Green
Tomato	Fruit borer	Parker, Bonus, VFN-8, Rick x Solan Gola (Lal, 1985)
Brinjal	Shoot and fruit borer	Sm 202, Sm 17-4, Pant Samrat, Pusa Purple Cluster
	Leafhopper	BI-7, 435-20
Cabbage	Aphid	Red Rock Mammoth, Red Drum Head, Glory, Early Queen, Express Mail, Rainy Princess, All Season, Red Pickling
Cauliflower	Head borer	ES-97, ES-96, Kartiki, KW-5, KW-8, KW-10, Kuwari, Kathmandu Local, Early Patna, EMS-30, PSK-16
Chilli	Aphid, jassid, mosaic, leaf curl	Pusa Jawala

BPH- Brown Plant Hopper; WBPH- White Backed Plant Hopper; GLH- Green Leaf Hopper

- b. **Cultural methods:** Routine agronomic practices can be utilized for minimizing pest infestation by slight modification in timing or method of their application. These function as a preventive method. The field operations right from field preparation to harvesting or post-harvesting can reduce the population of one pest species or other.
 - i) **Clean cultivation:** Destruction of crop residues helps to prevent the carryover of pests to the next season's crop, for example, the destruction of cucurbit vines and stubbles of rapeseed mustard after crop season kills the population of pumpkin beetle and painted bug, respectively. Similarly, the disposal of stubbles in rice, maize, and sorghum reduces the carryover of stem borers.
 - ii) **Tillage:** Deep ploughing of fields exposes the soil-inhabiting stages of pests, such as armyworms, cutworms, borers, white grubs, termites, and mole cricket to vagaries of nature and thus helps to reduce pest

infestation in crops (Fig. 5.15). The raking of bunds of rice fields destroys the egg pods of grasshoppers. Similarly raking the soil at the base of the mango tree destroys the eggs of the mango mealybug.



Fig. 5.15: Tillage and irrigation

- iii) **Sowing time:** The change in sowing/planting time is aimed at disturbing synchrony between host and pest populations. In rice, early planting (June-July) results in low populations of gall midge, leaf folder, and armyworm, while late planting (August) reduces the incidence of grasshopper, gundhi bug, and brown plant hopper (BPH). Early sowing of cucurbits escapes the attack of the pumpkin beetle. However, a change in planting time should be adopted based on the relative importance of different pests in the area. As far as possible, crops should be planted around the same time in a locality as staggered planting increases pest incidence. If there is an incidence of pests in the nursery then it should be preferably tackled in the nursery itself as it is easier to contain a pest problem at a small scale.
- iv) **Intercropping:** Intercropping helps in reducing the incidence of certain pests by making the microclimate less favourable for them. It also hinders the free movement of pests among the plants of the same species. Intercropping of cabbage with tomato reduces the infestation of the diamondback moth in cabbage. Intercropping of chickpea with wheat decreases the incidence of pod borer in chickpea, while intercropping of mustard with wheat causes a decline in the aphid population on rapeseed mustard.
- v) **Crop rotation:** Crops sharing the same pests should not be grown in a rotation. It has been found that with rice-wheat rotation, the pests of rice and wheat are showing more incidences. Similarly, multiple cropping of rice results in outbreaks of pests.
- vi) **Trap crops:** Trap crops protect by preventing the pests from reaching

the main crop. The trap crop must be more attractive to the pests and less economical than the main crop. Mustard as a trap crop with cabbage reduces the incidence of diamondback moth, leaf webber, webworm, and aphids. Marigold planted with tomato as a trap crop are highly effective against the tomato fruit borer.

- vii) Nutrient and water management:** Fertilizers should be applied in a balanced manner. Excess use of nitrogen intensifies the incidence of several pests, such as plant hoppers and leaf folders in rice. Most of the time farmers only apply nitrogen and not phosphorus and potash. Potash induces resistance in plants against pests. Likewise, excess use of water should also be avoided. In rice, it has been found that continuous flooding of fields is as good as alternate wetting and drying. Drying the field at some intervals helps reduce the incidence of planthoppers in rice.
- viii) Harvesting:** Crops should be harvested at the optimum time. These should be harvested in such a manner that pests inside the stubbles get destroyed. The harvesting of rice at ground level destroys larvae of stem borer, thereby reducing pest survival for the next season.
- ix) Mechanical and physical methods:** The removal of infected plant parts and conspicuous pest stages manually or mechanically can reduce pest incidence to a certain extent. Dead hearts, whiteheads, silver shoots, aphid-infested shoots, bored brinjal, and *bhindi* fruits should be picked and destroyed. Similarly, egg masses, hairy caterpillars, and beetles can also be collected and killed by dipping in kerosene. Light traps can be used for attracting and killing moths of stem borers, leaf folders, case worms, *etc.* (Fig. 5.16). Although mechanical methods are labour-intensive these prove very effective if practiced at the initiation of pest incidence. Likewise, pheromone traps carrying female moths' smell can also be used to attract and destroy male moths (Fig. 5.17).



Fig. 5.16: Light trap



Fig. 5.17: Pheromone trap

- c. Biological control:** There is a rich diversity of natural enemies, *viz.* predators, parasitoids, and pathogens in fields, and these help to reduce the pest incidence if not harmed by broad-spectrum pesticides. The egg parasitoids of *Trichogramma* are very effective against the stem borers,

fruit borers, and bollworms. Likewise, there are several effective larval and pupal parasites. Predators, such as coccinellids, syrphids, spiders, dragonflies, ground beetles, and rove beetles also devour pest populations in the field (Fig. 5.18). The vertebrate predators, like fish, ducks, and frogs also consume pests in the rice ecosystems. Likewise, bacteria (Bt), fungi, and viruses (NPV) also kill pests. Artificial releases of natural enemies are also carried out to combat these pests (Table 5.2).



Fig. 5.18: Spider- predator of pests

Table 5.2: Biological suppression of crop pests with released predators, parasitoids / sprayed pathogens

Pest	Natural enemy	Release/application rate
Rice yellow stem borer & leaf folder	<i>Trichogramma japonicum</i> & <i>T. chilonis</i>	Inundative release @50,000-100,000/ha/week for 5 weeks
Sugarcane pyrilla	<i>Epiricania melanoleuca</i>	Periodic release @8000-10,000 cocoons or 800,000-1000,000 eggs/ha
Maize stem borer	<i>T. chilonis</i>	Inundative releases @ 100,000/ha/3 days - 5 times (Singh, 2002)
Tobacco caterpillar	<i>Telenomus remus</i> (egg parasitoid)	3 fortnightly releases @ 40,000/ha
Cotton bollworms	<i>T. chilonis</i> (Bio C1, Bio C6)	@ 1,50,000/ha/7-10 days during egg laying period
	<i>Chrysopa carnea</i>	@ 2 larvae/plant during peak egg hatching
Apple San Jose scale	<i>Encarsia perniciosi</i> or <i>Aphytis</i> sp.	@ 2000/infested tree
	<i>Chilocorus infernalis</i>	@20/infested tree

Apple wooly aphid	<i>Aphelinus mali</i>	-
Codling moth	<i>T. embrophagum</i>	@2000/tree during active egg laying period
Mealybugs & scale insects	<i>Cryptolaemus montrouzieri</i>	@10-50 beetles/plant
Cottony cushion scale	<i>Rodolia cardinalis</i>	-
Anar butterfly <i>Deudorix isocrates</i>	<i>T. chilonis</i>	Inundative release @50,000/ha/week
Coconut back headed caterpillar <i>Opisina arenosella</i>	<i>Goniozus nephantidis</i>	Released at fortnightly interval during peak availability of pest
Subabul psyllid <i>Heteropsulla cubana</i>	Coccinellid, <i>Curinus coeruleus</i>	Inoculative release
Tomato fruit borer <i>Helicoverpa armigera</i>	<i>T. chilonis</i> / <i>T. pretiosum</i>	Inundative release @50000/ha/week
Spider mites	<i>Amblyseius</i> spp.	-
Tomato fruit borer	Ha NPV (nuclear polyhedrosis virus)	@250 larval equivalents (LE)/ha
Tobacco caterpillar <i>Spodoptera litura</i>	SI NPV	@250 LE/ha
Sugarcane shoot borer <i>Chilo infuscatellus</i>	Granulosis virus (GV)	-
Cotton bollworms Coffee berry borer	<i>Beauveria bassiana</i>	-
Sugarcane pyrilla Rhiniceros beetle	<i>Metarhizium anisopliae</i>	-
Tomato fruit borer Tobacco caterpillar Castor semilooper	<i>Nomuraea rileyi</i>	-
Lepidopteran (moths and butterflies) pests	<i>Bacillus thuringiensis</i> var. Kurstaki	-

Farmers must differentiate between pests and useful insects and should conserve the useful ones. Efforts should be made to enhance natural enemies by providing artificial foods and shelter. The application of broad-spectrum pesticides poses the greatest hazard to the natural enemies of pests. The preference should be given to selective pesticides, such as neem, Bt, NPV, diflubenzuron, propargite, cartap, etc. which do not harm non-target organisms.

In the absence of selective pesticides, even non-selective pesticides can be applied to achieve the same purpose through suitable formulations. The granular formulations of pesticides pose less danger to natural enemies of pests.

- d. Chemical control:** Pesticides have a definite role to play in pest management. However, their application should be need-based and in a judicious manner. Pesticide consumption can be greatly reduced, and these can be made safer for the environment by adopting improved techniques and proper timing of application, *e.g.* planthoppers are controlled more effectively if a spray is directed at the base of rice plants. Regular monitoring of pests should be carried out and pesticides should only be applied if the population has reached the economic threshold level. The ETL-based pesticide use causes a substantial reduction in the pesticide requirement. If pest infestation begins in the nursery then it should be nipped there itself, because it is easier to manage pests at a small scale and this will also not pollute the environment. The perusal of various control methods reveals their effectiveness, in general, against pests. Depending upon the pest problem, the relevant and mutually compatible methods can be selected and integrated to achieve pest suppression. Following is an example of the integration of different control methods for managing rice pests.

Integrated management of rice insect pests

Cultivation of resistant cultivars like Vajram, Chaitanya, Nagarjuna, Pratibha, Manasarovar and other high-yielding varieties/ cultures like Badava Mahsuri and IET 7251.

- Deep ploughing and raking of bunds to kill soil-dwelling pest stages and grasshopper egg pods
- Altering planting time, preponing, or postponing depending on pests in the area
- Clipping of seedling tips at the time of transplanting to destroy stem borer and hispa eggs
- Dipping seedlings in chlorpyrifos for protection against stem borer and planthoppers
- Formation of alleyways in the field for proper air circulation
- Synchronous planting to avoid food availability to pests for a longer period
- Balanced use of fertilizers with emphasis on recommended N dose
- Field sanitation- removal of excess nurseries and weeds from the field
- Water management- draining of water from the field at certain intervals
- Conservation of natural enemies through the need-based and selective use of pesticides.

In this way, pest management packages can be formulated against key pests of other crops. The adoption of IPM packages will help to reduce dependence on pesticides and pests can be managed in an eco-friendly and sustainable manner.

Contrary to the belief, the pesticide is not a bad word in pest management. Pesticides have to play a definite role in pest suppression. However, their use has to be judicious and need-based. The need for pesticides should be determined through regular monitoring and economic threshold levels.

Check Your Progress Exercise 4

Note: a) Use the spaces given below for your answers.

b) Check your answer with those given at the end of the unit.

1. Define EIL.

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2. Mention the ways by which resistant cultivars prove useful in pest management.

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3. Enumerate important cultural methods of pest control.

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5.6 PESTICIDE RESIDUES AND CONSEQUENCES

The amount of pesticide, its derivatives, and adjuvants found in or on a treated substrate, such as, plant, animal, or soil after various intervals of time is called “residue”. The amount of pesticide initially deposited on the treated surface may be termed a “deposit”. The residue is expressed as parts per million (ppm) or billion (ppb) or even trillion (ppt) or even at lower levels. Some portion of the deposited pesticide applied on the plant surface slowly gets weathered and whatsoever amount penetrates the system is termed as “effective residue”. Residue found in the substrate at the time of harvest is called “harvest time residue”.

After application, the degradation of pesticides takes place due to physical factors, like light, temperature, moisture, rainfall, and wind velocity, and biological factors, such as animals, plants, and microorganisms. Degradation refers to any change in the molecular structure of the pesticide or its products.

Terms, such as decomposition and breakdown are analogues to degradation.

The disappearance of residues refers to the apparent loss of the parent pesticide. Initially, the rate of disappearance is fast and is known as dissipation. This may also be called as the first phase of disappearance. Later the rate of disappearance slows down and there is a slow decrease in the amount of residue. This is known as persistence. The rate of disappearance of residues of pesticides can be expressed in terms of half-life, which is defined as the time required for half of the amount of initial deposit of pesticide to disappear.

5.6.1 Adverse Effects of Pesticide Residues

Pesticides are poisons and can cause carcinogenic, mutagenic, and teratogenic effects in animals. Some of the pesticides like the majority of chlorinated hydrocarbons undergo biomagnification as they move from lower trophic levels to higher trophic levels along the food chain, *e.g.* DDT, aldrin, endrin, *etc.* are lipophilic and after entering the living system these keep depositing in fatty tissues without being excreted. This way these chemicals reach a million times higher concentrations in organisms like big fish, birds, and human beings, which are at the higher end of the food chain than their concentration in water. When these concentrations reach lethal levels, they may prove fatal. Keeping in view the adverse effects of pesticides on humans, animals, and the environment as a whole, these should be used judiciously and commodities treated with them should only be consumed or used when residue levels are at the non-harmful levels. To safeguard the health of humans and animals, safety measures like acceptable daily intake (ADI), waiting periods, maximum residual limits (MRL), *etc.* are fixed and implemented.

5.6.2 Safety Precautions against Residues

Following are the important safety precautions against the pesticide residues:

- a. **Waiting period:** The waiting period is the time that must elapse between pesticide application and crop harvest or use. Maximum residual limits (MRLs) are defined as the maximum concentration of pesticide residue likely to occur in or on food and feeding stuff after the use of pesticides according to Good Agricultural Practice (GAP). The GAP refers to the application of pesticides in line with the product label recommendations and in keeping with local environmental and other conditions. These are expressed as milligrams (mg) of residue per kilogram of food/animal feeding stuff. MRL levels are set by the Codex Alimentarius Commission, a United Nations agency managed and funded jointly by the World Health Organization (WHO) and the Food and Agriculture Organization (FAO).
- b. **Maximum residue limit (MRL):** Where there is an approved use of the compound on a particular crop or for pesticide residues on imported produce (import tolerance), the MRL is generally set at a value determined from field trials. In field trials, the crop is treated with the pesticide, and samples of the crop are analyzed to determine residue

levels. MRLs can typically be less than a milligram (mg) of pesticide residue in a kilogram of food (1mg/kg or less) to up to 5 mg/kg.

However, where there is neither approved use nor an 'import tolerance', a residue level is set at the 'Limit of Determination (LOD)'. This is an effective 'zero' level of residues reflecting the lowest level at which reliable quantitative analysis can be performed. A LOD MRL can also be set where an approved use is not expected to result in residues and this has been demonstrated from the field trial data. The LOD MRLs are usually between 0.01 and 0.05 mg/kg.

MRLs are intended primarily as a check that the pesticide is being used correctly and to assist international trade in treated produce. MRLs are primarily trading standards but they also help ensure that residue levels do not pose unacceptable risks for consumers.

The MRL is not a health-based exposure limit and exposure to residues over an MRL does not necessarily imply a health risk. This is because the use of a pesticide would not be allowed if the proposed MRL resulted in long-term and short-term exposure to pesticide residues in the human diet above safety limits, such as the Acceptable Daily Intake (ADI) and Acute Reference Dose (ARfD).

c. Acceptable daily intake (ADI)

The ADI is the amount of a pesticide in food or drinking water that can be ingested (orally) daily over a lifetime without an appreciable health risk. ADIs are expressed in milligrams of the substance per kilogram of body weight per day (mg/kg body weight/day). An ADI value is based on current research, with long-term studies on animals and observations on humans. The studies are performed with several doses including high doses. First, a No Observable Adverse Effect Level (NOAEL) is determined based on these studies. The NOAEL refers to the amount of a pesticide that shows no toxic effects. In the case of several studies on different effects, the lowest NOAEL is usually taken. The NOAEL is then scaled down by a safety factor, conventionally 100, to account for the differences between test animals and humans and possible differences in sensitivity among humans.

d. Acute reference dose (ARfD)

The ARfD is defined as an amount of a pesticide in food or drinking water that can be ingested in 24 hours or less without appreciable health risks to the consumer based on all known facts at the time of the evaluation. This is expressed on a bodyweight basis. The ADI and ARfD are calculated before any pesticide approval is given. The MRL is not linked to the ADI or ARfD, and could result in dietary intakes considerably below these safety levels.

Check Your Progress Exercise 5

Note: a) Use the spaces given below for your answers.
b) Check your answer with those given at the end of the unit.

1. What is pesticide residue?

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2. What sort of effects can pesticide residues inflict on organisms?

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3. Define the waiting period.

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4. Differentiate between ADI and ARfD.

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5.7 LET US SUM UP

After studying this unit, you have learned that:

- Since man started agriculture, he has been disturbing the balance of nature by reducing environmental resistance through ecologically unsound methods of cultivation. Even any slight decrease in any of the processes of 'environment resistance' results in a population explosion of an insect species and the consequent epidemics. Man's interference in ecosystems thus causes the intensification of insects and diseases on crops.
- Rapid and accurate diagnosis of insect pests and diseases is necessary before we can adopt proper control measures. It is the first step towards

the study of any pest and its management. Diagnosis is largely done based on the characteristic symptoms expressed by the affected plant.

- Sole reliance on pesticides created several problems, such as the development of resistant pests, pest outbreaks, mortality of useful organisms, adverse effects on human health, and environmental degradation. As a result, integrated pest management came into vogue.
- Integrated pest management (IPM) refers to the use of available pest control methods in a unified manner to reduce the pest population below economic levels and ensure favourable economic, ecological, and economic consequences.
- In IPM, different methods of pest control, such as, resistant varieties, cultural methods, physical methods, natural enemies, and pesticides are integrated suitably to suppress the pest population without jeopardizing other components of the environment. Any pest management programme should preferably be based upon the use of resistant cultivars and natural enemies and other control methods should be blended with them for tackling the pest menace.
- Pesticides are poisons and can cause carcinogenic, mutagenic, and teratogenic effects in organisms. Therefore to safeguard human and animal health, safety measures like acceptable daily intake (ADI), waiting periods, maximum residual limits (MRLs), *etc.* are fixed and implemented.

5.8 KEYWORDS

Diagnosis: It is the identification of the nature and cause of anything.

Pest resurgence: When a resistance problem arises neither pesticides kill pests nor their natural enemies are present, and hence pest population explodes. This is known as pest resurgence.

Pest: A plant or animal detrimental to humans or human concerns (as agriculture or livestock production)

5.9 SUGGESTED READINGS/ REFERENCES

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5.10 ANSWERS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress Exercise 1

1. Due to interaction between an organism's biotic potential and environmental resistance.
2. Monoculture, multiple cropping, monogenic resistant cultivars, *etc.*

Check Your Progress Exercise 2

1. When the pest population suddenly attains a very high population, this is called a pest epidemic.
2. Human manipulation of the environment, foreign pests, *etc.*

Check Your Progress Exercise 3

1. To adopt a proper control method
2. Wheat
3. Mycoplasma like organisms
4. Hyperplasia is the abnormal increase in the size of a plant organ due to an increase in the number of cells while hypertrophy refers to the increased size of the organ due to an increase in the size of cells.

Check Your Progress Exercise 4

1. The EIL refers to the minimum pest population or damage that causes economic damage.
2. Enhanced efficiency of some natural enemies
3. Clean cultivation, tillage, sowing time, intercropping, crop rotation, *etc.*

Check Your Progress Exercise 5

1. The amount of pesticide, its derivatives, and adjuvants found in or on a treated substrate, such as a plant, animal, or soil after various intervals of time is called "residue".
2. Carcinogenic, mutagenic, and teratogenic effects.
3. The waiting period is the time that must elapse between pesticide application and crop harvest.
4. The ADI is the amount of a pesticide in food or drinking water that can be ingested (orally) daily over a lifetime without an appreciable health risk. The ARfD is defined as an amount of pesticide in food or drinking water that can be ingested in 24 hours or less without appreciable health risks to the consumer based on all known facts at the time of the evaluation.

5.11 TERMINAL QUESTIONS

1. Explain the main causes of pest intensification on crops.
2. Describe major factors responsible for pest epidemics.
3. Write a short note on Koch's postulates.
4. Discuss principles of pest management.
5. Explain pest management approaches.
6. Write a short note on the maximum residue limit (MRL).

